|| Shri Hari ||

BCSE 0101: Digital Image Processing

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Module I

Introduction and Fundamentals: Motivation and Perspective, Applications, Components of Image Processing System, Element of Visual Perception, A Simple Image Model, Sampling and Quantization, Some Basic Relationships between Pixels.

Intensity Transformations and Spatial Filtering: Introduction, Some Basic Intensity Transformation Functions, Histogram Processing, Histogram Equalization, Histogram Specification, Local Enhancement, Enhancement using Arithmetic/Logic Operations – Image Subtraction, Image Averaging, Basics of Spatial Filtering, Smoothing - Mean Filter, Order Statistics Filters, Sharpening – The Laplacian.

Filtering in the Frequency Domain: Fourier Transform and the Frequency Domain, Basis of Filtering in Frequency Domain

Introduction and Fundamentals

A major portion of the information received by a human from the environment is **visual**

Life started on earth about 3.7 billion years ago

Can you think of what big thing happened about 555 million years ago?

The animals got eyes !!

The process of *receiving* & *analyzing* visual information by digital computer is called **Digital Image Processing**

More specifically, DIP is

- Improvement of pictorial information for human interpretation.

- Processing of **image data** for storage, transmission, and representation for autonomous machine perception.

A picture is worth a thousand words.

The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes

Low Level Process	Mid Level Process	High Level Process
Input: Image **Output:** Image	**Input:** Image **Output:** Attributes	**Input:** Attributes **Output:** Understanding
Examples: Noise removal, image sharpening	**Examples:** Object recognition, segmentation	**Examples:** Scene understanding, autonomous navigation

In this course we will stop here

DIP has a broad spectrum of applications

Remote sensing via satellites & other spacecrafts

Radar, sonar & acoustic image processing

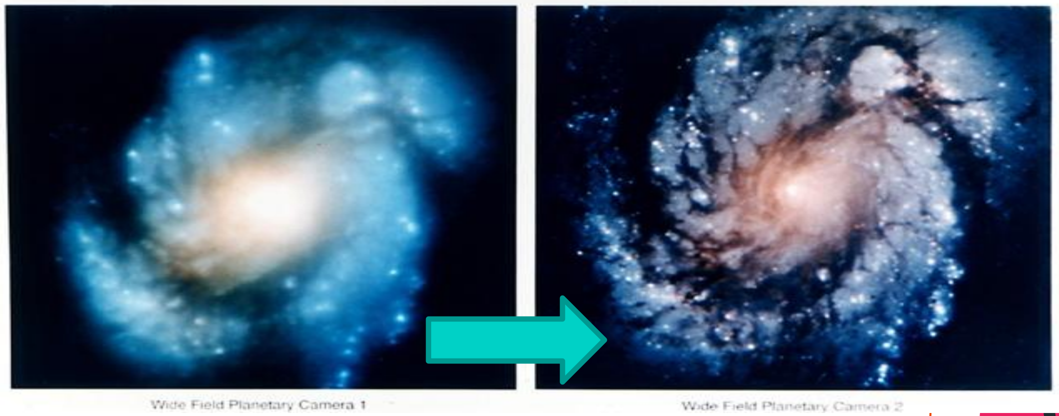
Medical processing

Image transmission & storage for business applications

Automated inspection of industrial parts

THE HUBBLE TELESCOPE

- Launched in 1990 the Hubble telescope can take images of very distant objects
- However, an incorrect mirror made many of Hubble's images useless
- Image processing techniques were used to fix this



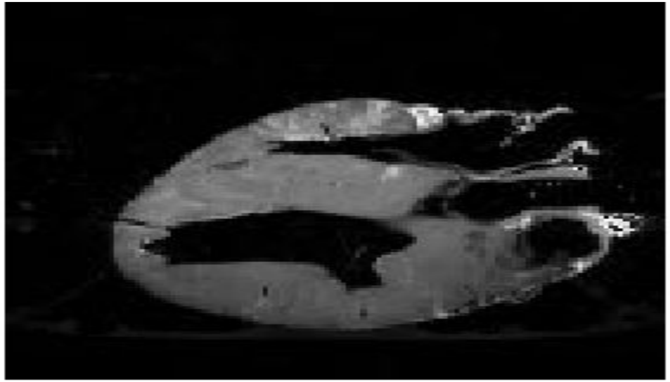
ARTISTIC EFFECTS

Artistic effects are used to make images more visually appealing, to add special effects and to make composite images

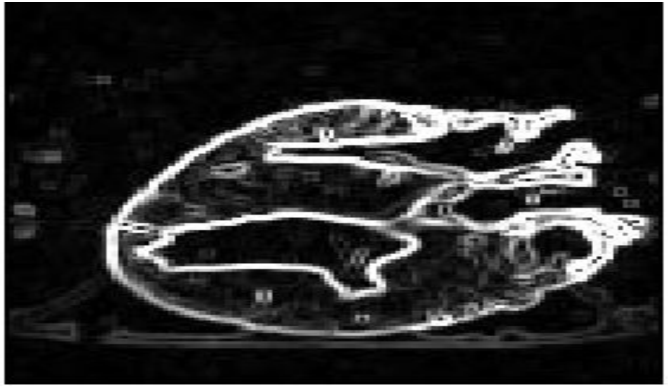
MEDICINE

Take slice from MRI scan of canine heart, and find boundaries between types of tissue

- Image with gray levels representing tissue density
- Use a suitable filter to highlight edges

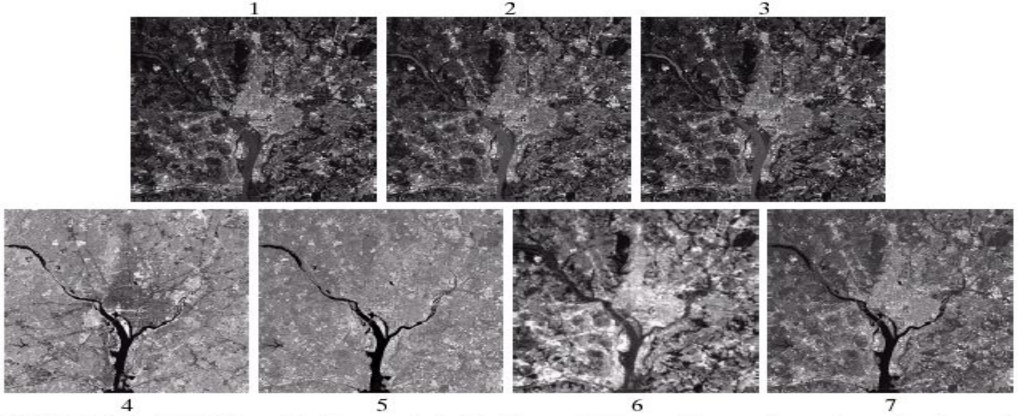


Original MRI Image of a Dog Heart



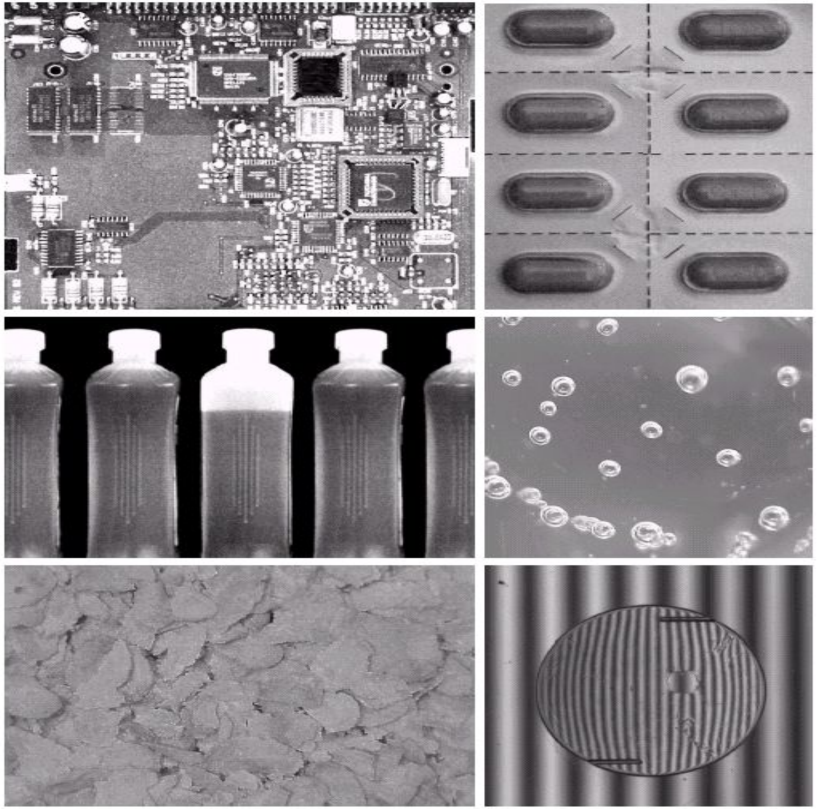
Edge Detection Image

- Digital image processing techniques are used extensively to manipulate satellite imagery
- Terrain classification
- Meteorology



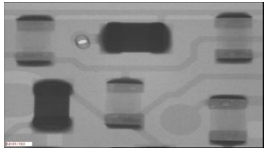
INDUSTRIAL INSPECTION

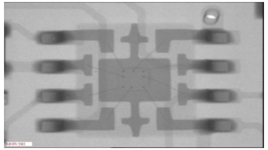
- Human operators are expensive, slow and unreliable
- Industrial vision systems are used in all kinds of industries

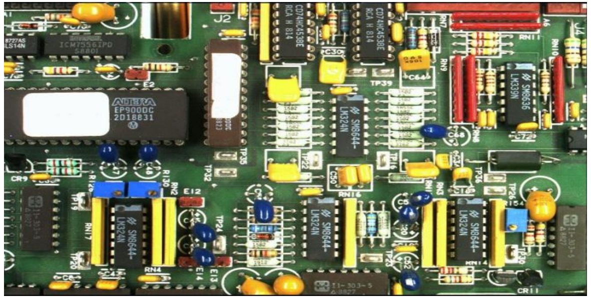


PCB INSPECTION

- Printed Circuit Board (PCB) inspection
  - Machine inspection is used to determine that all components are present and that all solder joints are acceptable
  - Both conventional imaging and x-ray imaging are used

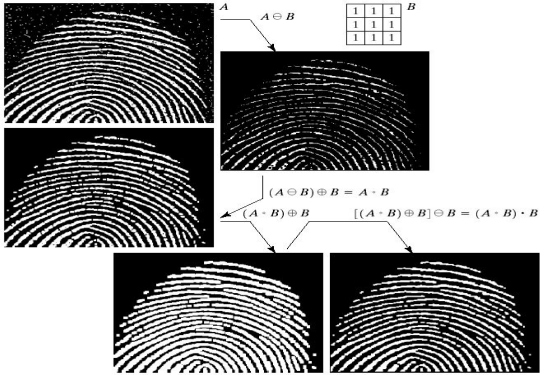






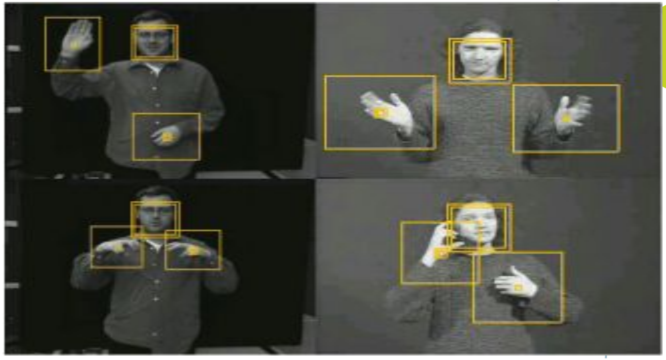
LAW ENFORCEMENT

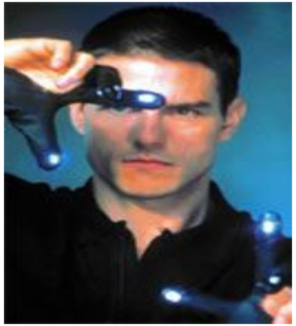
- Image processing techniques are used extensively by law enforcers
  - Number plate recognition for speed cameras/automated toll systems
  - Fingerprint recognition
  - Enhancement of CCTV images



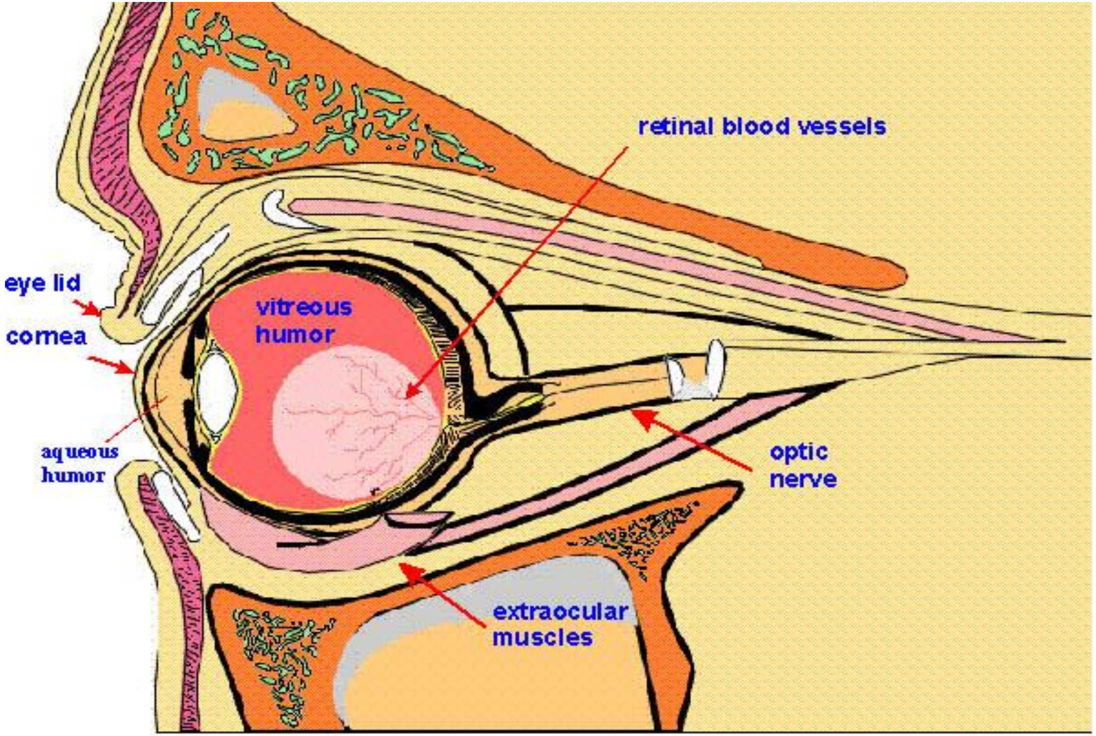
HCI

- Try to make human computer interfaces more natural
  - Face recognition
  - Gesture recognition
- These tasks can be extremely difficult





Elements of Visual Perception



DI
P

Mathematica
l

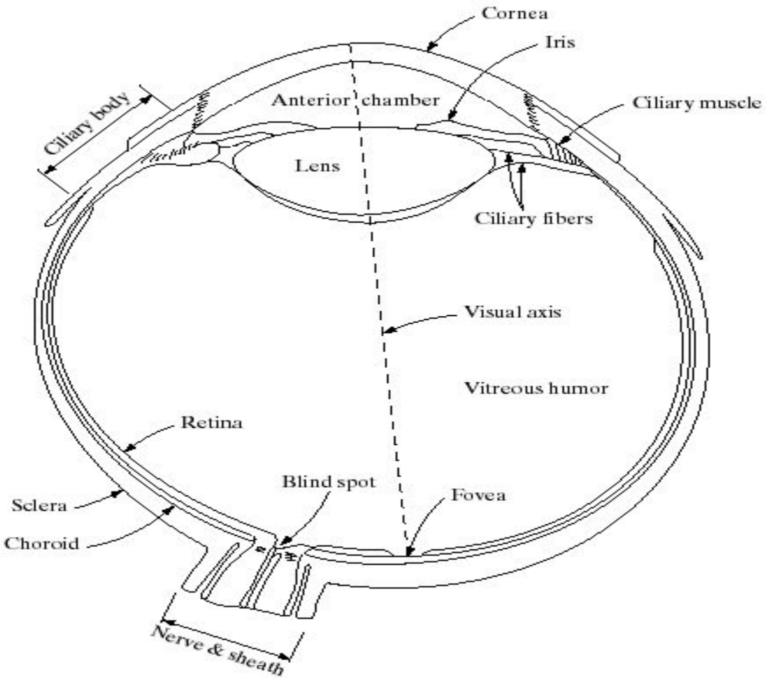
Probabilistic

Formulations

However, the choice of technique is often based on subjective, visual judgment

Hence, understanding Human Visual Perception
VERY IMPORTANT !!!

Human Visual Perception



Cross Section of Human Eye

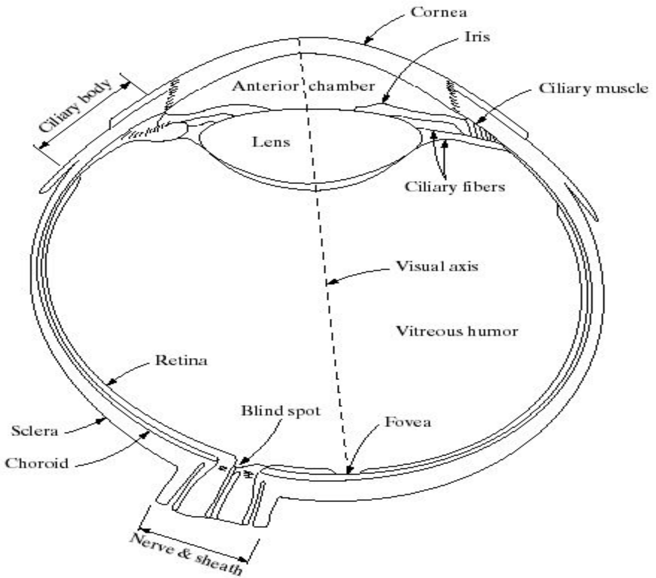
- Avg diameter : 20mm
- 3 membranes enclose the eye

1. Cornea & sclera
2. Choroid
3. Retina

1. Cornea & Sclera

Cornea

- Tough & Transparent tissue
- Covers the anterior surface of the eye

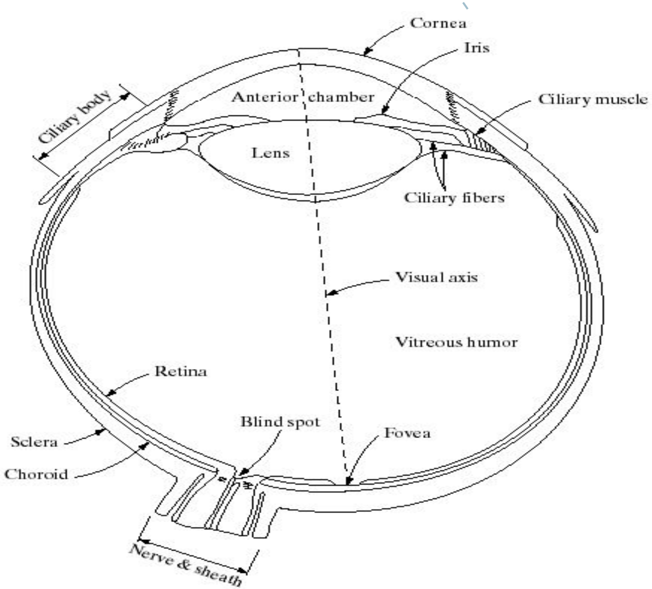


Sclera

- Continuous with cornea
- Opaque membrane
- Encloses the remainder of the optic globe

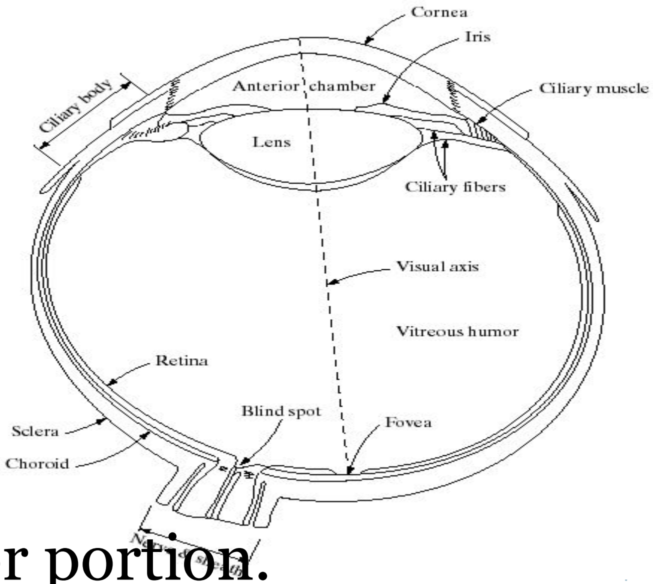
2. Choroid

- Choroid contains blood vessels for eye nutrition; it is heavily pigmented to reduce extraneous light entrance and backscatter.
- Divided into the ciliary body & the iris diaphragm, which controls the amount of light that enters the pupil (2 mm ~ 8 mm).



- Lens made up of fibrous cells & is suspended by fibers that attach it to the ciliary body.
- It is slightly yellow and absorbs approx. 8% of the visible light spectrum.

3. Retina

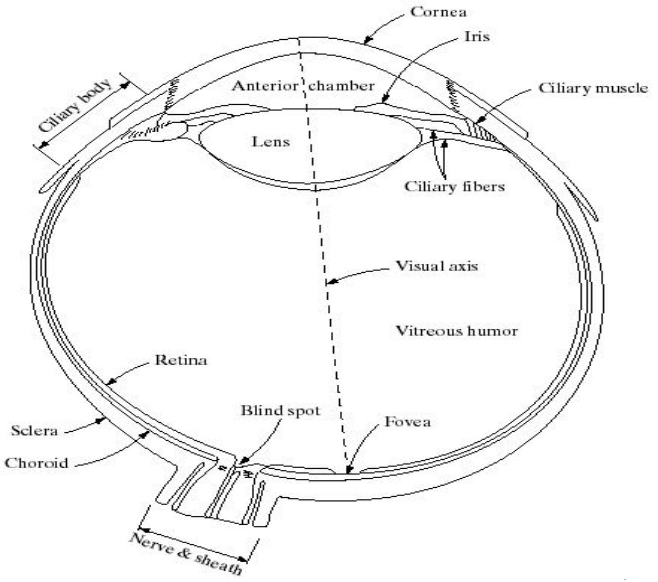


Light from an object is imaged on the retina

- The retina lines the entire posterior portion.
- Discrete light receptors are distributed over the surface of the retina:
  - cones (6-7 million per eye) and
  - rods (75-150 million per eye)

Cones

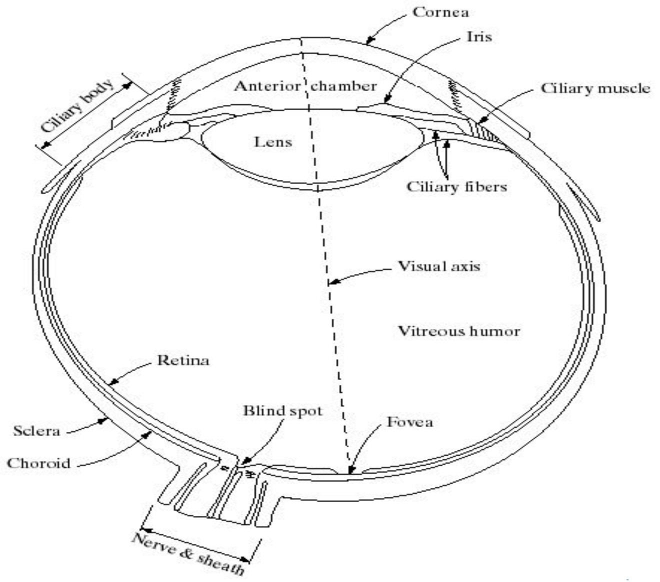
- Cones are located in the fovea and are sensitive to color.
- Each one is connected to its own nerve end.
- Cone vision is called photopic (or bright-light vision).



Muscles controlling the eye rotate the eye ball until the image of an object of interest falls on the fovea.

Rods

- Rods are distributed over the retinal surface
- Rods give a general, overall picture of the field of view and are not involved in color vision.
- Several rods are connected to a single nerve and are sensitive to low levels of illumination (*scotopic* or dim-light vision).



Objects seen by moon light appear as colourless forms because only rods are stimulated.

BLIND-SPOT EXPERIMENT

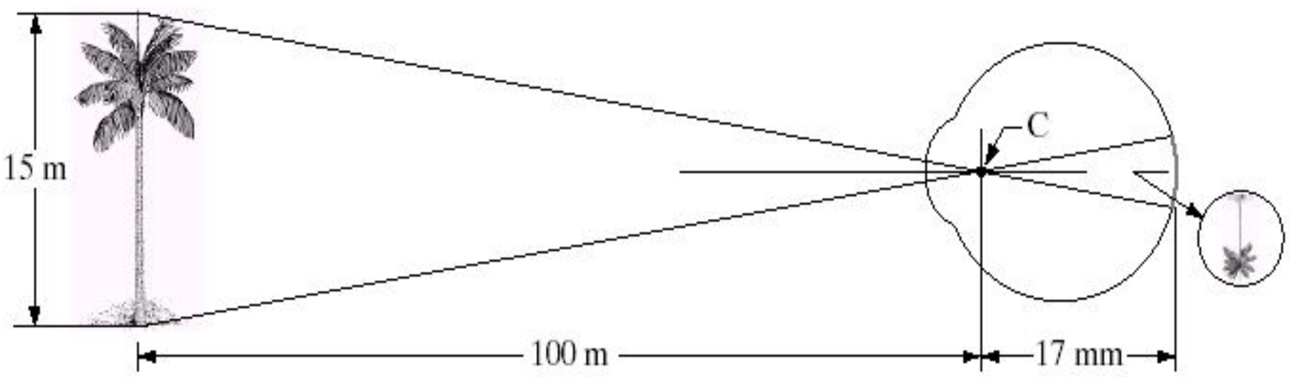
- Draw an image similar to that below on a piece of paper (the dot and cross are about 6 inches apart)

- Close your right eye and focus on the cross with your left eye
- Hold the image about 20 inches away from your face and move it slowly towards you
- The dot should disappear!

Image formation in the Eye

- The range of focal lengths is approximately 14 mm to 17 mm, the latter taking place when the eye is relaxed and focused at distances greater than about 3 m
- Retinal image is reflected primarily in the fovea. Perception takes place by relative excitation of light receptors
- Receptors transform radiant energy into electrical impulses which are decoded by the brain

Graphical representation of the eye looking at a palm tree. Point *C* is the optical center of the lens.



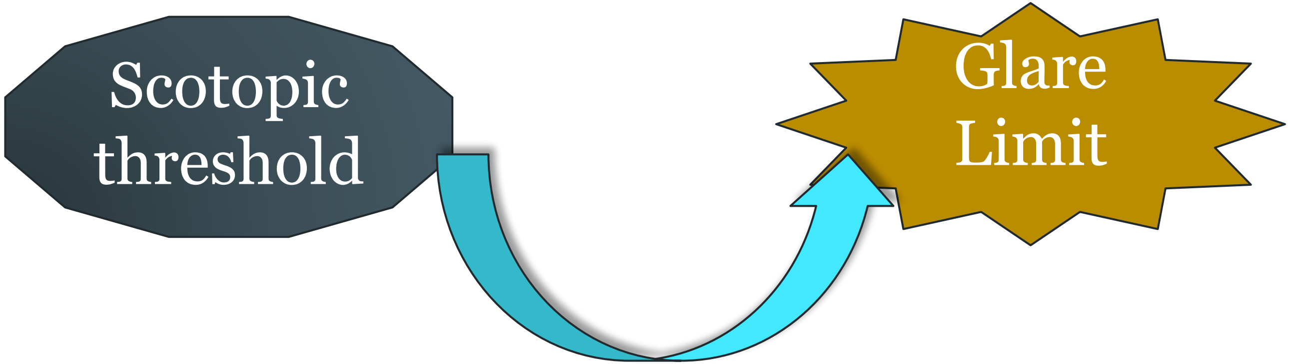
Size of retinal image (h)

15 / 100 = h / 17

h = 2.55 mm

Brightness Adaptation & Discrimination

Range of light intensity levels to which human visual system (HVS) can adapt of the order of 10^{10}

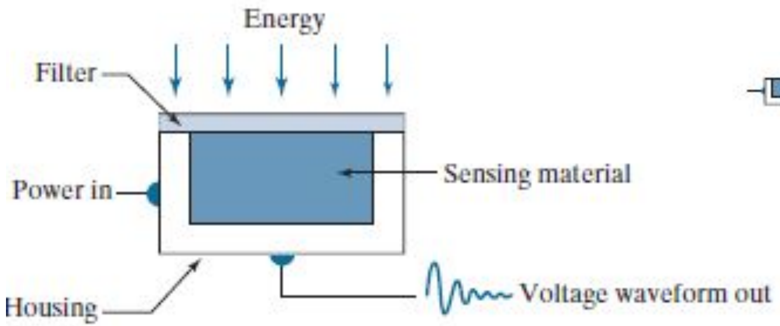


Subjective brightness (i.e. intensity as perceived by the HVS) is a **logarithmic function** of the light intensity incident on the eye.

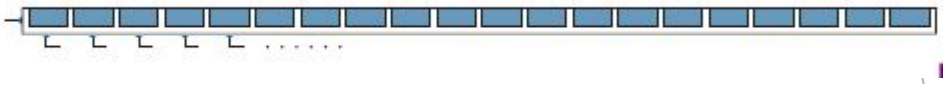
Range of adaptable light intensity levels - 10^{10}

- The HVS cannot operate over such a range simultaneously
- Accomplishes this large variation by changing its overall sensitivity, a phenomenon known as **brightness adaptation.**
- The total range of distinct intensity levels the eye can discriminate simultaneously is rather small

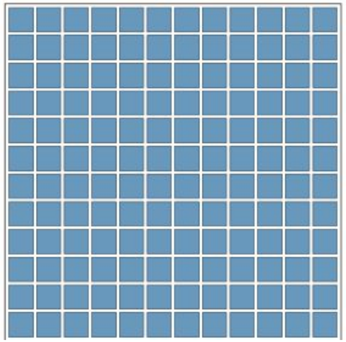
Image Acquisition



Single sensing element.
E.g. photodiode, whose output is a voltage proportional to light intensity.

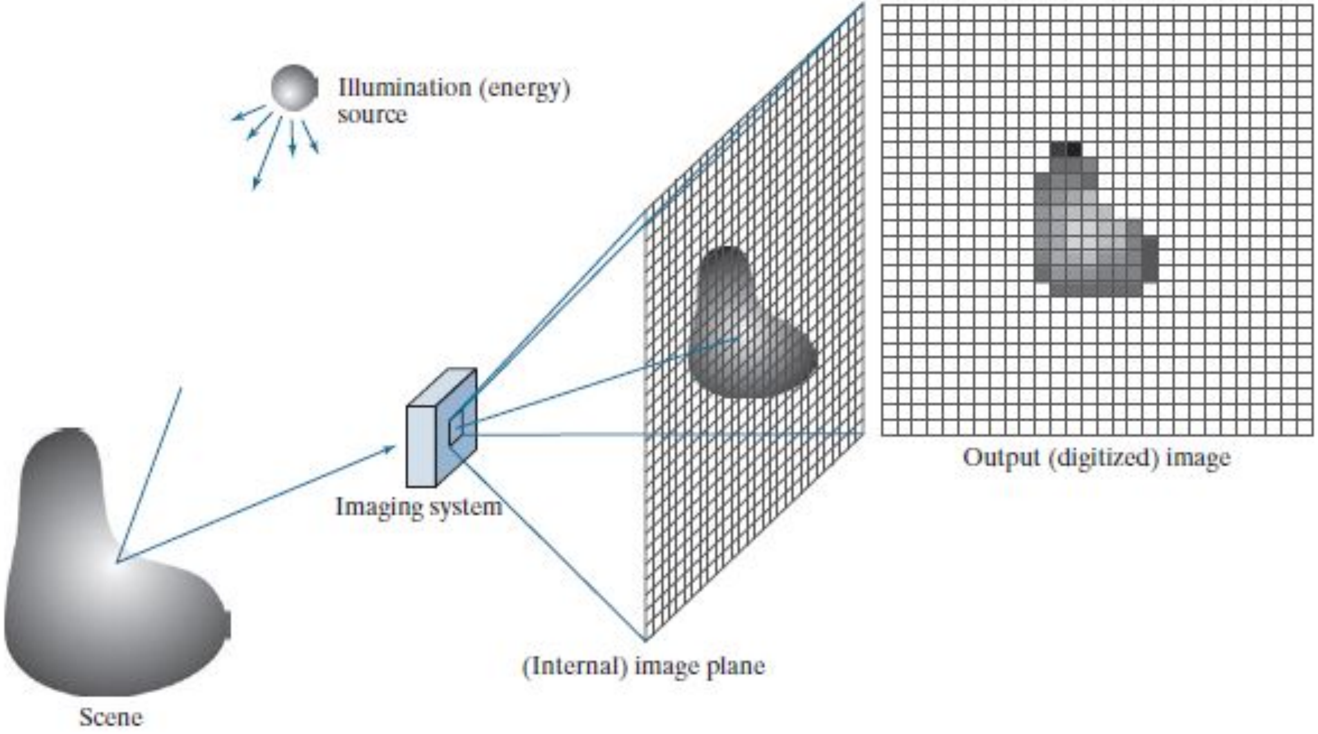


Line Sensor



Array Sensor

A SIMPLE IMAGE FORMATION MODEL



Function f(x, y)

Characterized by two components:

1. ***Illumination i(x, y)*** - the amount of source illumination incident on the scene being viewed, and
2. ***Reflectance r(x, y)*** - the amount of illumination reflected by the objects in the scene.

The two functions combine as a product to form $f(x, y)$

$$\mathbf{f(x, y) = i(x, y)r(x, y)}$$

where

$$0 \leq i(x, y) < \infty$$

$$0 \leq r(x, y) \leq 1$$

Typical values of i(x, y)

- On a sunny day, illumination on earth's surface is 90,000 lm/m²
- On a cloudy day it is 10,000 lm/m²
- Full moon yields 0.01 lm/m²
- Commercial office yields 1000 lm/m²

Typical values of r(x, y)

- for black velvet – 0.01
- Stainless steel – 0.65
- Flat white wall paint – 0.90
- Snow – 0.93

Digital Image Representation

Gray scale images can be represented as a function of two variables, **f(x, y)**,
where f(x, y) is the brightness (or grayness or intensity) of the image at the coordinate (x, y)

In colour image, the intensity is measured in 3 wavelengths (red, green, blue), so

$$f(x, y) = [f_R(x, y), f_G(x, y), f_B(x, y)].$$

00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...
00000000	01110010	01100100	...
00011001	01000010	01100101	...
00011111	00110010	01100101	...

...

What we see

What the computer sees!!!

- **Digital image** is a representation of a two-dimensional image using ones and zeros (binary).

Image Sampling & Quantization

Numerous ways to acquire images

- o/p of most sensors is a continuous voltage waveform whose amplitude & spatial behaviour are related to the physical phenomenon (e.g. brightness) being sensed.
- For an image to be processed by a computer, it must be represented by an appropriate discrete data structure (e.g. Matrix)
- Two important processes to convert continuous analog image into digital image – **Sampling & Quantization**

Discretization

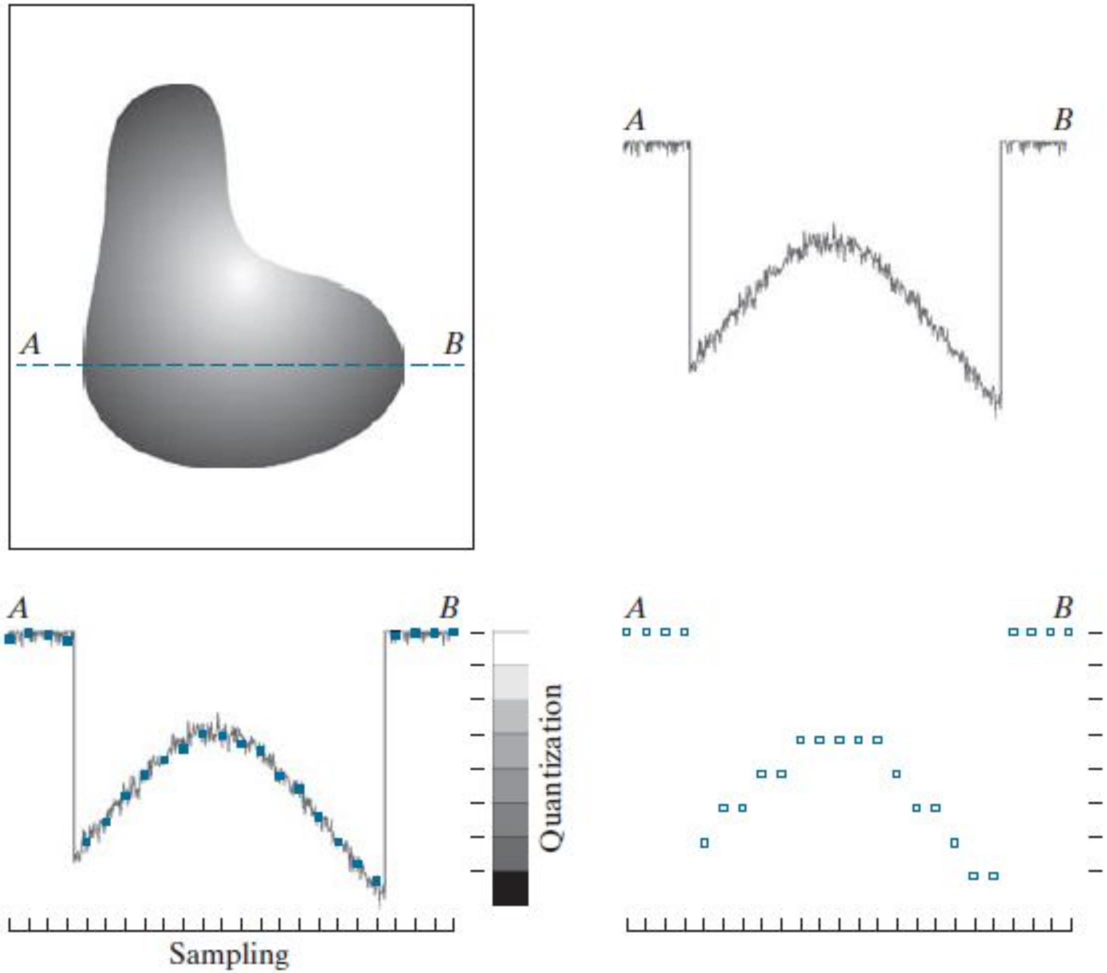
Process in which signals or data samples are considered at regular intervals.

Sampling

It is the discretization of image data in spatial coordinates.

Quantization

It is the discretization of image intensity (gray level) values.

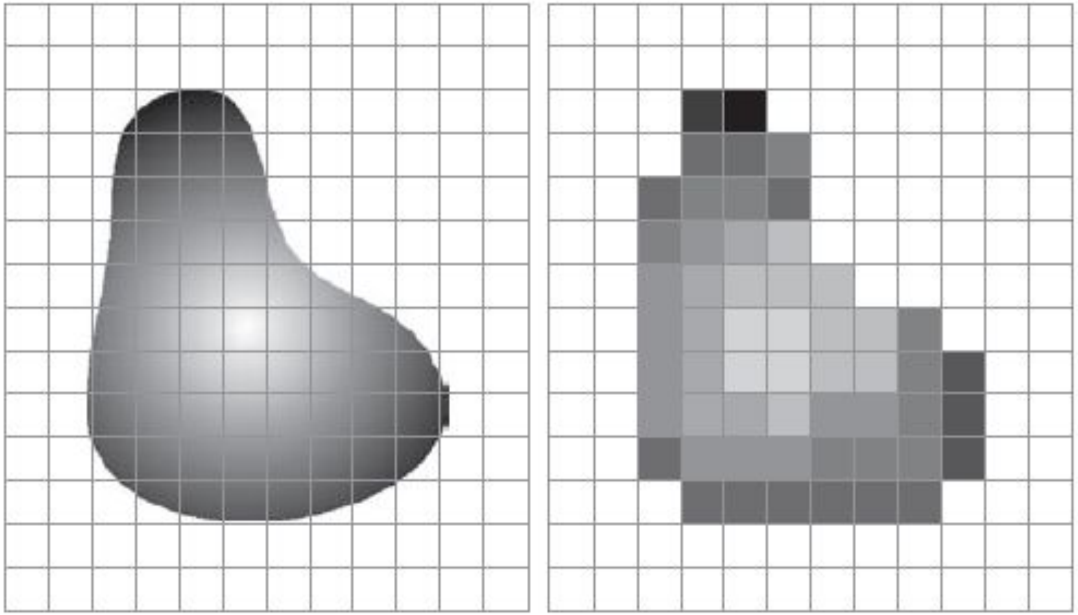


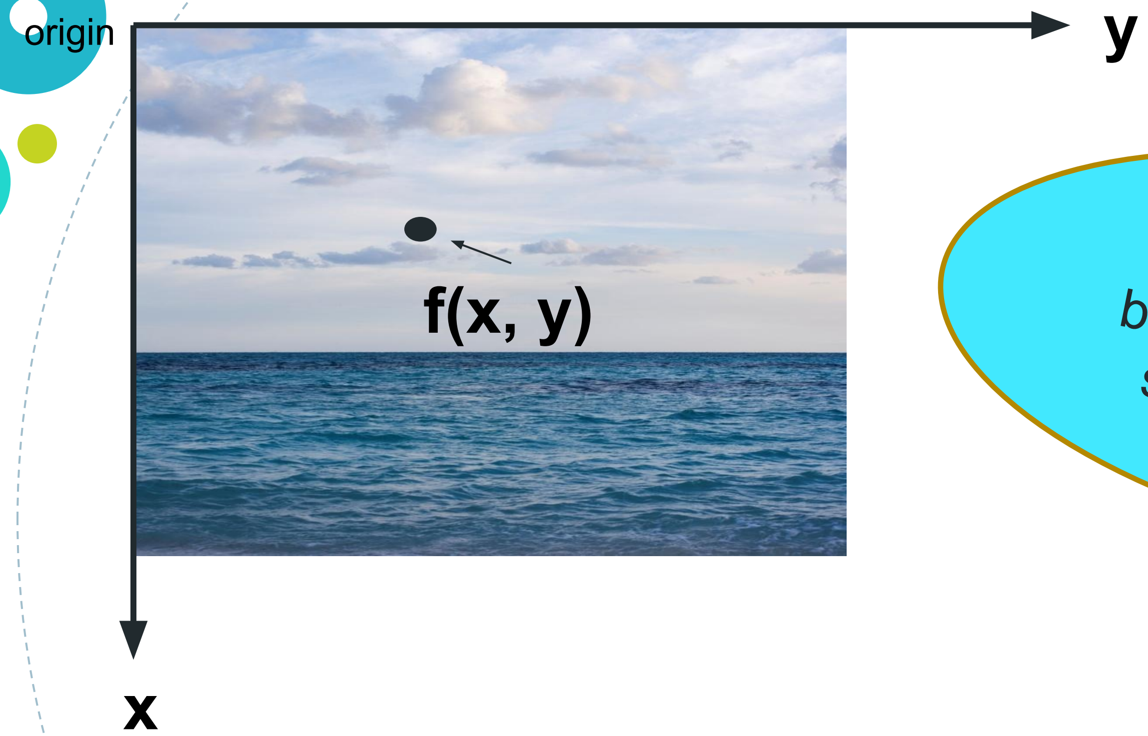
a	b
c	d

FIGURE 2.16 (a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

a b

FIGURE 2.17
(a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



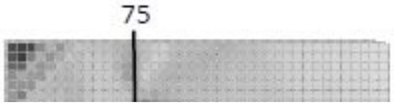


A digital image is an image f(x, y) that has been discretized in both spatial coordinates & brightness.

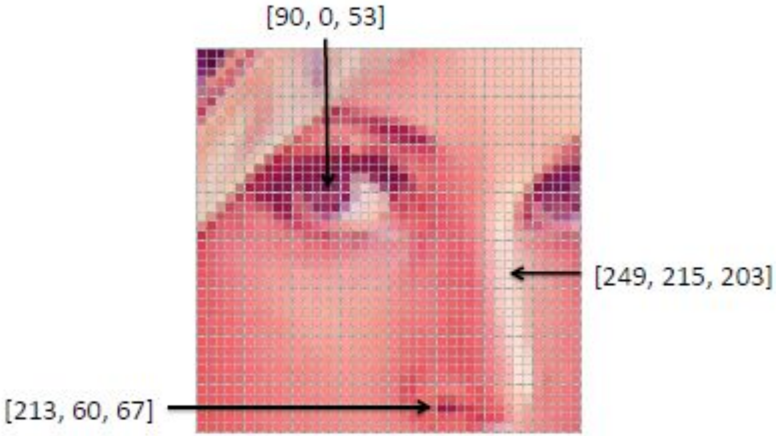
A digital image can be considered as a matrix, where the matrix element value identifies the gray level at that point.

Pixels

- An image contains discrete number of pixels
- Pixel value:
  - "grayscale" (or "intensity"): [0,255]



  - "color"
  - – RGB: [R, G, B]
  - – Lab: [L, a, b]
  - – HSV: [H, S, V]



f(x, y) gives the **intensity** at position (*x, y*)

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \ldots & f(0,N-1) \\ f(1,0) & f(1,1) & \ldots & f(1,N-1) \\ \ldots & \ldots & \ldots & \ldots \\ f(M-1,0) & f(M-1,1) & \ldots & f(M-1,N-1) \end{bmatrix}$$

For digital images M, N and L (no. of gray levels) are mostly considered as powers of 2. Suppose

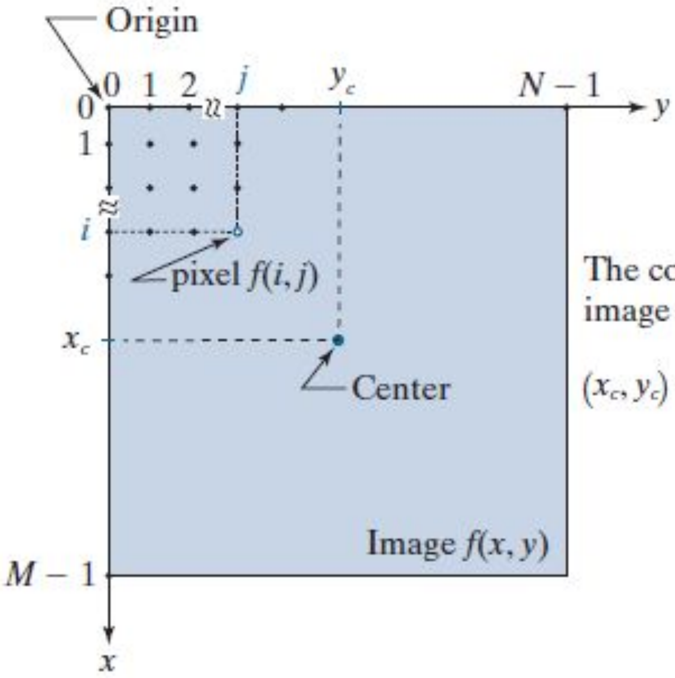
$M = 2^9$, $N = 2^{10}$ & $L = 2^k = 2^8$

Total number of bits b required to represent an image???

b = 512 x 1024 x 8 = 41,94,304 bits

Coordinates of the Centre of the Image?

FIGURE 2.19
Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between x and y and the rows (r) and columns (c) of a matrix.



The coordinates of the image center are

$$(x_c, y_c) = \left(\text{floor}\left(\frac{M}{2}\right), \text{floor}\left(\frac{N}{2}\right)\right)$$

k-bit image
An image having 2^k gray levels

An image having 256 gray levels is called a 8-bit image

$b = N^2 k$

Number of storage bits for various values of N and k.

N/k	1 (L = 2)	2 (L = 4)	3 (L = 8)	4 (L = 16)	5 (L = 32)	6 (L = 64)	7 (L = 128)	8 (L = 256)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

8-bit images of size 1024 x 1024 or higher
Storage Requirement Not Insignificant !!!!

The intensity of a monochrome image at any coordinate (x, y) is called the gray level (I) of the image at that point

Gray Scale
$[L_{min}, L_{max}]$

$$I = f(x, y)$$

$$L_{min} \leq I \leq L_{max}$$

Common Practice

Shift the interval numerically to the interval [0, L-1].

I = 0 is considered black &

I = L-1 is considered white on the gray scale.

Resolution of an Image

Spatial Resolution

It is the smallest discernible detail in an image

Gray-level Resolution

It is the smallest discernible change in the gray level of an image.

L-level digital image of size M x N

Spatial resolution – M x N pixels

Gray-level resolution – L levels

- **Dots per inch (dpi)**
  - E.g. newspapers are printed with a resolution of 75 dpi, magazines at 133 dpi, glossy brochures at 175 dpi
- **Pixels per inch (PPI)**

200 x 200

100 x 100

50 x 50

25 x 25

Measuring Spatial Resolution

Samsung Galaxy S22 Ultra

Expected Launch in India: Feb 2022

Rs. 87,999 (Expected Price)

This product has not been announced yet, and is based on unofficial preli...

What does it mean?

Key Specs

See Full

Android v11

Performance	Display	Camera	Battery
Octa core (2.9 GHz, Si...	6.8 inches (17.27 cm)	108 + 12 + 10 + 10 MP...	5000 mAh
Samsung Exynos 2100	516 PPI, Dynamic AM...	LED Flash	Fast Charging
12 GB RAM	120 Hz Refresh Rate	40 MP Front Camera	USB Type-C Port

Samsung Galaxy S22 Ultra

Expected Launch in India: Feb 2022

Display Type	Dynamic AMOLED
Screen Size	6.8 inches (17.27 cm)
Resolution	1440 x 3200 pixels
Aspect Ratio	20:9
Pixel Density	516 ppi

What is meant by Screen size?

Pixels per inch along the diagonal?

How many pixels along the diagonal of the screen?

$$\mathbf{d} = \sqrt{(1440)^2 + (3200)^2}$$

$$= 3509.07 \approx 3509$$

$$\mathbf{d}_{\mathbf{PPI}} = \frac{3509}{6.8} = 516.02 \approx 516$$

Fixed Gray-level
Varying number of samples

A 1024 × 1024, 8-bit image subsampled down to size 32 × 32 pixels. The number of allowable gray levels was kept at 256.

(a) & (b)
virtually impossible to tell apart. The level of detail lost is too fine.

(c)
V slight fine checkerboard pattern in the borders & some graininess throughout the image

(d), (e), (f)
These effects much more visible.

(a) 1024 × 1024, 8-bit image. (b) 512 × 512 image resampled into 1024 × 1024 pixels by row and column duplication. (c) through (f) 256 × 256, 128 × 128, 64 × 64, and 32 × 32 images resampled into 1024 × 1024 pixels.

Checkerboard Effect

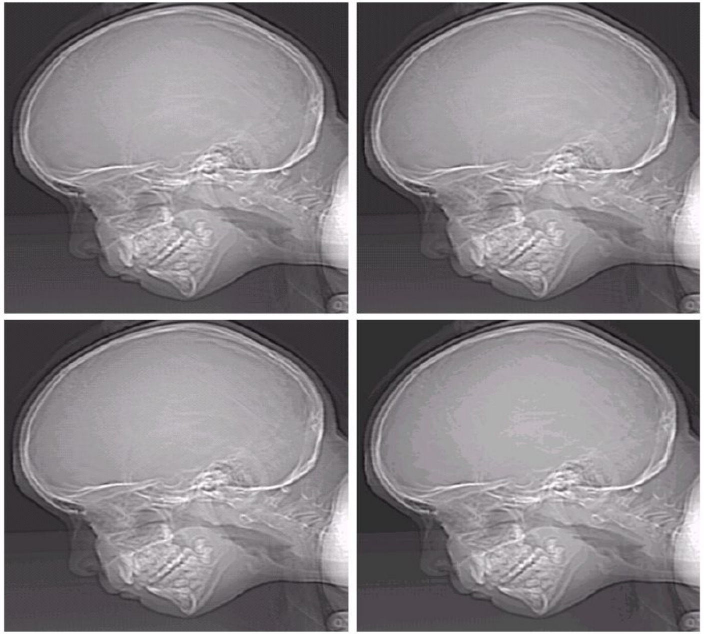
When the no. of pixels in an image is reduced keeping the no. of gray levels in the image constant, fine checkerboard patterns are found at the edges of the image. This effect is called the checker board effect.

Measuring Gray Level Resolution

- Common practice to refer to the number of bits used to quantize intensity as the "**intensity resolution**."
- For example, it is common to say that an image whose intensity is quantized into 256 levels has 8 bits of intensity resolution.

Varying Gray-level
Fixed number of samples

False contouring

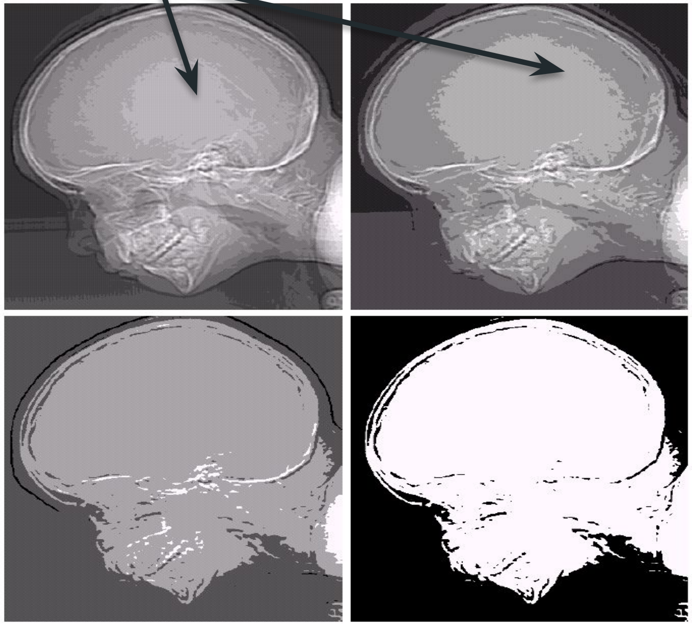


a b
c d

FIGURE 2.21
(a) 452 × 374, 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

e f
g h

(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



False Contouring

When the no. of gray-levels in the image is low, the foreground details of the image merge with the background details of the image, causing ridge like structures.This degradation phenomenon is known as false contouring.

(The ridges resemble topographic contours in a map)

Baud rate

A common measure of transmission for digital data. It is the no. of bits transmitted per second.

Q) Generally transmission is accomplished in packets consisting of a start bit, a byte of information & a stop bit.

How many minutes would it take to transmit a 1024 x 1024 image with 256 gray levels using a 56K baud rate?

Total no. of bits needed to represent the image:

1024 x 1024 x 8

No. of packets required :

1024 x 1024

Total no. of bits that need to be transferred:

1024 x 1024 x [8 + 2]

Total time required to transmit over 56K baud modem:

$$\frac{1024 \times 1024 \times 10}{56000}$$

= 187.25 sec or 3.1 min

SOME BASIC RELATIONSHIPS BETWEEN PIXELS

Neighbours of a Pixel

NW	N	NE
W	X	E
SW	S	SE

X_3	X_2	X_1
X_4	p	X_0
X_5	X_6	X_7

$N_4(p)$

Pixel p at coordinates (x, y) has 4 horizontal & vertical neighbours.

(x-1, y) (x+1, y) (x, y-1) (x, y+1)

$N_D(p)$

Pixel p at coordinates (x, y) has 4 diagonal neighbours.

(x-1, y-1) (x+1, y-1) (x-1, y+1) (x+1, y+1)

$N_4(p)$ + $N_D(p)$ = $N_8(p)$

- If pixel p lies on the boundary, then some of the neighbours of p will lie outside the digital image.
- The set of image locations of the neighbors of a point p is called the neighborhood of p.
- The neighborhood is said to be closed if it contains p.
- Otherwise, the neighborhood is said to be open.

Adjacency

V

The set of gray-level values used to define adjacency

In a gray-scale image, V typically contains more than one elements. E.g. in an image of 265 gray-levels, V could be any subset of these 256 values.

In a binary image, V = {1} for finding the adjacency of a pixel with value 1.

Types of Adjacency

I) 4-adjacency

Two pixels p & q with values from V are 4-adjacent if q is in the set $N_4(p)$.

II) 8-adjacency

Two pixels p & q with values from V are 8-adjacent if q is in the set $N_8(p)$.

III) m-adjacency (mixed adjacency)

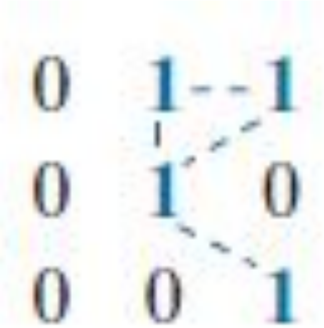
Two pixels p & q with values from V are m-adjacent if

a) q is in the set $N_4(p)$, or
b) q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V.

Mixed adjacency is a modification of 8-adjacency. It is used to eliminate ambiguities that may arise when 8-adjacency is used.

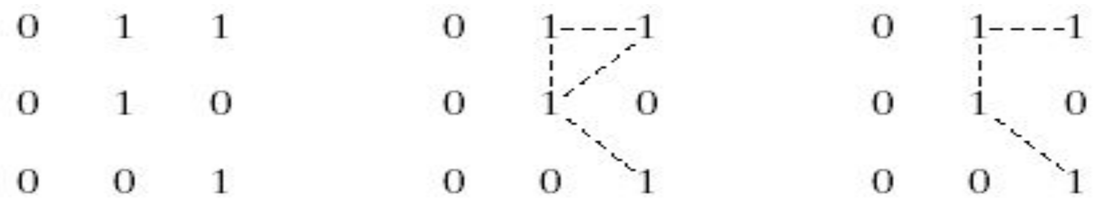
0	1	1
0	1	0
0	0	1

Find 8-adjacency and m-adjacency of the pixel in the centre.
Note: V = {1}



8 adjacency

m - adjacency



(a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) *m*-adjacency.

Path can be 4-, 8- or m-path, depending upon the type of adjacency specified.

Path between NE & SE pixels

8-path shown in (b) & m-path shown in (c)

Note: No ambiguity in m-path

Digital Path or Curve

From pixel p with coordinates (x_0, y_0)
to pixel q with coordinates (x_n, y_n)

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

where pixels (x_i, y_i) & (x_{i-1}, y_{i-1}) are adjacent for $1 \le i \le n$

n is the length of the path

If $(x_0, y_0) = (x_n, y_n)$ then the path is closed

Connected in S

- S – subset of pixels in an image.

Pixels p & q are said to be *connected in S*, if there exists a path between them consisting entirely of pixels in S.

For any pixel p in S, *the set of pixels* that are connected to it in S, is called the *connected component of S.*
If S has only one connected component, then S is called a *connected set.*

Region

R – subset of pixels in an image.

4 R is a ***region*** of an image if **R is a connected set**

4 R_i & R_j are ***adjacent*** if their union forms a connected set; otherwise the regions are said to be *disjoint*.

1	1	0	0	1	1
1	1	0	1	0	1
1	1	1	0	1	1

Are the two regions adjacent?

- Suppose an image contains K disjoint regions, R_k , k =1, 2,…, K, none of which touches the image border.
- Let R_u denote the union of all the K regions.
- Let $(R_u)^C$ denote its complement
- We call all the points in R_u the foreground, and all the points in $(R_u)^C$ the background of the image.

Boundary

- The boundary (aka border or contour) of a region R is the set of pixels in R that have one or more neighbours that are not in R.
- If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns of the image.

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Is it a boundary pixel?

Boundary

another definition

Inner border of the region -

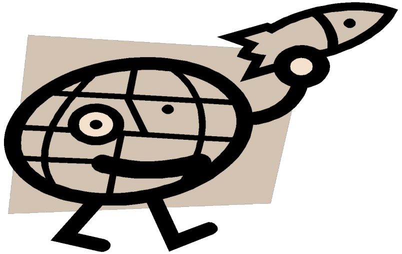
set of pixels in R that are adjacent to pixels in the complement of R.

Outer border - which is the corresponding border in the background. This distinction is important in the development of border-following algorithms. Such algorithms usually are formulated to follow the outer boundary in order to guarantee that the result will form a closed path.

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

Edge vs Boundary

Boundary of a region
forms a closed path
"Global Concept"



Edges are formed from pixels with derivative values that exceed a preset threshold. It is thus a local concept that is based on a measure of intensity-level discontinuity at a point.

Distance Measures

Pixel	Coordinate
p	(x, y)
q	(s, t)

Euclidean distance between p & q

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$

City-block distance between p & q

$$D_4(p, q) = |x - s| + |y - t|$$

Chessboard distance between p & q

$$D_8(p, q) = \max(|x - s| , |y - t|)$$

What will be the shape of the figure which consists of all pixels which are at a distance of **≤ 3** from pixel (3, 3), when the distance measure is

- Euclidean ?
- City block ?
- Chessboard ?

			(3,3)			

	0	1	2	3	4	5	6
0	(0, 0)						
1							
2							
3				(3,3)			
4							
5							
6							

			3.0			
	2.8	2.2	2.0	2.2	2.8	
	2.2	2.0	1.0	2.0	2.2	
3.0	2.0	1.0	**0.0**	1.0	2.0	3.0
	2.2	2.0	1.0	2.0	2.2	
	2.8	2.2	2.0	2.2	2.8	
			3.0			

Euclidean Distance
Disk shape

			3.0			
		3.0	2.0	3.0		
	3.0	2.0	1.0	2.0	3.0	
3.0	2.0	1.0	**0.0**	1.0	2.0	3.0
	3.0	2.0	1.0	2.0	3.0	
		3.0	2.0	3.0		
			3.0			

City-block Distance
Diamond shape

3.0	3.0	3.0	3.0	3.0	3.0	3.0
3.0	2.0	2.0	2.0	2.0	2.0	3.0
3.0	2.0	1.0	1.0	1.0	2.0	3.0
3.0	2.0	1.0	**0.0**	1.0	2.0	3.0
3.0	2.0	1.0	1.0	1.0	2.0	3.0
3.0	2.0	2.0	2.0	2.0	2.0	3.0
3.0	3.0	3.0	3.0	3.0	3.0	3.0

Chessboard Distance
Square shape

- The *D_4 and D_8 distances between p and q are independent of any paths* that might exist between these points because these distances involve only the coordinates of the points.
- In the case of *m-adjacency, however, the D_m distance between* two points is defined as the shortest *m-path between the points*.
- *In this case, the* distance between two pixels will depend on the values of the pixels along the path, as well as the values of their neighbors.
- For instance, consider the following arrangement of pixels and assume that *p, p2 , and p4 have a value of 1, and that p1 and p3 are 0*

	p_3	p_4
p_1	p_2	
p		

$D_m(p, p_4) = ?$

If value of $p_1 = 1$

$D_m(p, p_4) = ?$

Intensity Transformations and Spatial Filtering

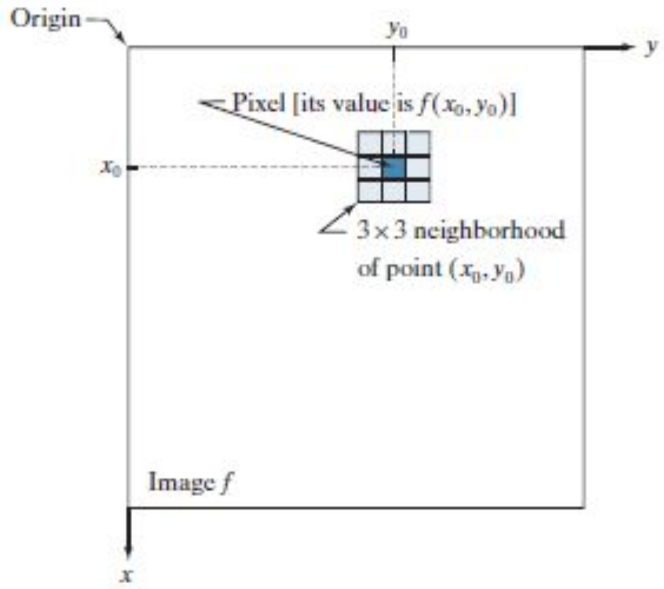
We will now see some Image Processing Techniques implemented in **Spatial Domain** (the plane containing the pixels of an image)

$$g(x, y) = T[\, f(x, y)]$$

Output Image

Operator on f defined over a neighbourhood of point (x, y)

Input Image



The point (x_o, y_o) *is an arbitrary location in the image*

- The small region shown is a *neighborhood of (x_o, y_o).*
- Move the center of the neighborhood (x_o, y_o), from pixel to pixel, and apply the operator *T to the pixels in the neighborhood* to yield an output value at that location.
- For any specific location (x_o, y_o), the value of the output image *g* at those coordinates is equal to the result of applying T to the neighborhood with origin at (x_o, y_o) in f.

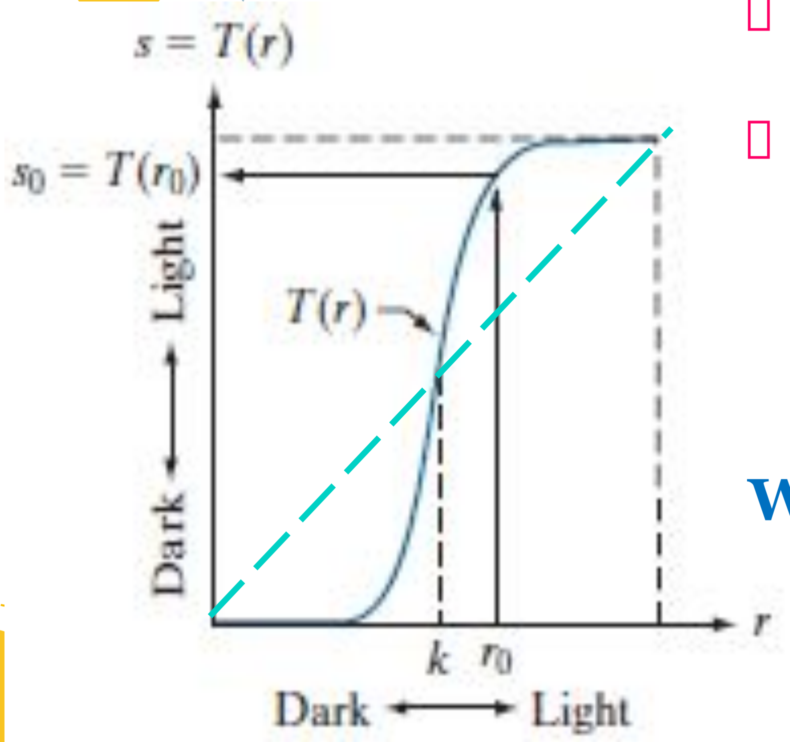
- Suppose that the neighborhood is a square of size 3 × 3
- Operator T *is defined as "compute* the average intensity of the pixels in the neighborhood."
- **Find g(2, 2)**

200	200	200	200	100
200	100	100	200	100
200	100	**100**	200	100
200	100	200	200	100
200	100	100	100	100

g(2, 2) = (100 + 100 + 200 + 100 + 100 + 200 + 100 + 200 + 200) / 9

= **144**

Intensity Transformation Function
(aka called a gray-level, or mapping)



- The smallest possible neighborhood is of size 1 × 1.
- In this case, *g depends only* on the value of *f at a single point (x, y)*

$$s = T(r)$$

What is this transformation doing?

Contrast Stretching

Intensity transformations & Spatial filtering have a broad range of applications.

One important application

Image Enhancement

Enhanced, but we are not going to do such enhancements!!

Principal Objective Of Enhancement

To process an image so that the result is more suitable than the original image for a specific application

- Enhance otherwise hidden information
- Filter important image features
- Discard unimportant image features

Basic Gray Level Transformations

Amongst the simplest of all image enhancement techniques

$$s = T(r)$$

- **Linear transformations**
  - Image negative
- **Non-linear transformations**
  - Logarithmic transformation
  - Power Law (Exponential) transformation
- **Piecewise-linear transformations**
  - Contrast stretching
  - Gray-level slicing
  - Bit plane slicing

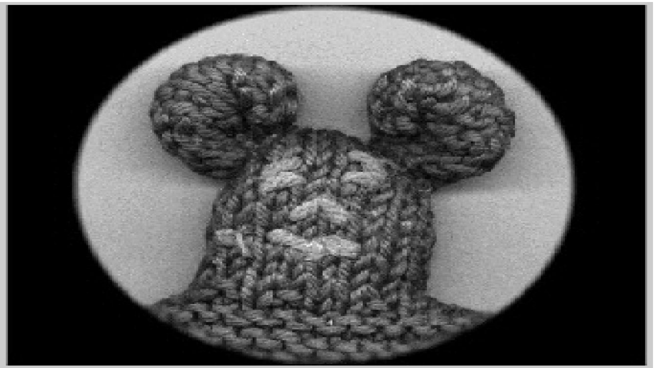
1. Image Negatives

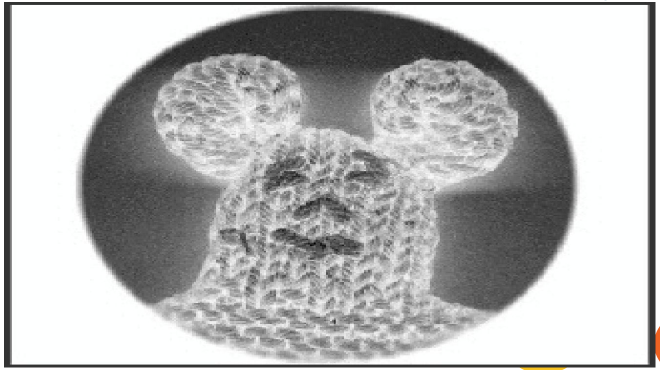
- Gray levels of the original image in the range of [0, L-1]
- Then the negative of an image can be obtained by reversing the intensity levels, the expression for which is given as:

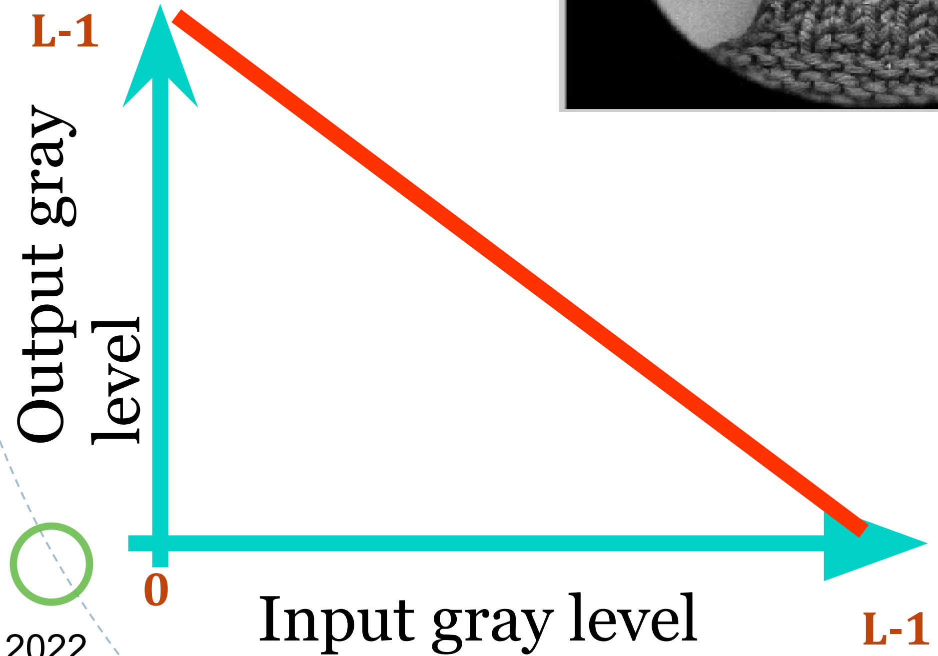
$$s = L - 1 - r$$

- This type of processing is suited for enhancing white or gray detail embedded in dark regions of an image, especially when black areas are dominant.

Image Negatives







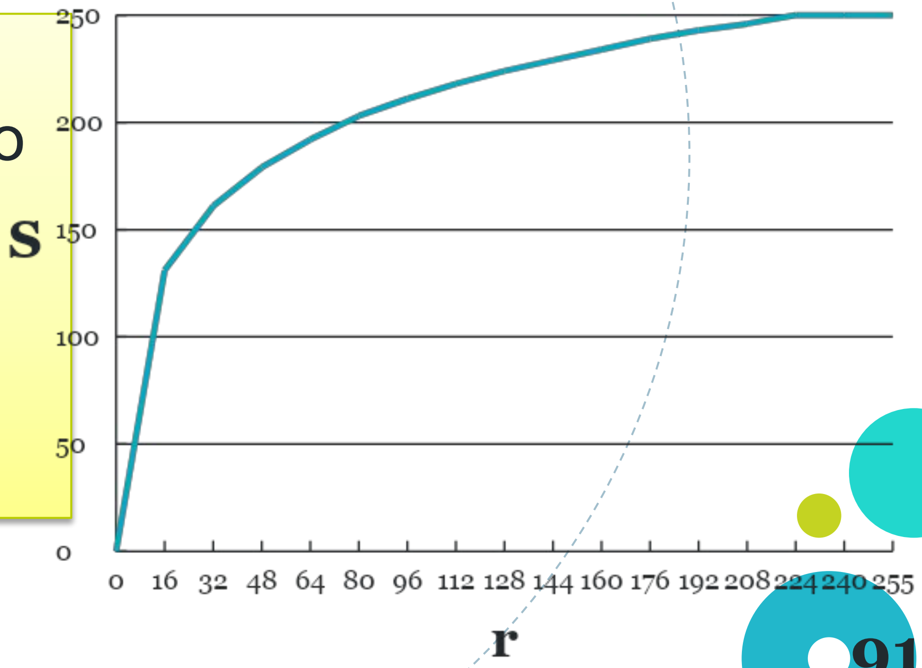
2. Log Transformations

General form:

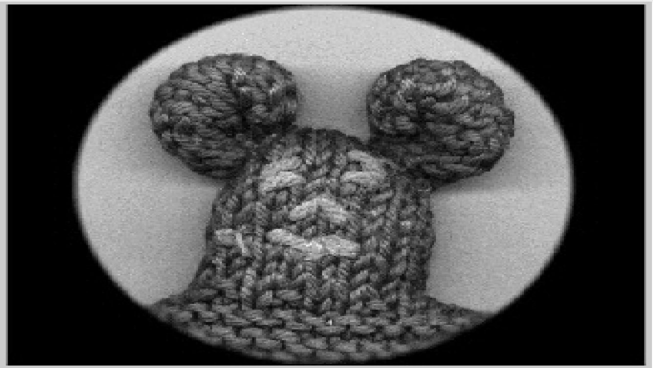
$$s = c \log(1 + r)$$

where c is a constant & $r \geq 0$.

- ✔ Maps a narrow range of low gray levels in the i/p image into wider range of o/p levels.
- ✔ Opposite is true at higher values of i/p levels.



This type of processing is used to expand the values of dark pixels in an image while compressing the higher level values.

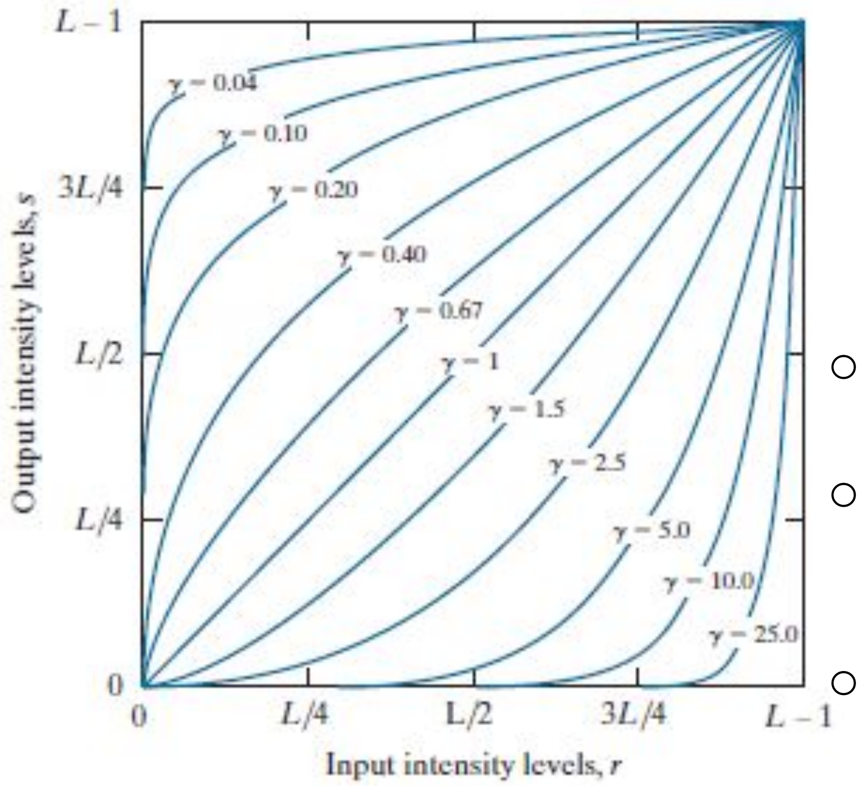


InvLog

Log

Ja

3. PowerLaw (Gamma) Transformations



General form:

$$s = c\ r^{\gamma}$$

Where c & γ are positive constants.

When γ < 1

- map a narrow range of dark i/p values into wider range of o/p values.
- Opposite is true for higher i/p values.

When γ > 1

- Exactly the opposite effect as those generated with values of γ < 1.

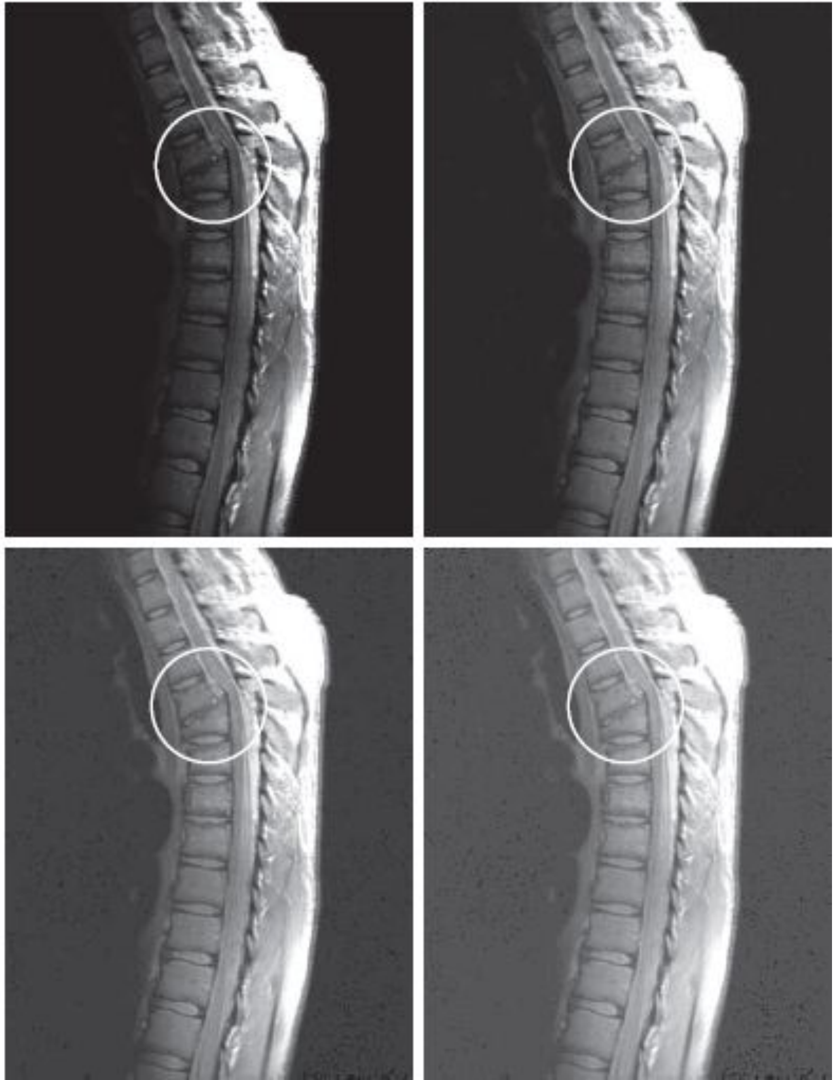
Varying *gamma* (γ) obtains family of possible transformation curves

$\gamma > 1$

- Compresses dark values
- Expands bright values

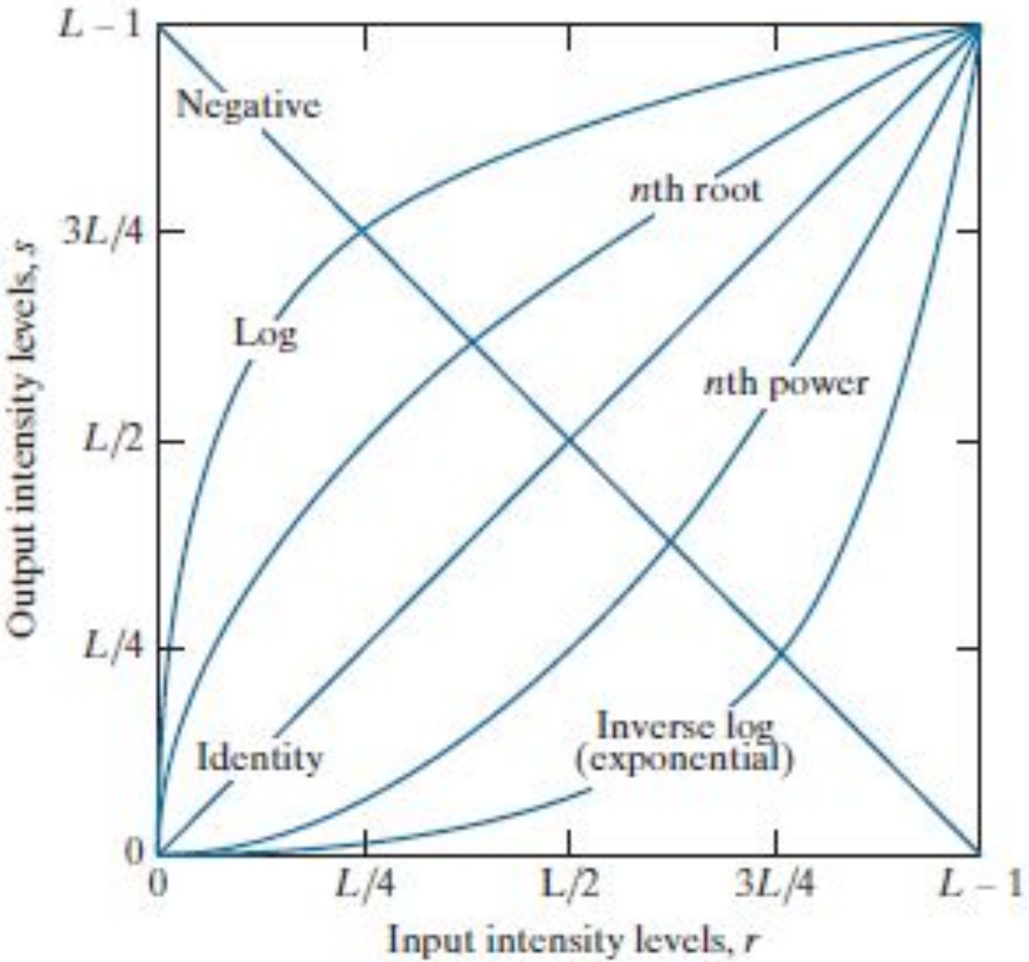
$\gamma < 1$

- Expands dark values
- Compresses bright values



(a) Magnetic resonance image (MRI) of a fractured human spine (the region
of the fracture is enclosed by the circle).
(b)–(d) Results of applying the Gamma Transformation with $c = 1$ *and* $\gamma = 0.6$, 0.4, and 0.3, respectively.

Some basic intensity transformation functions.
Each curve was scaled independently so that all curves would fit in the same graph.
The interest here is on the shapes of the curves, not
on their relative values.



Piecewise-Linear Transformation Functions

A complementary approach to the previous methods

- Form of piecewise functions can be arbitrarily complex.
- Practical implementation of some important transformations can be formulated only as piecewise functions

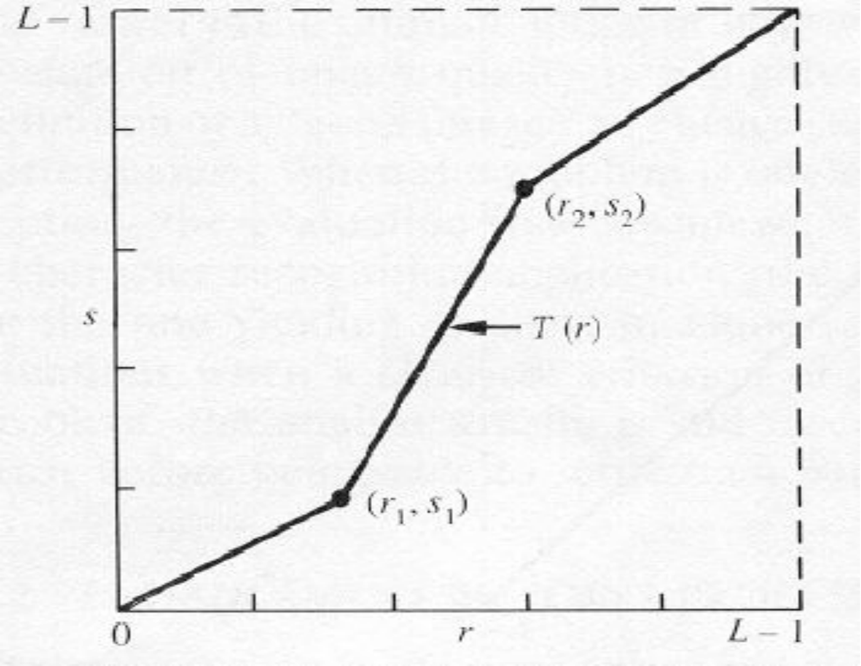
- Their specification requires considerably more user i/p.

4. Contrast Stretching

- Simplest of piecewise linear functions.
- Basic idea – to increase the dynamic range of the gray levels in the image being processed.

During image acquisition, low contrast images may result due to

- ♦ Poor illumination
- ♦ Lack of dynamic range in image sensor
- ♦ Wrong setting of the lens aperture



Typical transformation function used for contrast stretching

The locations of (r_1,s_1) and (r_2,s_2) control the shape of the transformation function

If $r_1 = s_1$ and $r_2 = s_2$?

- If $r_1 = s_1$ and $r_2 = s_2$ the transformation is a **linear function** and produces no changes.

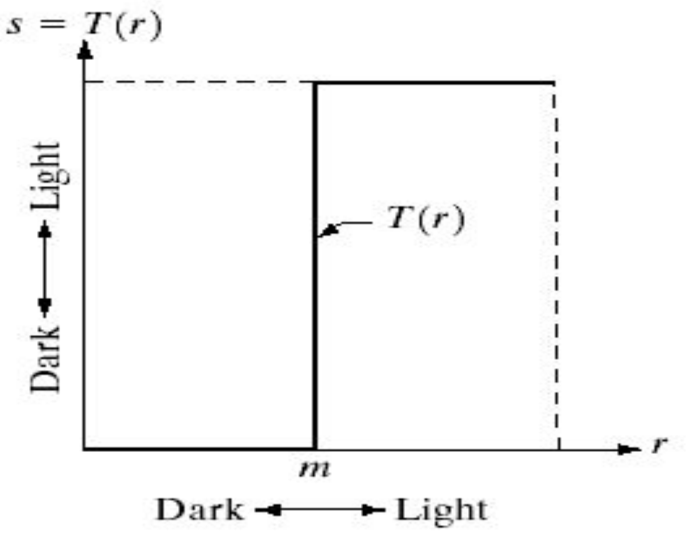
If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L-1$?

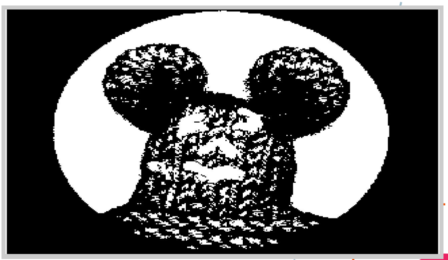
- If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L-1$, the transformation becomes a **thresholding function** that creates a binary image.

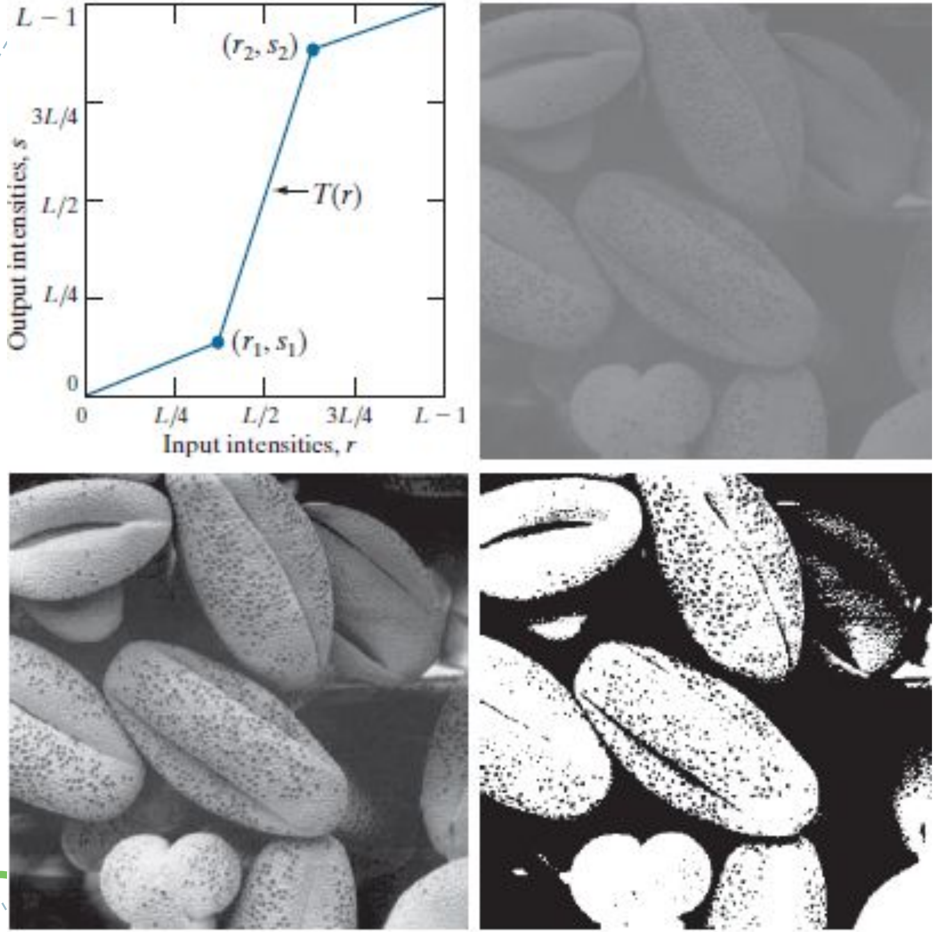
Thresholding Function

m = ‘threshold level’

$$s = \begin{cases} L-1 & \text{if } r > m \\ 0 & \text{otherwise} \end{cases}$$







Contrast stretching.

(a) Piecewise linear

Transformation function.

(b) A low contrast electron

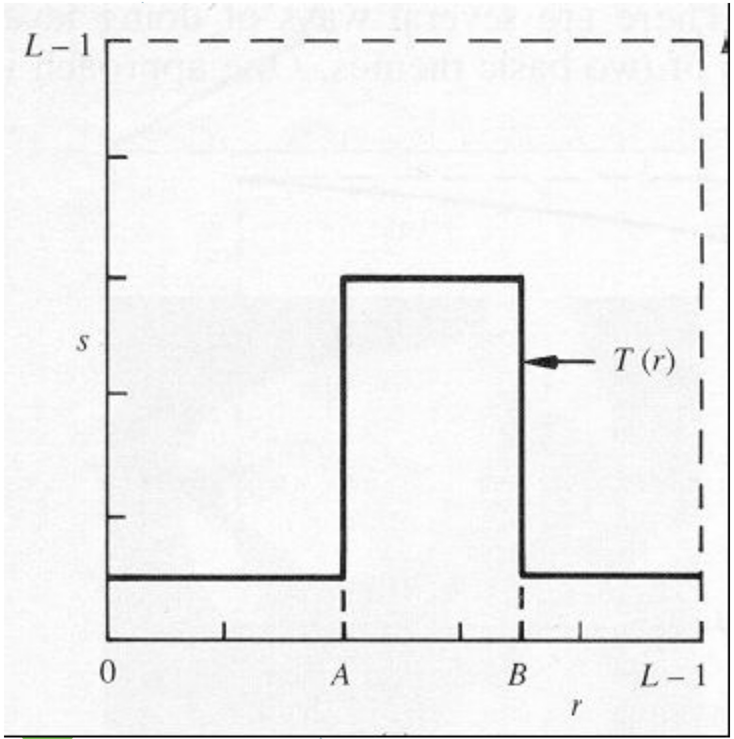
microscope image of pollen, magnified 700 times.

(c) Result of contrast stretching.

(d) Result of thresholding.

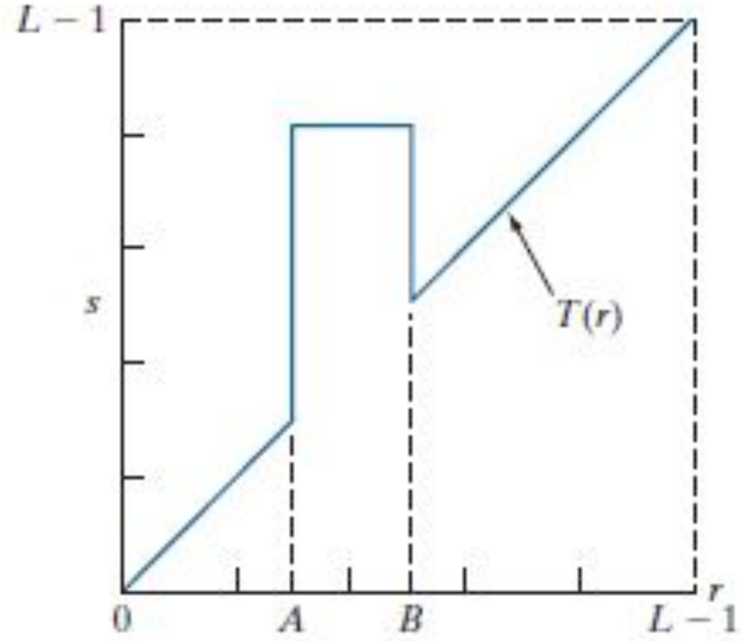
5. Gray Level / Intensity-Level Slicing

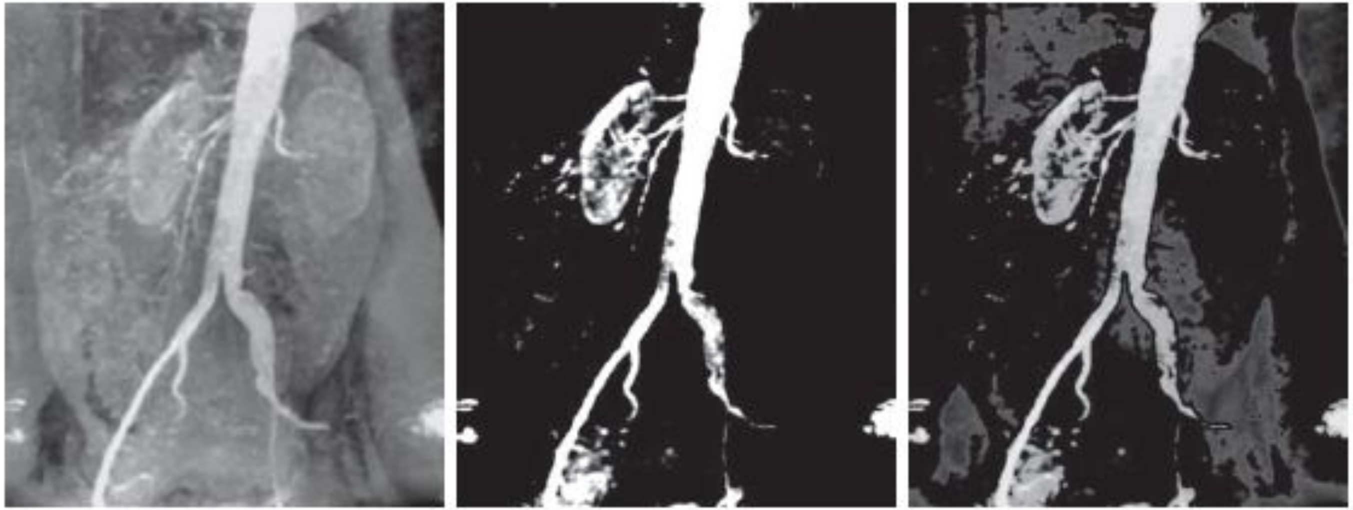
Used to highlight a specific range of gray values



One way is to display a high value for all gray levels in the range of interest and a low value for all other gray levels (binary image).

What should we do to brighten the desired range of gray levels and leave all other intensities unchanged?

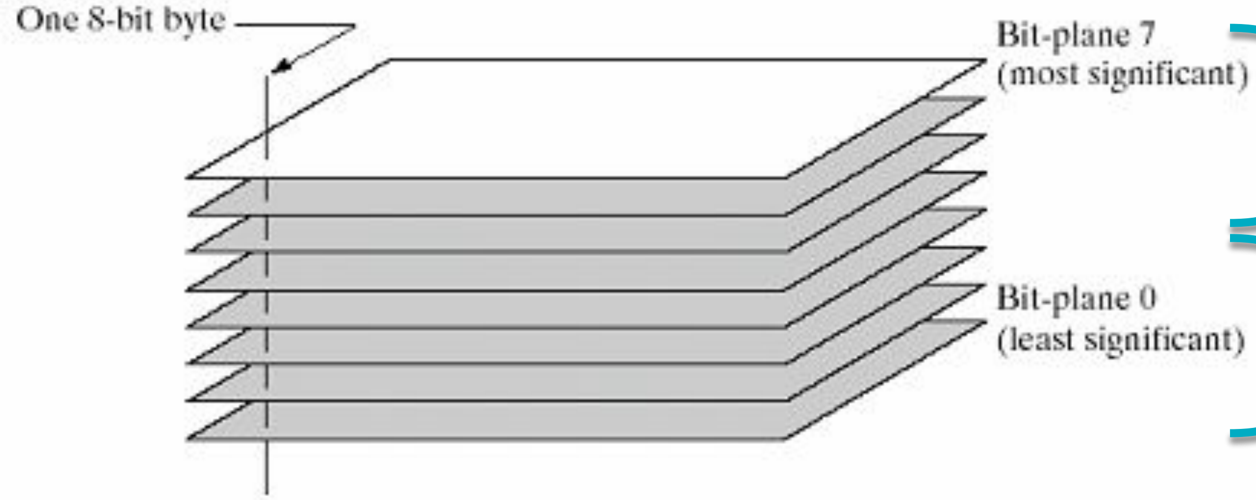




a b c

(a) Aortic angiogram.
(b) Result of using a slicing transformation, with the range of intensities of interest selected in the upper end of the gray scale.
(c) Result of using the transformation with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved.

6. Bit Plane Slicing



Visually significant data

Subtle details

Extracts the information of a single bit plane

Instead of highlighting gray-level ranges, highlighting the contribution made to the total image appearance by specific bits might be desired.

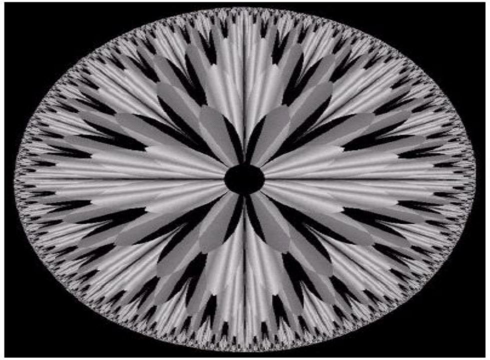


FIGURE 3.13 An 8-bit fractal image. (A fractal is an image generated from mathematical expressions). (Courtesy of Ms. Melissa D. Binde, Swarthmore College, Swarthmore, PA.)

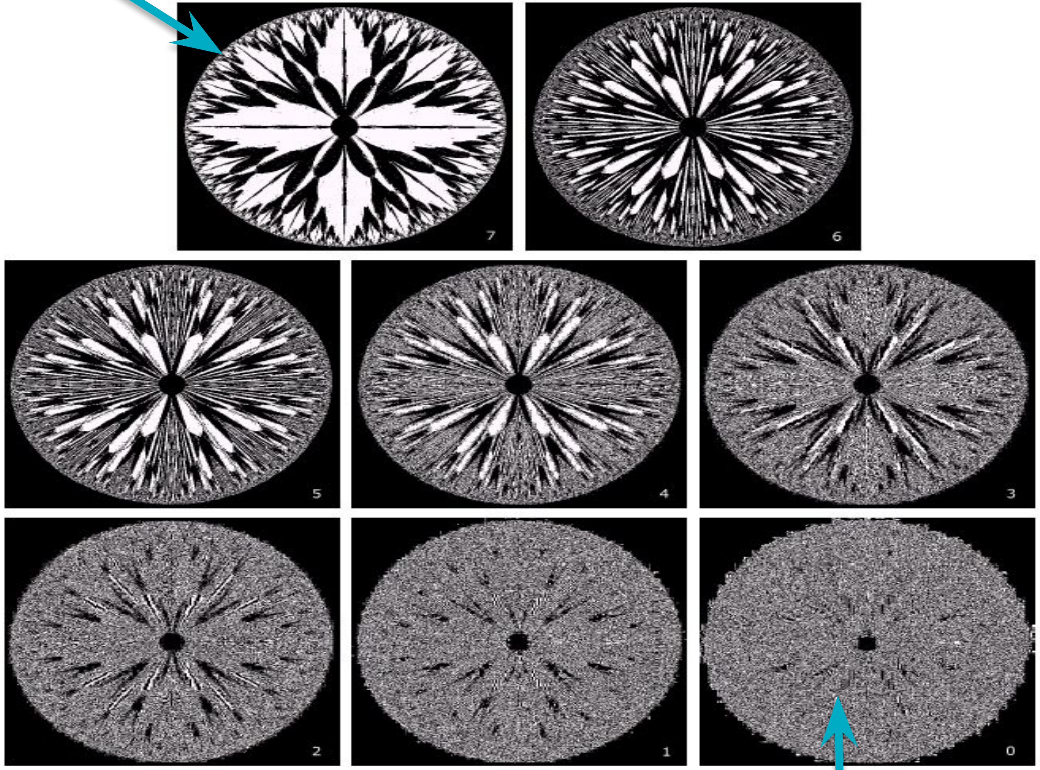


FIGURE 3.14 The eight bit planes of the image in Fig. 3.13. The number at the bottom, right of each image identifies the bit plane.

HISTOGRAMS

The **histogram** of a digital image with grey levels from 0 to L-1 is a discrete function

$$\mathbf{h(r_k) = n_k}$$

or for a **normalized Histogram** it is

$$p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$$

where:

- r_k is the k-th grey level
- n_k is the no. of pixels in the image with that grey level
- M, N are the no. of rows & columns in the image
- k = 0, 1, 2, …, L-1

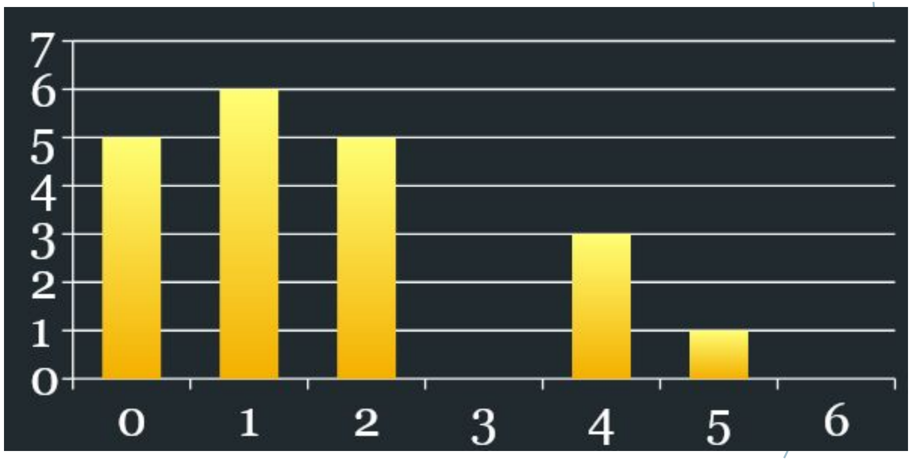
Histogram

The (intensity or brightness) histogram shows how many times a particular grey level (intensity) appears in an image.

For example, 0 - black, 255 – white

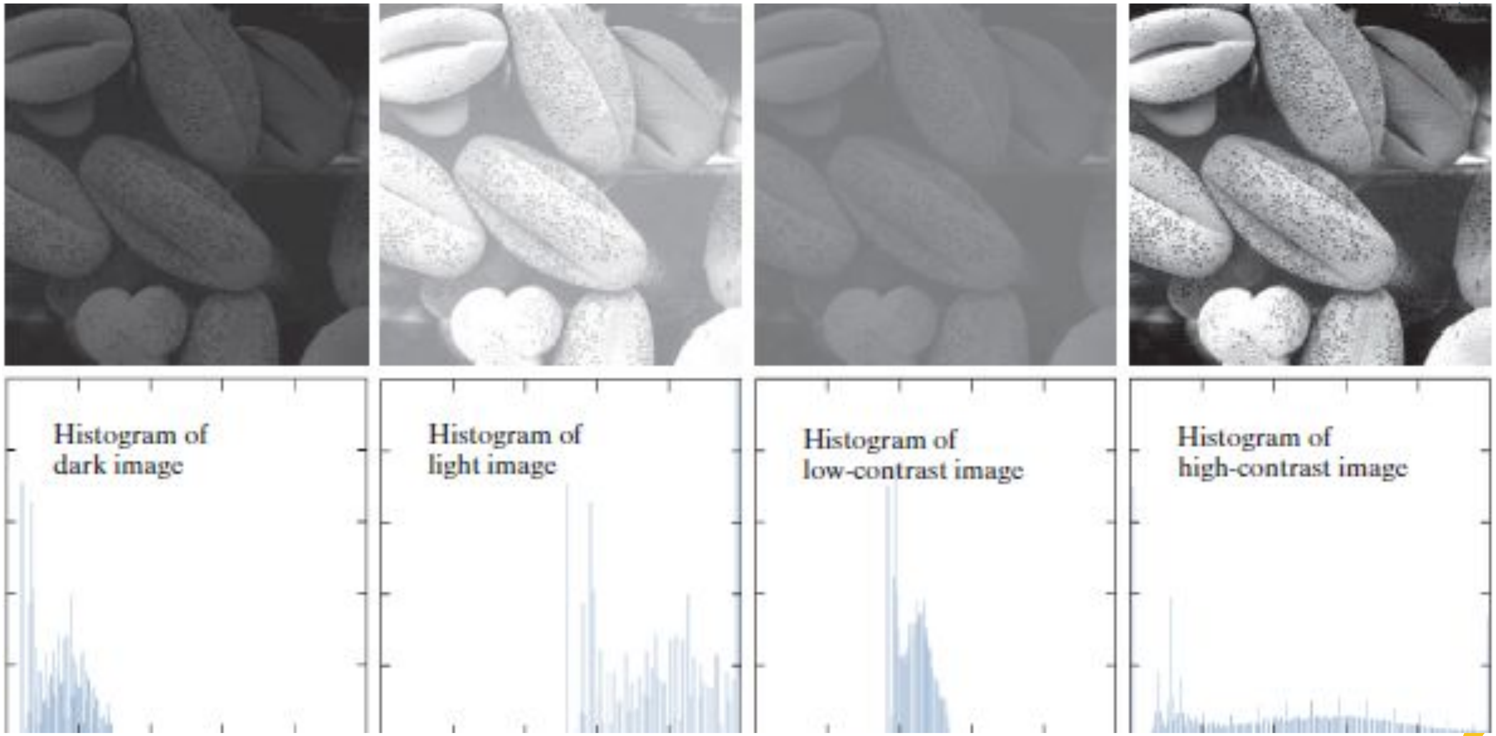
0	1	1	2	4
2	1	0	0	2
5	2	0	0	4
1	1	2	4	1

Image



Histogram

Histogram shape is related to image appearance



Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k *and the vertical axis are values of* $p(r_k)$.

Normalized Histogram

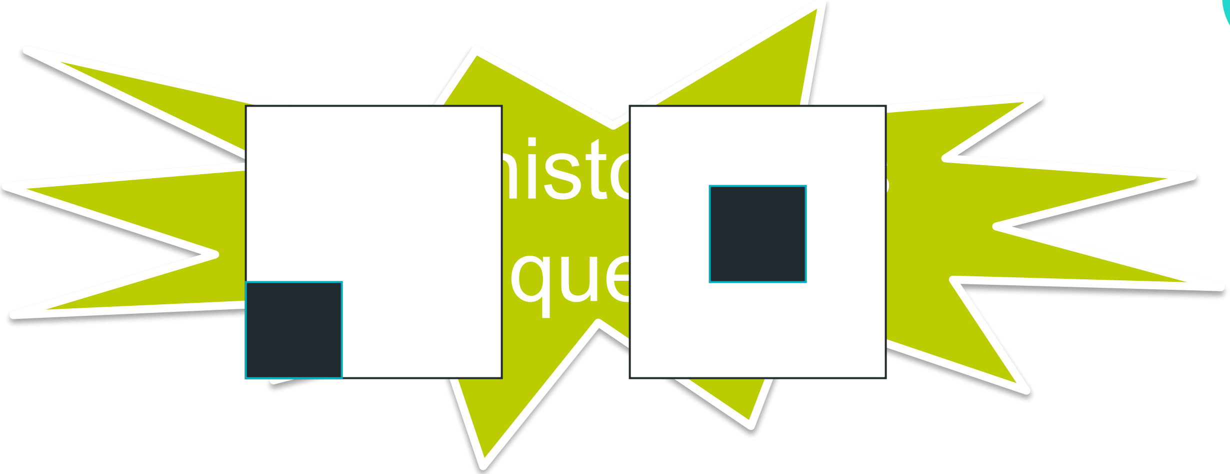
probability of occurrence of intensity level r_k

$$p(r_k) = n_k/MN$$

- Sum of all components = 1

For example: 8-bit grey-scale image

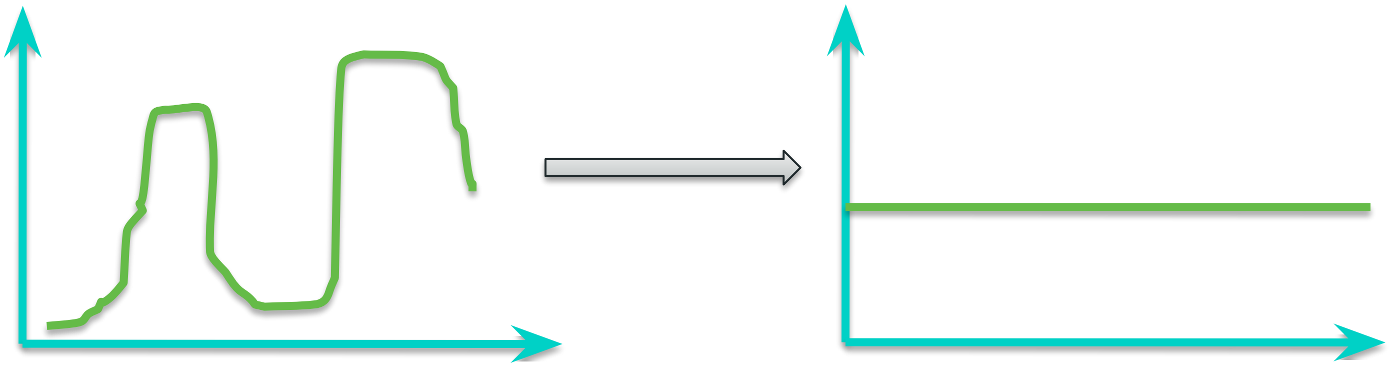
- $p(0) = 0.1$ 10% of pixels are black
- $p(\ldots) = \ldots$
- $p(128) = 0.05$ 5% of pixels are med-grey
- $p(\ldots) = \ldots$
- $p(255) = 0.0$ 0% white pixels



Histograms of both the images are same !!

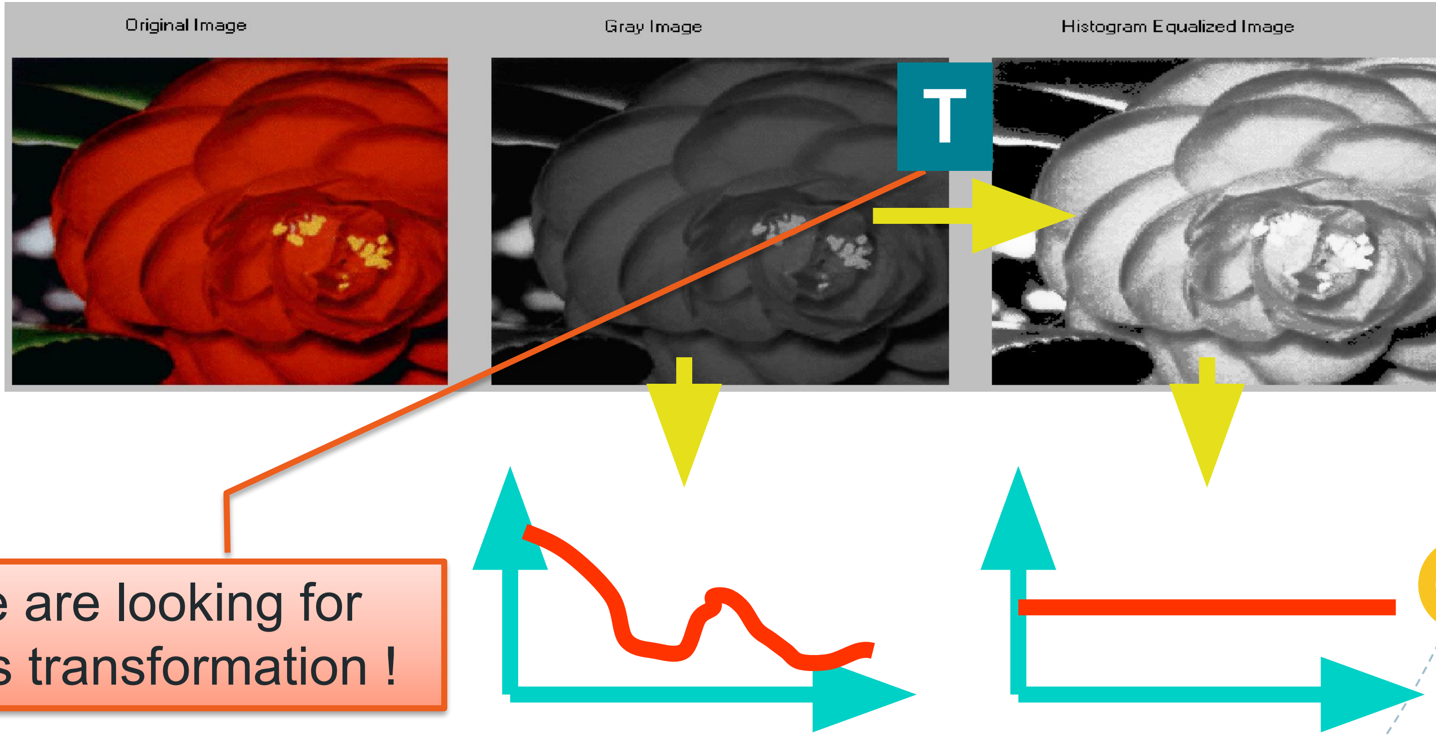
A gray scale transformation for contrast enhancement is usually found automatically using the **Histogram equalization** technique.

The aim is to create an image with equally distributed brightness levels over the whole brightness scale.



Histogram Equalization

Example:



We are looking for this transformation !

Histogram Equalization

Let

r - represent the gray levels, with $r = 0$ representing black & $r = L - 1$ representing white.

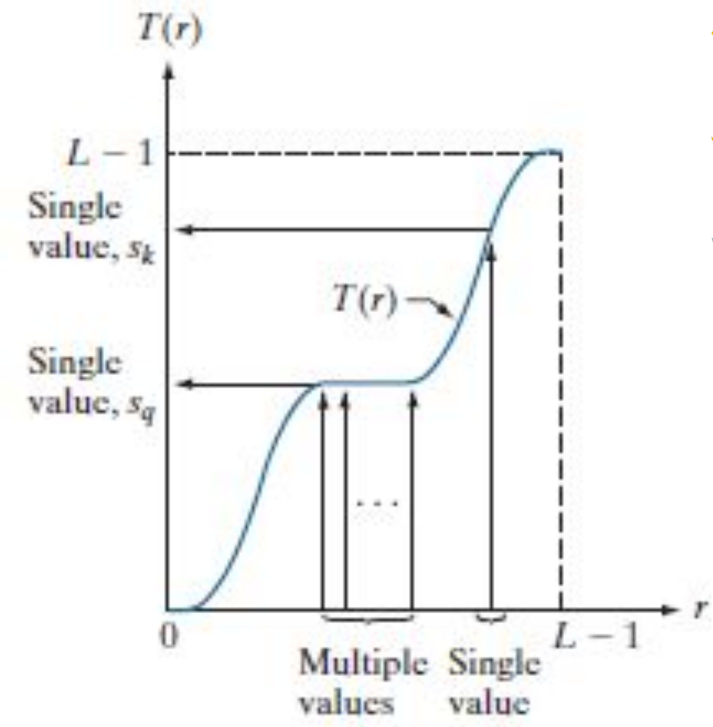
Consider the following transformation function, which produces a level s for every pixel value r in the original image

$$s = T(r) \quad 0 \leq r \leq L - 1$$

We assume that

a) T(r) is a **monotonic increasing function** in the interval $0 \le r \le L-1$

b) $0 \le T(r) \le L-1$ for $0 \le r \le L-1$.



Monotonic increasing function, showing how multiple values can map to a single value.

Discrete values:

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

$$= (L-1)\sum_{j=0}^{k} \frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j \qquad k=0,1,\ldots, L-1$$

M – no. of rows
N – no. of columns
MN – total no. of pixels in the image
L – no. of gray levels

Example: Histogram Equalization

Suppose that a 3-bit image (L=8) of size 64 × 64 pixels (MN = 4096) has the intensity distribution shown in following table.

Get the histogram equalization transformation function

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

I/p Gray Level (r_k)	no. of pixels (n_k)	$p(r_k) = n_k/MN$	Σ	(L-1)Σ (s_k)	O/p Gray Level
0	790	0.19	0.19	1.33	1
1	1023				
2	850				
3	656				
4	329				
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

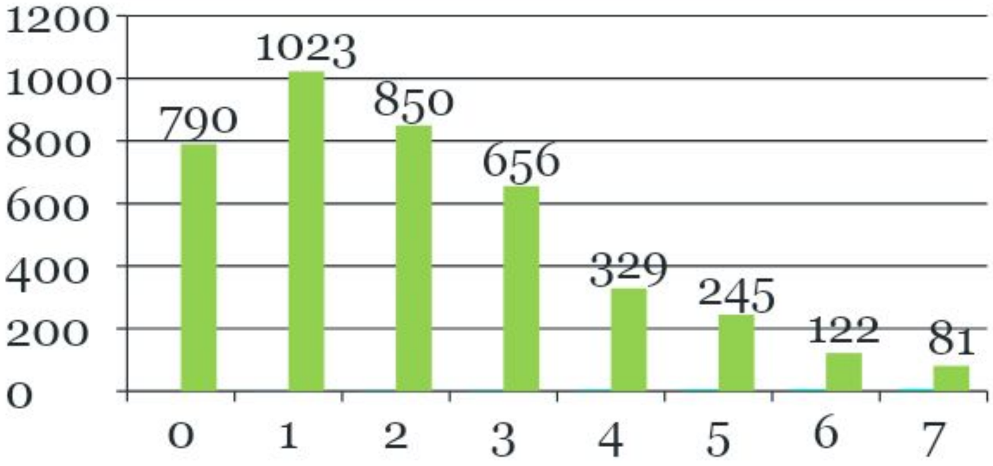
Discrete values:

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

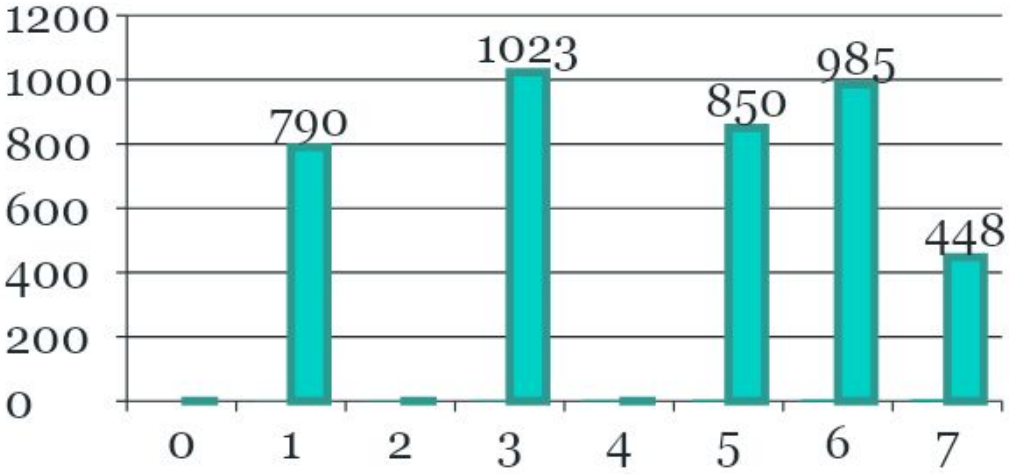
Grey level	n_k	$p_r(r_k)$	s_k
0	790	0.19	1
1	1023	0.25	3
2	850	0.21	5
3	656	0.16	6
4	329	0.08	6
5	245	0.06	7
6	122	0.03	7
7	81	0.02	7

What will be the histogram of the new image?

Equalized Grey level	n_k
0	0
1	790
2	0
3	1023
4	0
5	850
6	656 + 329 = 985
7	245 + 122 + 81 = 448



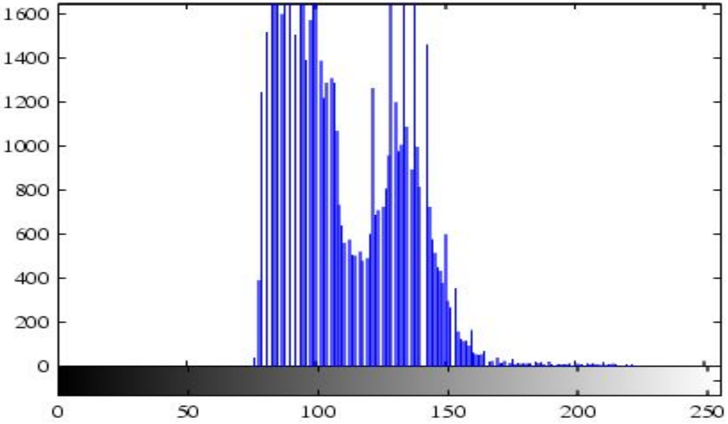
Histogram of the dark image

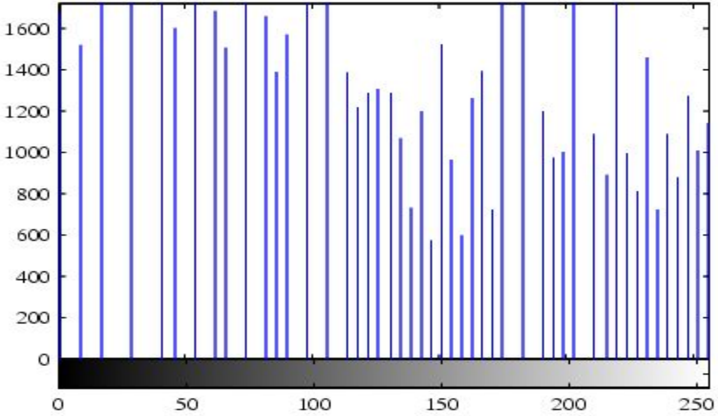


Equalized Histogram of the image

Jar

Histogram equalization





Equalize the given histogram.

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	100	90	50	20	0	0	0	0

Old Gray Level	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
New Gray Level	**3**	**5**	**6**	**7**	**7**	**7**	**7**	**7**

Gray Level	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
No. of pixels	**0**	**0**	**0**	**100**	**0**	**90**	**50**	**20**

Dark Histogram

equalize

Flat Histogram (Approximately)

Equalize again ??

Histogram Matching / Specification

- There are applications in which attempting to base enhancement on a uniform histogram is not the best approach

- Sometimes it is useful to be able to specify the shape of the histogram that we wish the processed image to have.

- For a image, whose enhancement is to be done, we are **given an histogram, $G(z_k)$,** that actually shows **how the processed image's histogram should look** after applying the transformation function to the i/p image.

Equalize the i/p image

You will get s_k

Match s_k to z_k

Equalize the image whose histogram we want to match

You will get z_k

Steps for Histogram Matching

1. Equalize the histogram of the input image. Round the resulting values, s_k , to the integer range [0, L – 1].

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j)$$

2. Equalize the image whose histogram we want to match. Round the resulting values, v_k , to the integer range [0, L – 1]. Store the rounded values of G in a lookup table.

$$v_k = G(z_k) = \sum_{j=0}^{k} p_z(z_j)$$

3. For every value of s_k, $k = 0, 1, 2, \ldots, L-1$, use the stored values of G to find the corresponding value of z_q so that $G(z_q)$ is closest to s_k. Store these mappings from s to z. When more than one value of z_q gives the same match (i.e., the mapping is not unique), choose the smallest value by convention.

4. Form the histogram-specified image by mapping every equalized pixel with value s_k to the corresponding pixel with value z_q in the histogram-specified image, using the mappings

Example: Histogram Matching

- Suppose that a 3-bit image (L=8) of size 64 × 64 pixels (MN = 4096) has the intensity distribution shown in the following table (on the left).
- Get the histogram transformation function and make the output image with the specified histogram, listed in the table on the right.

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

1. Equalize the histogram of the input image. Round the resulting values, s_k , to the integer range [0, L – 1].

I/p Gray Level (r_k)	no. of pixels (n_k)	$p(r_k) = n_k/MN$	Σ	$(L-1)\Sigma$	O/p Gray Level (s)
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

2. Equalize the image whose histogram we want to match. Round the resulting values, v_k , to the integer range [0, L – 1]. Store the rounded values of G in a lookup table.

Gray Level (z_k)	$p(z_k) = n_k/MN$	Σ	(L-1)Σ	O/p Gray Level $G(z_k)$
0	0.00			
1	0.00			
2	0.00			
3	0.15			
4	0.20			
5	0.30			
6	0.20			
7	0.15			

3. For every value of s_k , k = 0,1, 2,..., L− 1, use the stored values of G to find the corresponding value of z_q so that G (z_q) is closest to s_k . Store these mappings from s to z. When more than one value of z_q gives the same match (i.e., the mapping is not unique), choose the smallest value by convention.

I/p Gray Level (r_k)	O/p Gray Level (s_k)
0	1
1	3
2	5
3	6
4	6
5	7
6	7
7	7

O/p Gray Level $G(z_k)$	Gray Level (z_k)
0	0
0	1
0	2
1	3
2	4
5	5
6	6
7	7

I/p Gray Level (r_k)	O/p Gray Level (s_k)
0	1
1	3
2	5
3	6
4	6
5	7
6	7
7	7

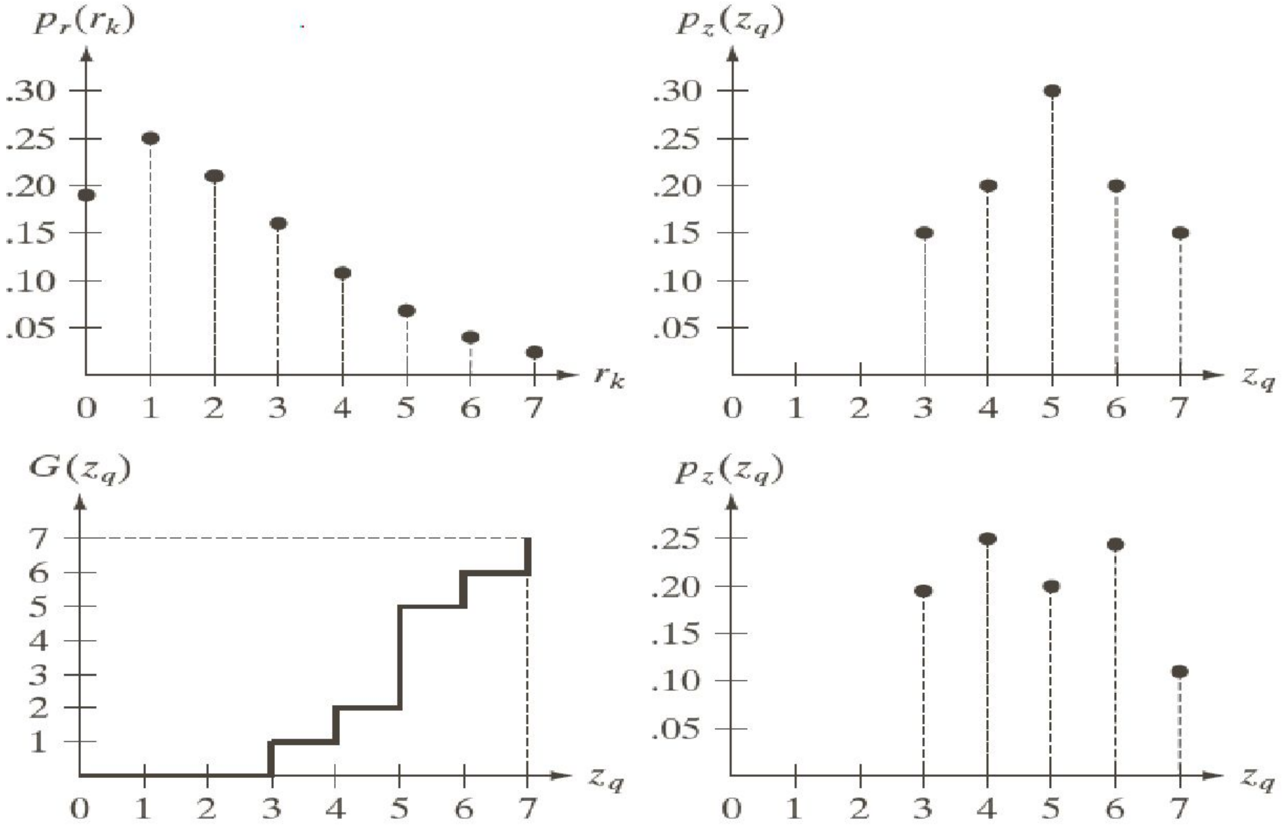
O/p Gray Level $G(z_k)$	Gray Level (z_k)
0	0
0	1
0	2
1	3
2	4
5	5
6	6
7	7

I/p Gray Level (r_k)		Gray Level (z_k)
0		0
1		1
2		2
3		3
4		4
5		5
6		6
7		7

Histogram Matching

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

r_k	z_q	No. of pixels
0	3	
1	4	
2	5	
3	6	
4	6	
5	7	
6	7	
7	7	



a	b
c	d

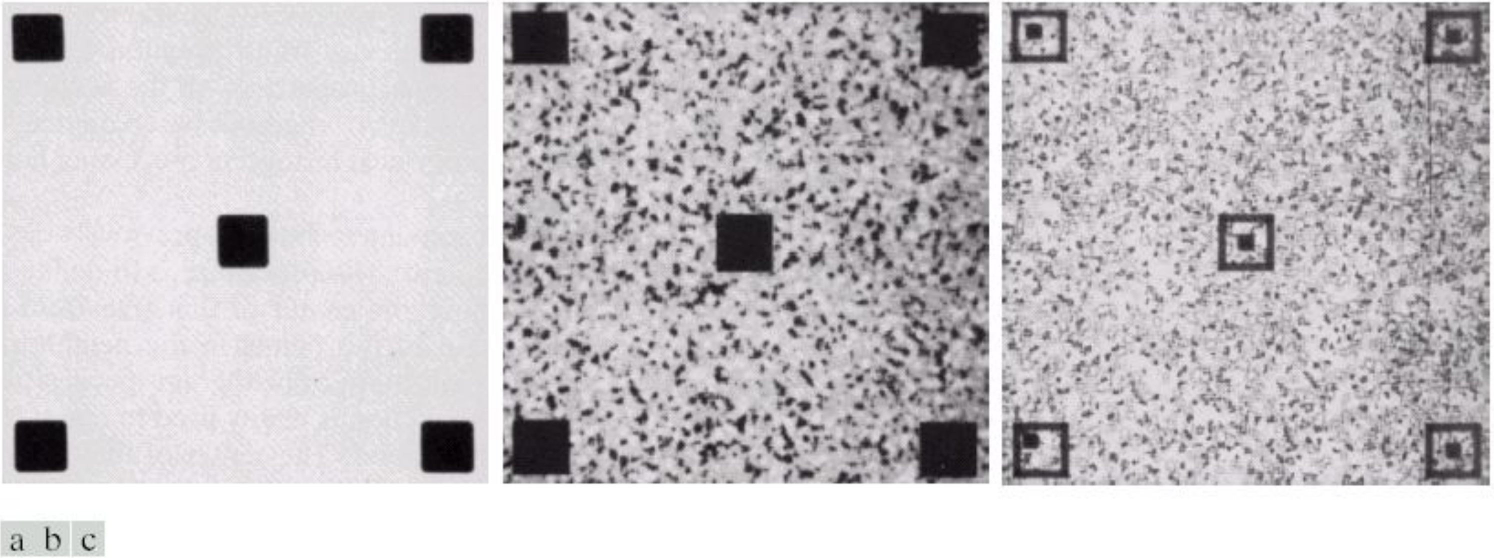
(a) Histogram of a 3-bit image. (b) Specified histogram.
(c) Transformation function obtained from the specified histogram.
(d) Result of performing histogram specification. Compare (b) and (d).

Local Enhancement

- The histogram processing methods discussed are *global*
- Suitable for overall enhancement, but generally fails when the objective is to enhance details over small areas in an image.
  - the number of pixels in small areas have negligible influence on the computation of global transformations.
- The solution is to devise transformation functions based on the intensity distribution of pixel neighborhoods.

Local Enhancement

- Define a neighborhood (e.g. a square or rectangle) and move its center from pixel to pixel.
- At each location, the histogram of the points in the neighborhood is computed. Either histogram equalization or histogram specification transformation function is obtained
- Map the intensity of the pixel centered in the neighborhood
- Move to the next location and repeat the procedure



(a) Original image.

(b) Result of global histogram equalization.

(c) Result of local histogram equalization using a 7 x 7 neighbourhood about each pixel.

Lab Assignment

- Construct the histogram of the following image.

0	1	2	3	4	5	6	7
0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
8	9	10	11	12	13	14	15

- What will be the effect on the histogram if we set to zero the
  - Higher order bit plane?
  - Lower order bit plane ?
- Take your grey scale image & construct its histogram.

Enhancement using Arithmetic/Logic Operations

Arithmetic/Logic operations are performed on the pixels of two or more images

Image Subtraction

$$g(x, y) = f(x, y) - h(x, y)$$

Obtained by computing the difference between all pairs of corresponding pixels from images f & h.

Subtraction enhances the differences between the two images

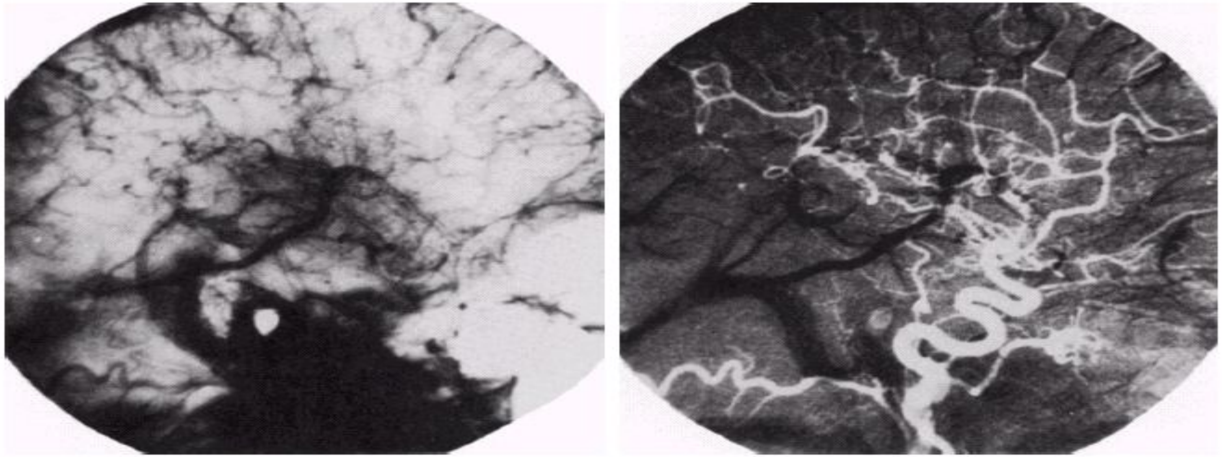
Mask Mode Radiography

One of the most successful commercial applications of image subtraction

Example: imaging blood vessels and arteries in a body. Blood stream is injected with a dye and X-ray images are taken before and after the injection

– f(x, y): image after injecting a dye
– h(x, y): image before injecting the dye (the mask)

• The difference of the 2 images yields a clear display of the blood flow paths.



a b

Enhancement by image subtraction. (a) Mask image. (b) An image (taken after injection of a contrast medium into the bloodstream) with mask subtracted out.

(a) Mask image
(b) Image (after injection of dye into the bloodstream) with mask subtracted out

Image Averaging

- A noisy image:

$$g(x, y) = f(x, y) + n(x, y)$$

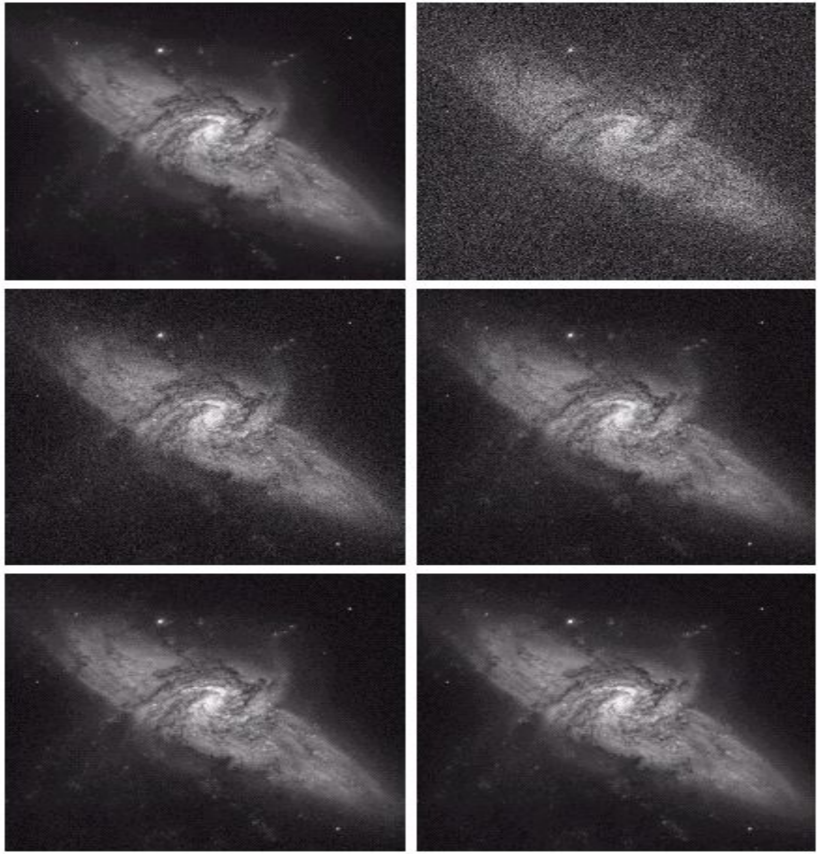
- Averaging M different noisy images:

$$\bar{g}(x, y) = \frac{1}{M}\sum_{i=1}^{M} g_i(x, y)$$

- As M increases, the variability of the pixel values at each location decreases.

- This means that g(x, y) approaches f(x, y) as the number of noisy images used in the averaging process increases.

An important application of image averaging is in the field of astronomy



a	b
c	d
e	f

(a) Image of Galaxy Pair NGC 3314. (b) Image corrupted by additive Gaussian noise with zero mean and a standard deviation of 64 gray levels. (c)–(f) Results of averaging $K = 8, 16, 64$, and 128 noisy images. (Original image courtesy of NASA.)

Image Division

Shading Correction is an important applications of image division

An imaging sensor produces image g(x, y), where

$$\underbrace{g(x, y)}_{\text{Sensed image}} = \underbrace{f(x, y)}_{\text{Perfect image}} \underbrace{h(x, y)}_{\text{Shading function}}$$

By dividing g(x, y) by h(x, y),
we can get the perfect image.

Shading correction.

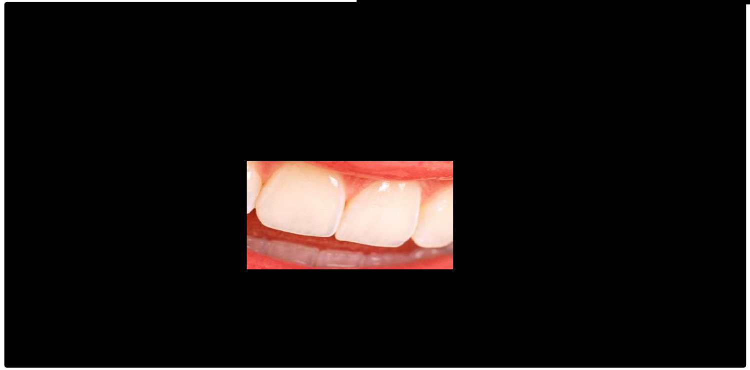
(a) Shaded test pattern.

(b) Estimated shading pattern.

(c) Division of (a) by (b).

Image Multiplication

An important use of image multiplication is in *masking operations, also called region of interest (ROI).*



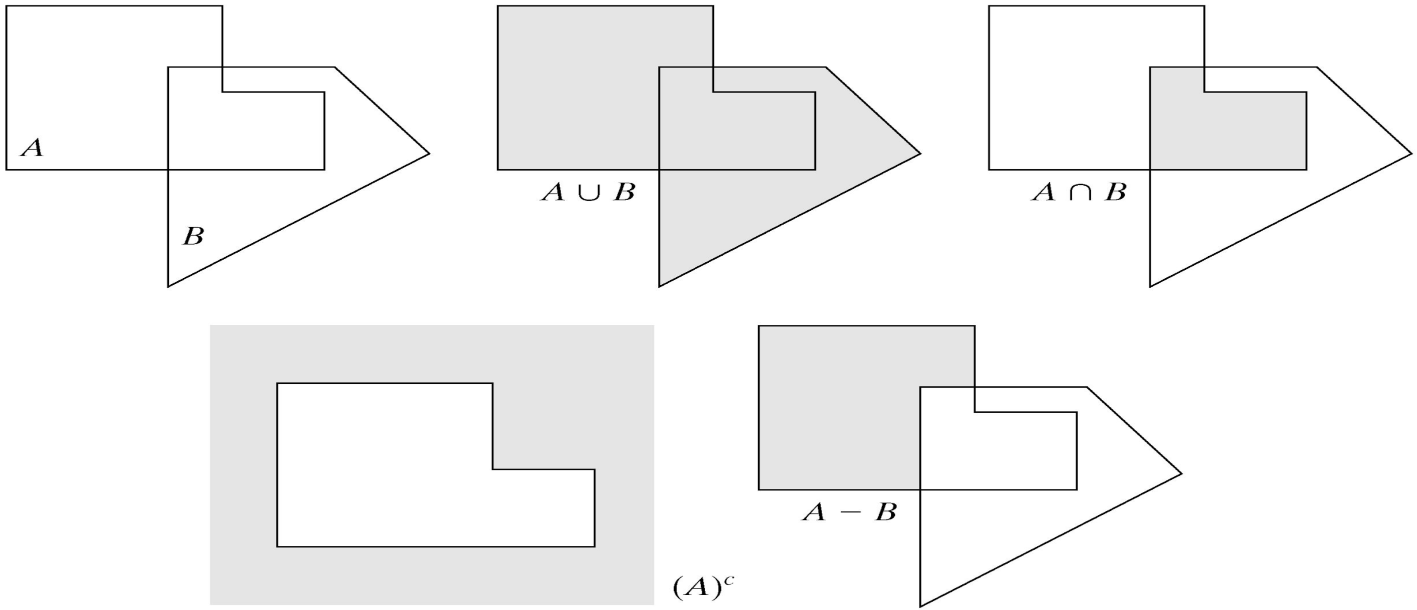
Points to note ...

- If the dynamic range is [0, L-1], the intensity values could go below 0 or above L-1 as a result of arithmetic operations.
- Given an image f, the following approach guarantees that the full range of an arithmetic operation between images is captured into a fixed number of bits.

1. $f_m = f - \min(f)$
   This creates an image whose minimum value is 0.
2. $f_s = (L\text{-}1)[f_m / \max(f_m)]$
   This creates a scaled image f_s, whose values are in the range of [0, L-1]

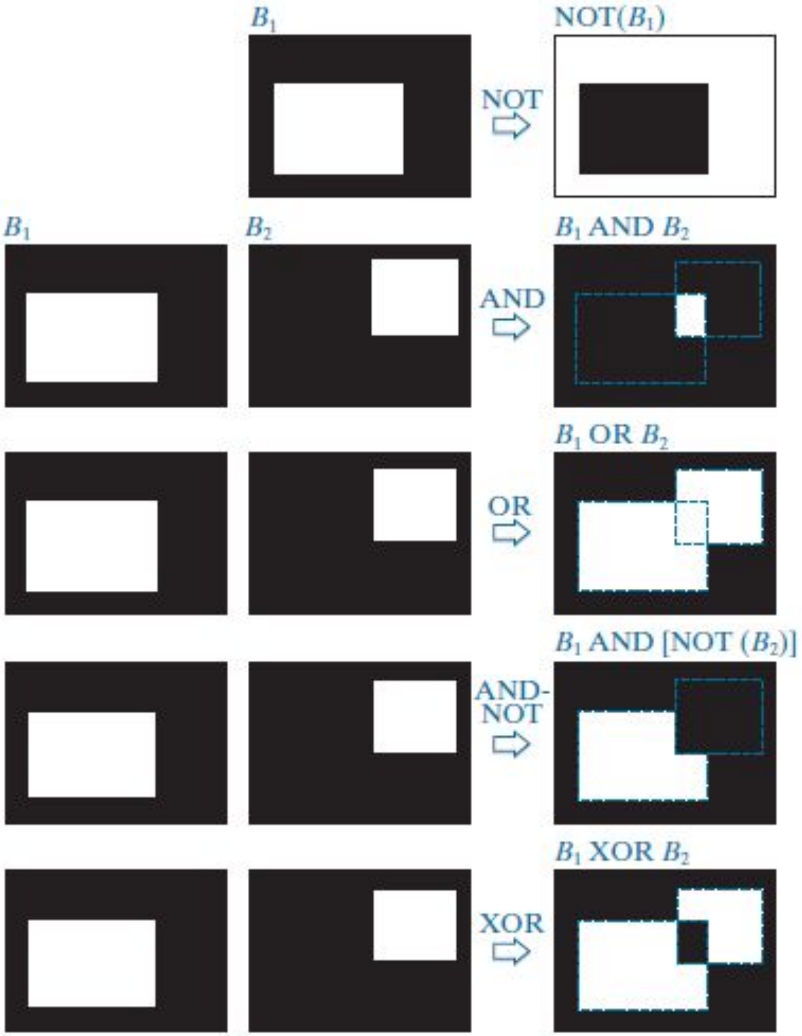
When performing division, a small no. should be added to the pixels of the divisor image to avoid division by 0.

Set Operations



Logical Operations

- Illustration of logical operations involving foreground (white) pixels.
- Black represents binary 0's and white binary 1's.
- The dashed lines are shown for reference only. They are not part of the result.



Spatial Filtering

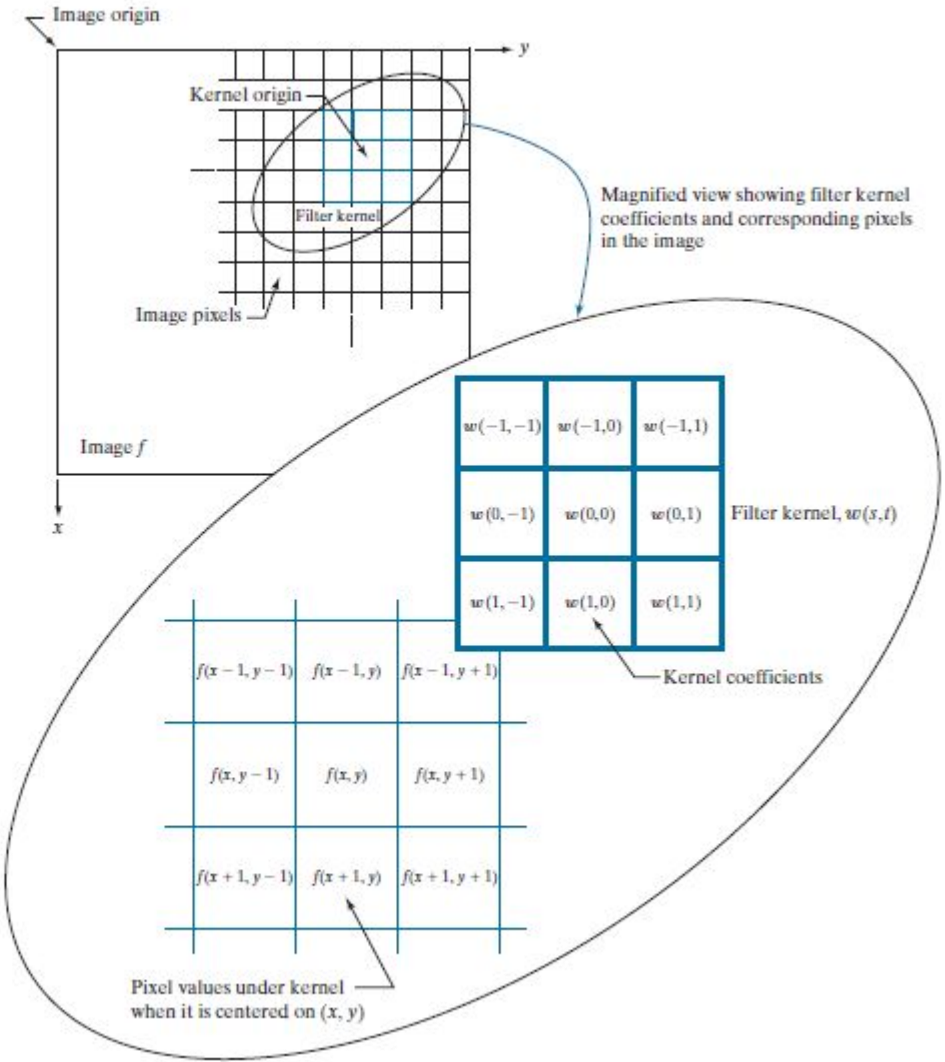
- Spatial filtering is the filtering operations that are performed directly on the pixels of an image
- If the operation performed on the image pixels is linear, then the filter is called a linear spatial filter.
- Otherwise, the filter is a nonlinear spatial filter.

THE MECHANICS OF LINEAR SPATIAL FILTERING

- A linear spatial filter performs a sum-of-products operation between an image f and a filter kernel, w.
- The kernel is an array whose size defines the neighborhood of operation, and whose coefficients determine the nature of the filter.
- At any point (x, y) in the image, the response, g(x, y), of the filter is the sum of products of the kernel coefficients and the image pixels encompassed by the kernel:

$$\begin{aligned} \mathbf{g(x, y)} = {} & w(-1,-1)\, f(x-1, y-1) + w(-1, 0)\, f(x-1, y) + w(-1, 1)\, f(x-1, y+1) \\ & + w(0,-1)\, f(x, y-1) + w(0, 0)\, f(x, y) + w(0, 1)\, f(x, y+1) \\ & + w(1,-1)\, f(x+1, y-1) + w(1, 0)\, f(x+1, y) + w(1, 1)\, f(x+1, y+1) \end{aligned}$$

- The mechanics of linear spatial filtering using a 3×3 kernel.
- The origin of the image is at the top left
- The origin of the kernel is at its center.
- Placing the origin at the center of spatially symmetric kernels simplifies writing expressions for linear filtering.



Note: Linear filtering

Coefficie[partially hidden] f(x, y), indicatin[partially hidden]

For a ma[partially hidden] 2a+1 and n=2b+1, [partially hidden] er. Then m and n are [partially hidden]

To generate a complete filtered image, the following equation must be applied for

$x = 0, 1, \ldots, M-1$

$y = 0, 1, \ldots, N-1$

In general, linear filtering of an image f of size M x N with a filter mask of size m x n is given by the expression

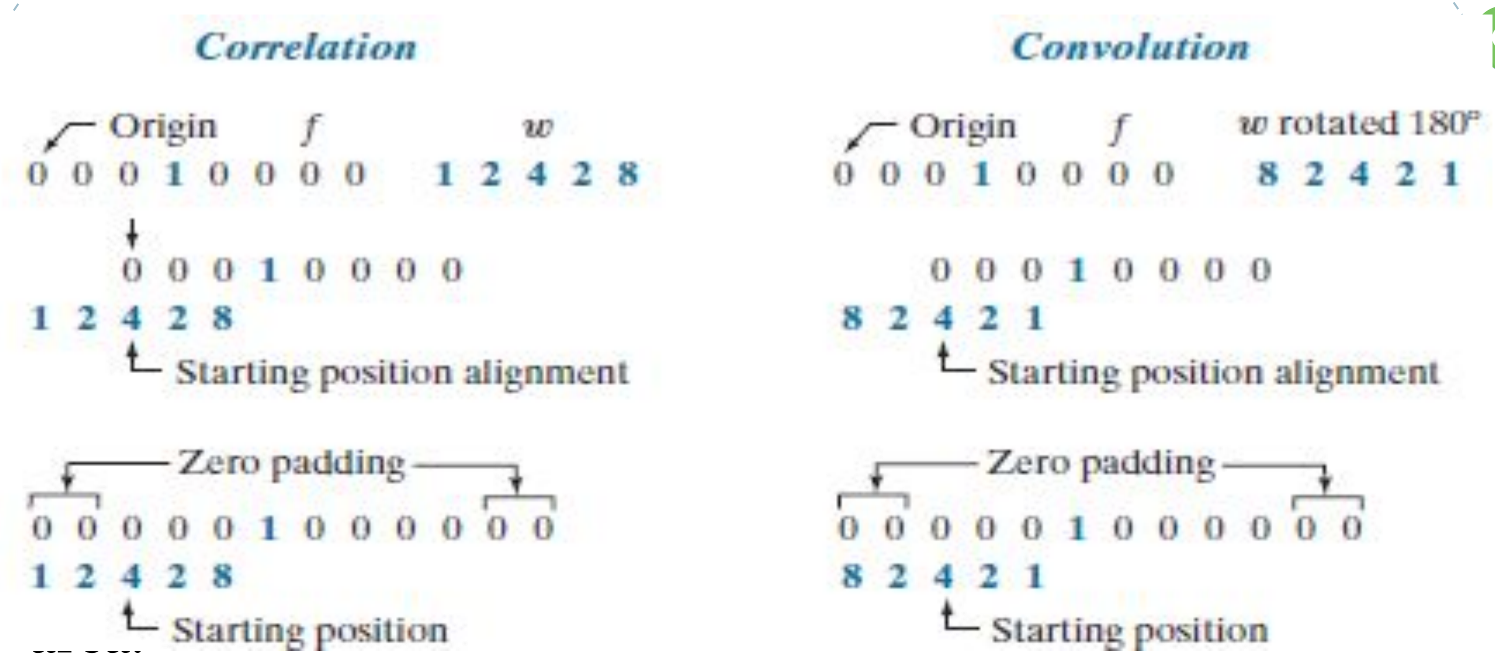
$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

a = (m – 1) / 2 & b = (n – 1) / 2

SPATIAL CORRELATION AND CONVOLUTION

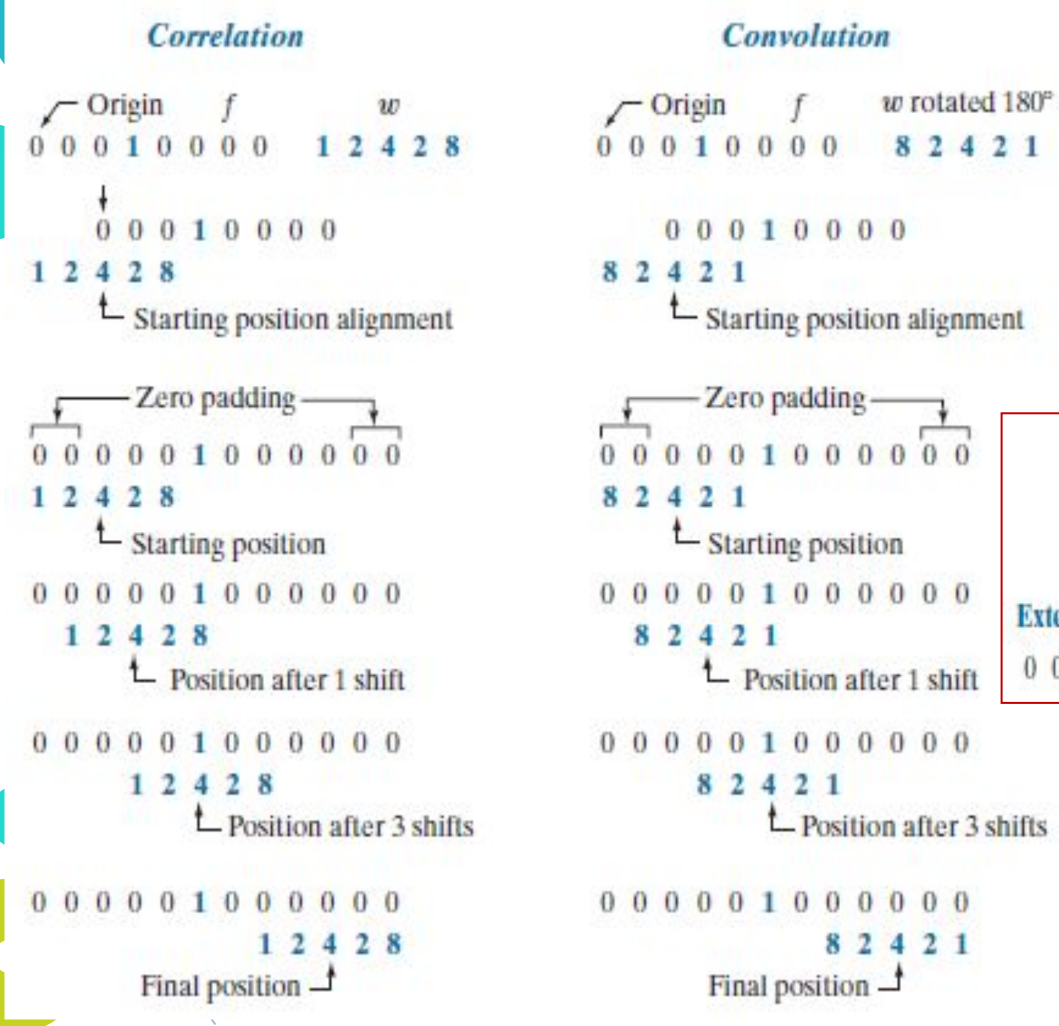
- Correlation consists of moving the center of a kernel over an image, and computing the sum of products at each location.
- The mechanics of spatial convolution are the same, except that the correlation kernel is rotated by 180°.
- When the values of a kernel are symmetric about its center, correlation and convolution yield the same result.
- Following is a 1-D illustration for Correlation

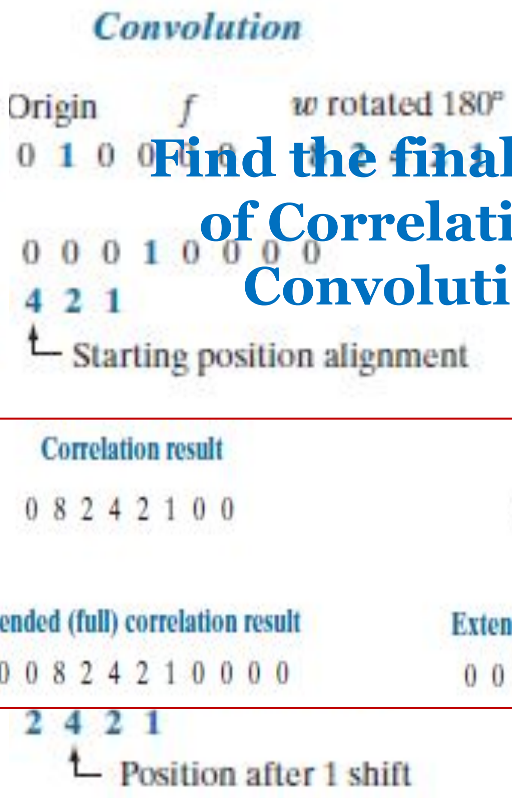
$$g(x) = \sum_{s=-a}^{a} w(s) f(x+s)$$



- area.
- A solution to this problem is to pad function f with enough 0's on either side.
- In general, if the kernel is of size 1 × m, we need (m – 1) / 2 zeros on either side of f in order to handle the beginning and ending configurations of w with respect to f.

What is the value of g(0) for Correlation & Convolution?





Find the final result of Correlation & Convolution.

Correlation result	Convolution result
0 8 2 4 2 1 0 0	0 1 2 4 2 8 0 0
Extended (full) correlation result	Extended (full) convolution result
0 0 0 8 2 4 2 1 0 0 0 0	0 0 0 1 2 4 2 8 0 0 0 0

f g(1) for
olution?

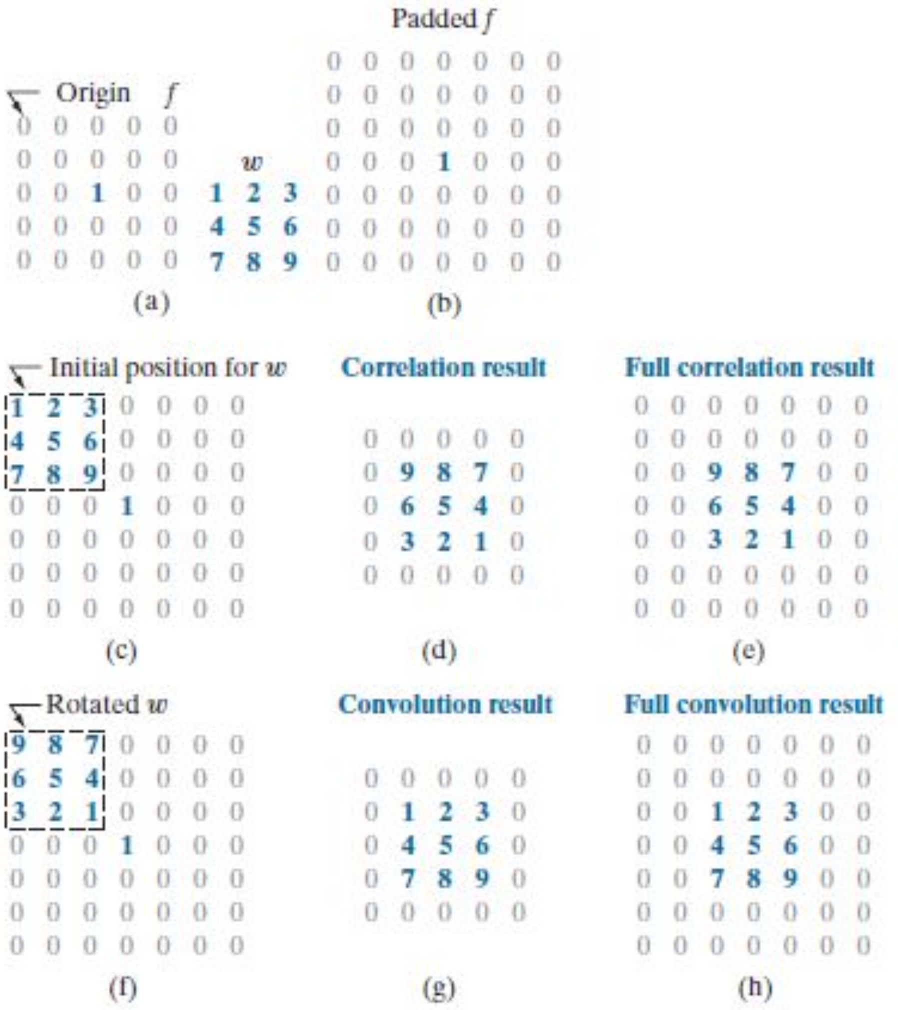
SPATIAL CORRELATION AND CONVOLUTION IN IMAGE

Correlation

$$(w * f)(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x + s, y + t)$$

Convolution

$$(w * f)(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x - s, y - t)$$



Correlation and Convolution of a 2-D kernel with an image

Smoothing Spatial Filters

Smoothing filters are used for blurring and for noise reduction.

- Blurring is used in preprocessing steps, such as removal of small details from an image prior to object extraction, and bridging of small gaps in lines or curves
- Noise reduction can be accomplished by blurring with a linear or non linear filter.

Types of Smoothing filters

There are 2 types of smoothing spatial filters

- Smoothing Linear Filters
- Order-Statistics Filters

Smoothing Linear Filters

aka Averaging Filters / Lowpass Filters

- Linear spatial filter is simply the average of the pixels contained in the neighborhood of the filter mask.
- Sometimes called “averaging filters”.
- The idea is replacing the value of every pixel in an image by the average of the gray levels in the neighborhood defined by the filter mask.

Two 3x3 Smoothing Linear Filters

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Standard average

$$\frac{1}{16} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

Weighted average

5x5 Smoothing Linear Filters

$$\frac{1}{25} \times$$

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Smoothing Linear Filters

The general implementation for filtering an M x N image with a weighted averaging filter of size m x n is given by the expression

$$g(x,y) = \frac{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t) f(x+s, y+t)}{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)}$$

Result of Smoothing Linear Filters

Original Image

[3x3]

[5x5]

[7x7]

Order-Statistics Filters

- Order-statistics filters are nonlinear spatial filters whose response is based on ordering (ranking) the pixels contained in the image area encompassed by the filter, and then replacing the value of the center pixel with the value determined by the ranking result.
- Best-known "median filter"

Process of Median filter

	10	15	20	
	20	100	20	
	20	20	25	

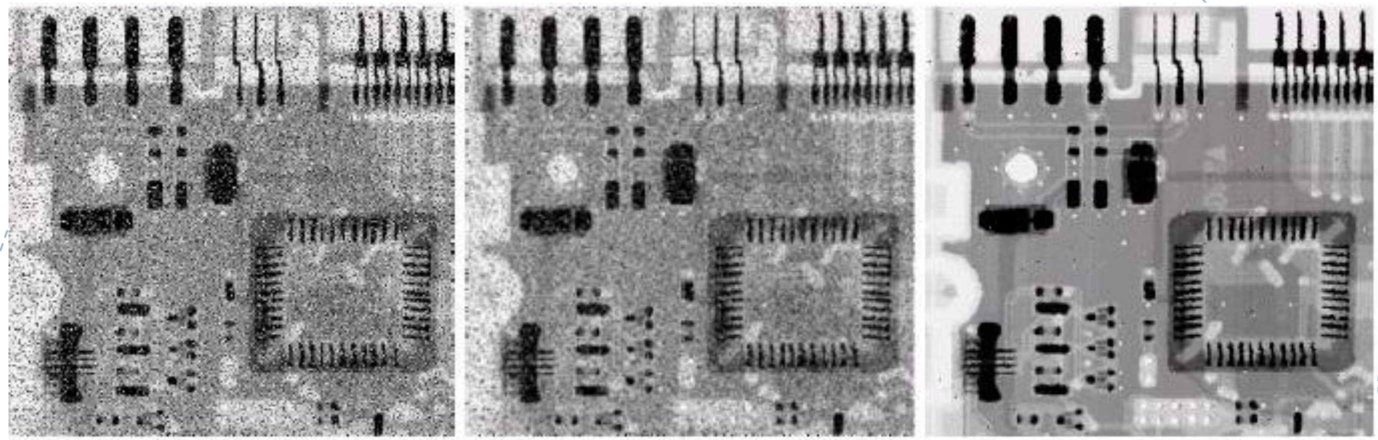
10, 15, 20, 20, 20, 20, 20, 25, 100

5th

- Crop region of neighborhood
- Sort the values of the pixel in the region
- In the M x N mask the median is (M x N +1) div 2

MEDIAN FILTERS ARE QUITE POPULAR …

- For certain types of random noise, they provide excellent noise reduction capabilities, with considerably less blurring than linear smoothing filters of similar size.
- They are particularly effective in the presence of *impulse noise* (*aka salt-&-pepper noise* because of its appearance as white & black dots superimposed on an image).



X-ray image of circuit board corrupted by salt-pepper noise

b)Noise reduction with a 3 x 3 averaging mask

c)Noise reduction with a 3 x 3 median filter

Median Filtering: Example

Max & Min Filters

Using the 100th percentile results gives us the ***max filter***, which is useful in finding the brightest points in the image. The response of a 3 x 3 max filter is given by

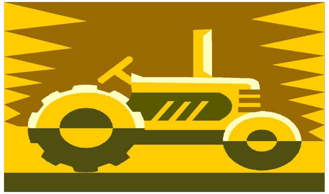
$$R = \max\{z_k \mid k = 1,2, \ldots, 9\}$$

The 0th percentile filter is the ***min filter***, used for the opposite purpose.

Sharpening Spatial Filters

The principal objective of sharpening is to highlight fine detail in an image or to enhance detail that has been blurred, either in error or as an natural effect of a particular method of image acquisition.

Uses of Image Sharpening



- Electronic printing
- Medical imaging
- Industrial inspection
- Autonomous guidance in military systems

- As we know that blurring can be done in spatial domain by **pixel averaging in a neighbors**

- **Averaging is analogous to integration**

- Therefore, logically it can be concluded that the **sharpening must be accomplished** by **spatial differentiation**.

Sharpening Filters based on first- and second-order derivatives

Definition for a first order derivative

4 Must be zero in areas of constant intensity

4 Must be nonzero at the onset of an intensity step or ramp

4 Must be nonzero along intensity ramps

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

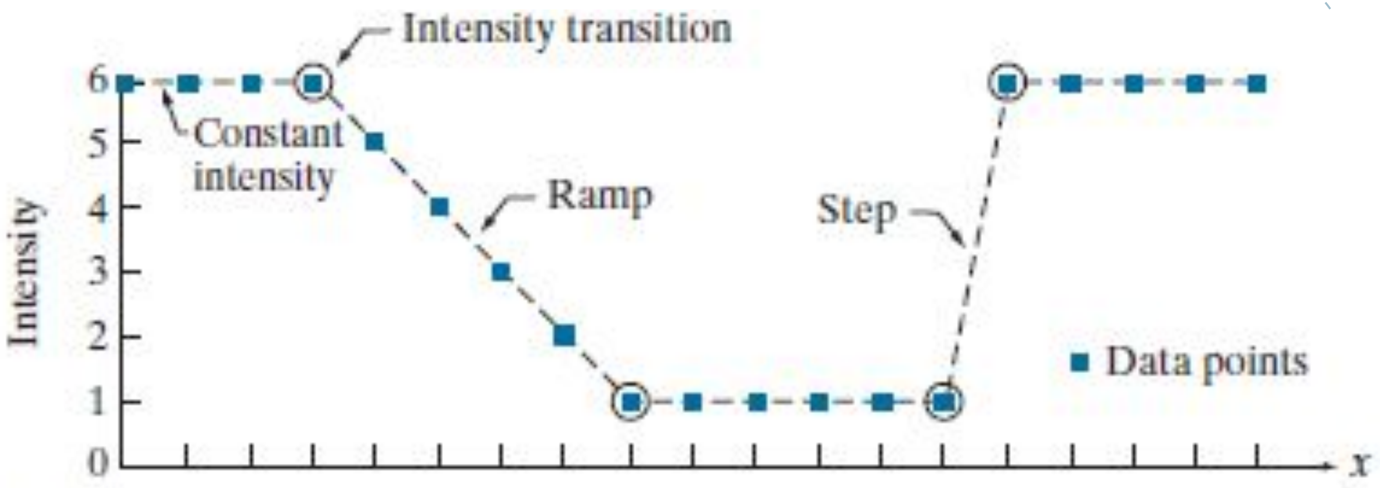
Definition for a second order derivative

4 Must be zero in areas of constant intensity

4 Must be nonzero at the onset and end of a intensity step or ramp

4 Must be zero along intensity

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x).$$



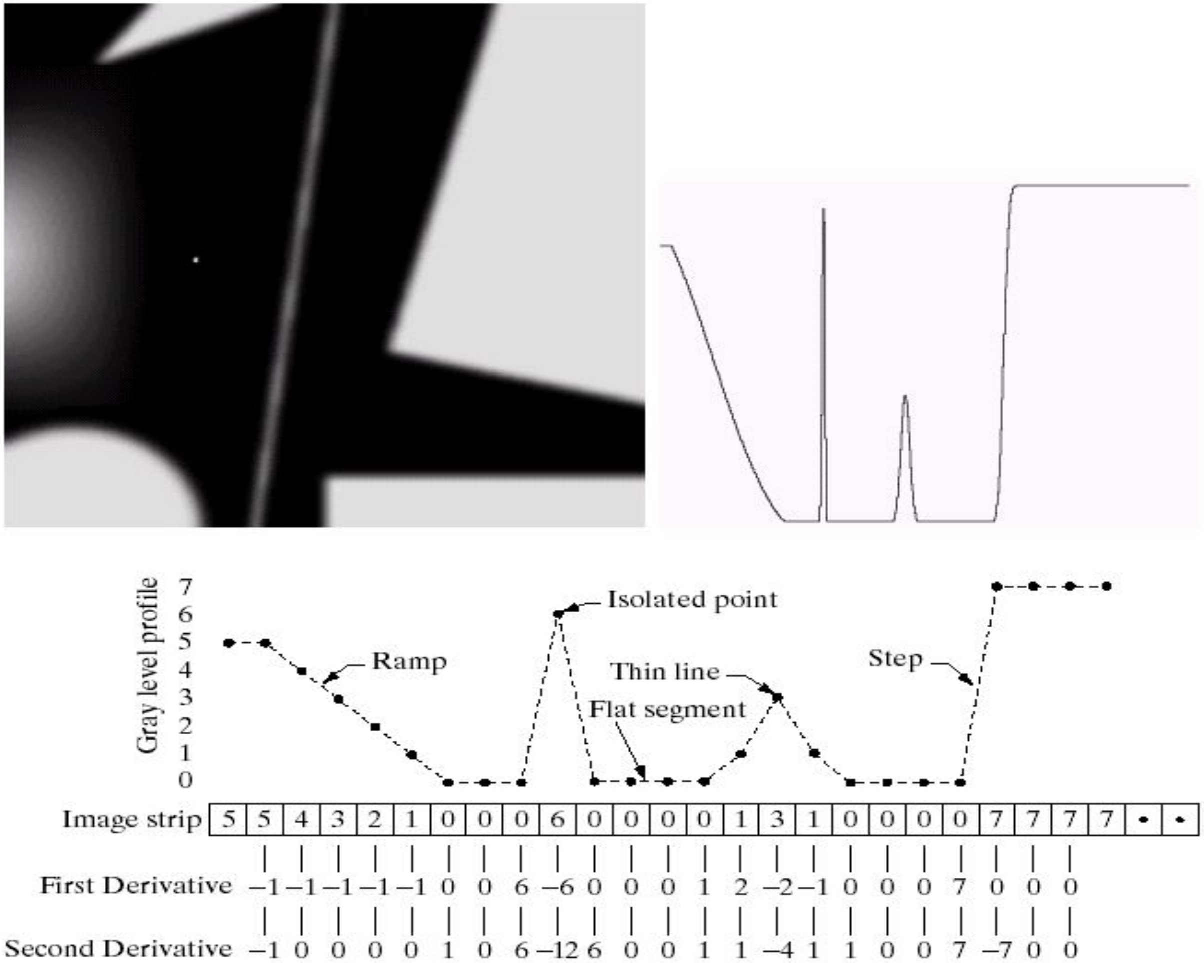
Values of scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative		0	0	−1	−1	−1	−1	−1	0	0	0	0	0	5	0	0	0	0	
2nd derivative		0	0	−1	0	0	0	0	1	0	0	0	0	5	−5	0	0	0	

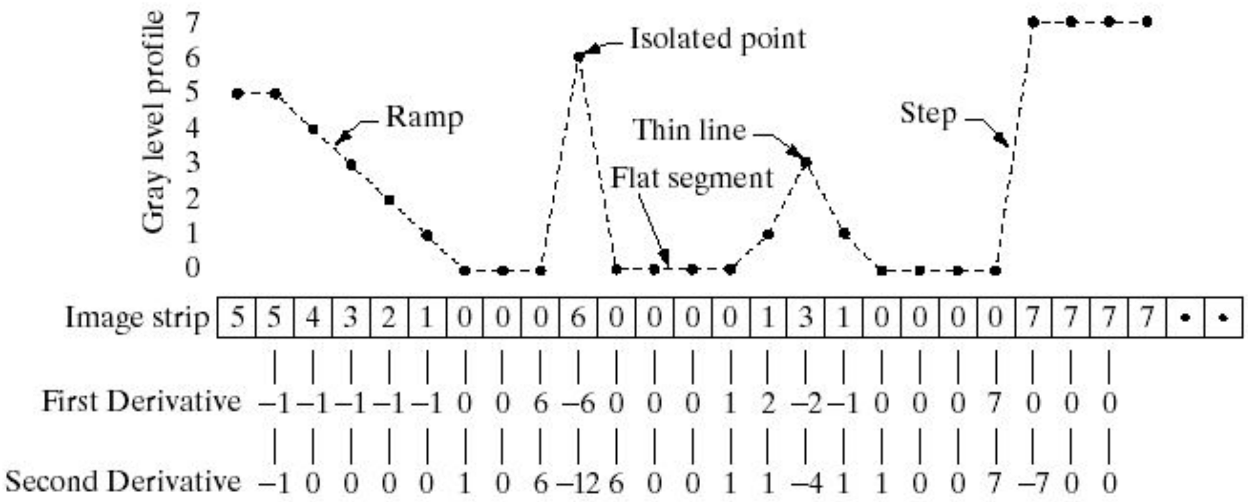
a) A section of a horizontal scan line from an image, showing ramp and step edges, as well as constant segments.

b) Values of the scan line and its derivatives.

a b
c

(a) A simple image. (b) 1-D horizontal gray-level profile along the center of the image and including the isolated noise point.
(c) Simplified profile (the points are joined by dashed lines to simplify interpretation).





Analysis

1) The 1st-order derivative is nonzero along the entire ramp, while the 2nd-order derivative is nonzero only at the onset and end of the ramp.
2) Edges in an image represent this type (ramp) of transition. Therefore, **1st makes thick edge and 2nd makes thin, much finer edges**
3) The response at and around the point is much stronger for the 2nd- than for the 1st-order derivative.

Summary

- First-order derivatives generally produce thicker edges in an images.
- Second-order derivatives have a stronger response to fine detail (sa, thin lines or isolated points).
- First-order derivatives generally have a stronger response to a gray-level step.
- Second-order derivatives produce a double response at step changes in gray level

THE LAPLACIAN

USING THE SECOND DERIVATIVE FOR IMAGE SHARPENING

The approach basically consists of the following steps:

1. Defining a discrete formulation of the second-order derivative.
2. Constructing a filter mask based on that formulation.

- The filter is isotropic: response of the filter is independent of the direction of discontinuities in an image.
- Simplest 2-D isotropic second order derivative is the Laplacian:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 \mathbf{f}}{\partial \mathbf{x}^2} = \mathbf{f}(\mathbf{x}+1,\mathbf{y}) + \mathbf{f}(\mathbf{x}-1,\mathbf{y}) - 2\mathbf{f}(\mathbf{x},\mathbf{y})$$

$$\frac{\partial^2 \mathbf{f}}{\partial \mathbf{y}^2} = \mathbf{f}(\mathbf{x},\mathbf{y}+1) + \mathbf{f}(\mathbf{x},\mathbf{y}-1) - 2\mathbf{f}(\mathbf{x},\mathbf{y})$$

$$\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$$

0	1	0
1	-4	1
0	1	0

This mask gives an isotropic result for rotations in increments of 90°.

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

a b c d

(a) Laplacian kernel

(b) Kernel used to implement an extension of this equation that includes the diagonal terms.

(c) and (d) Two other Laplacian kernels.

Effect of Laplacian Operator

As it is a derivative operator

- it highlights gray-level discontinuities in an image
- it deemphasizes regions with slowly varying gray levels

Tends to produce images that have

- grayish edge lines and other discontinuities, all superimposed on a dark,
- featureless background.

-1	-1	-1
-1	8	-1
-1	-1	-1

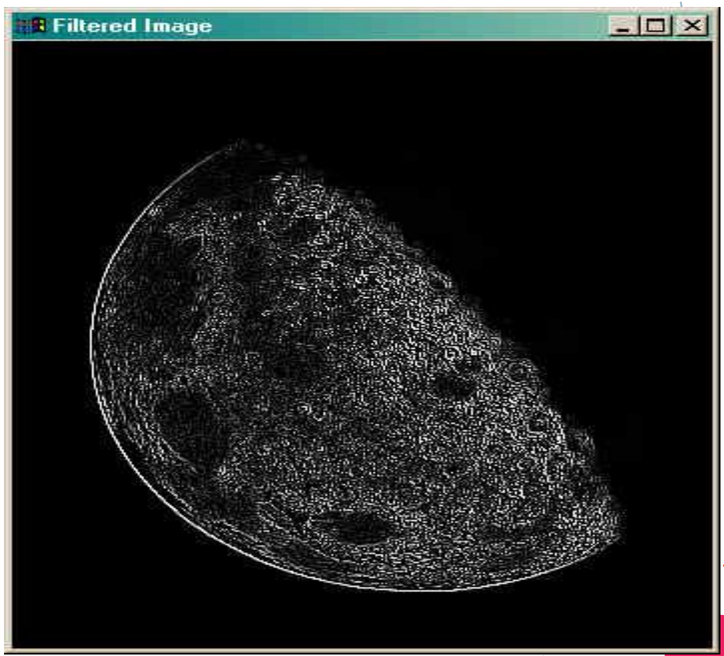


Image of the north pole of the moon

Background features can be "recovered" while still preserving the sharpening effect of the Laplacian operator simply by adding the original & the Laplacian images.

$$g(x,y) = \begin{cases} f(x,y) - \nabla^2 f(x,y) & \textit{If the center coefficient is negative} \\ f(x,y) + \nabla^2 f(x,y) & \textit{If the center coefficient is positive} \end{cases}$$

Where f(x, y) is the original image

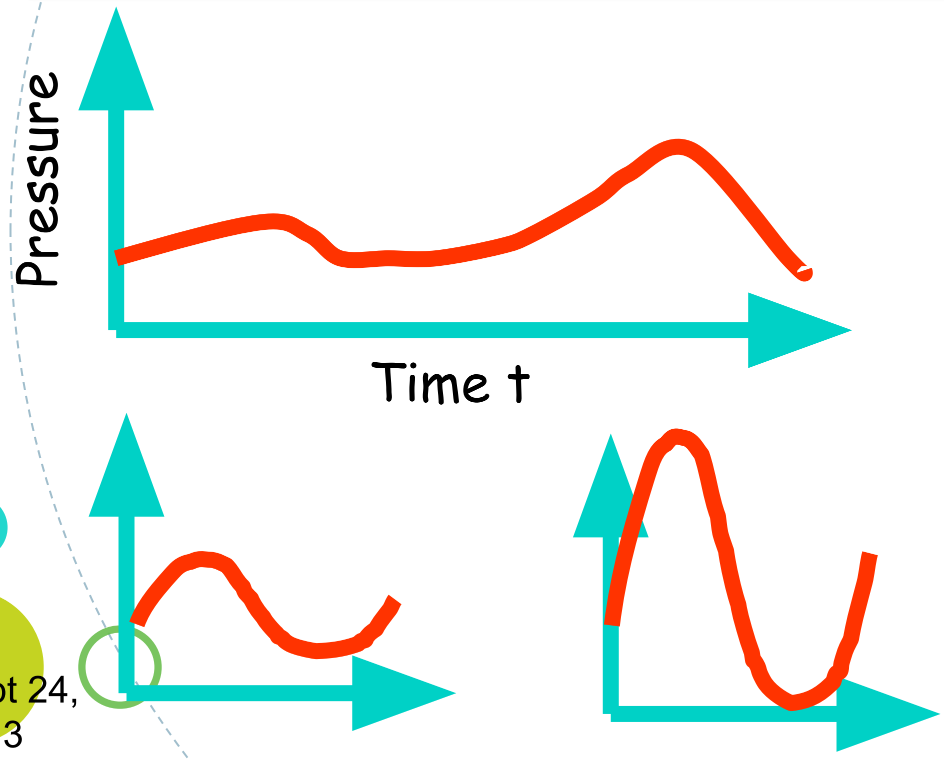
$\nabla^2 f(x,y)$ is Laplacian filtered image

g(x, y) is the sharpened image

The details in this image is clearer & sharper than in the original image

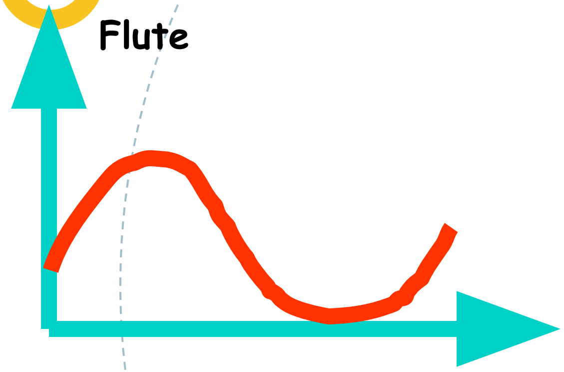
Filtering in the Frequency Domain

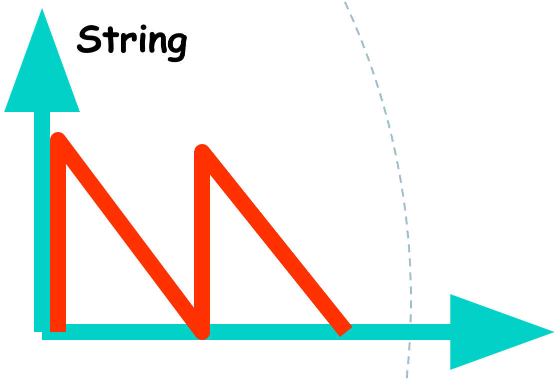
Let's first forget about images, and look at SOUND. SOUND: One dimensional function of changing (air-) pressure in time

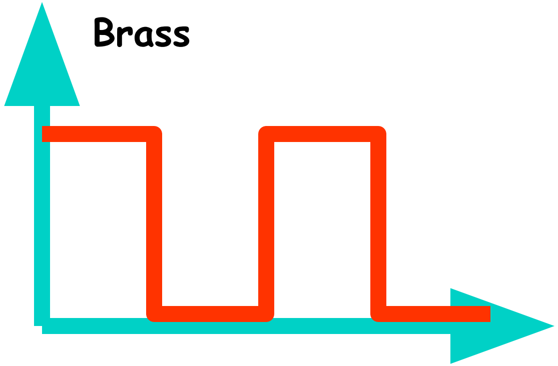


If the function is periodic, we perceive it as sound with a certain frequency (else it's noise).
The *frequency* defines the pitch.
The amplitude of the curve defines the volume.

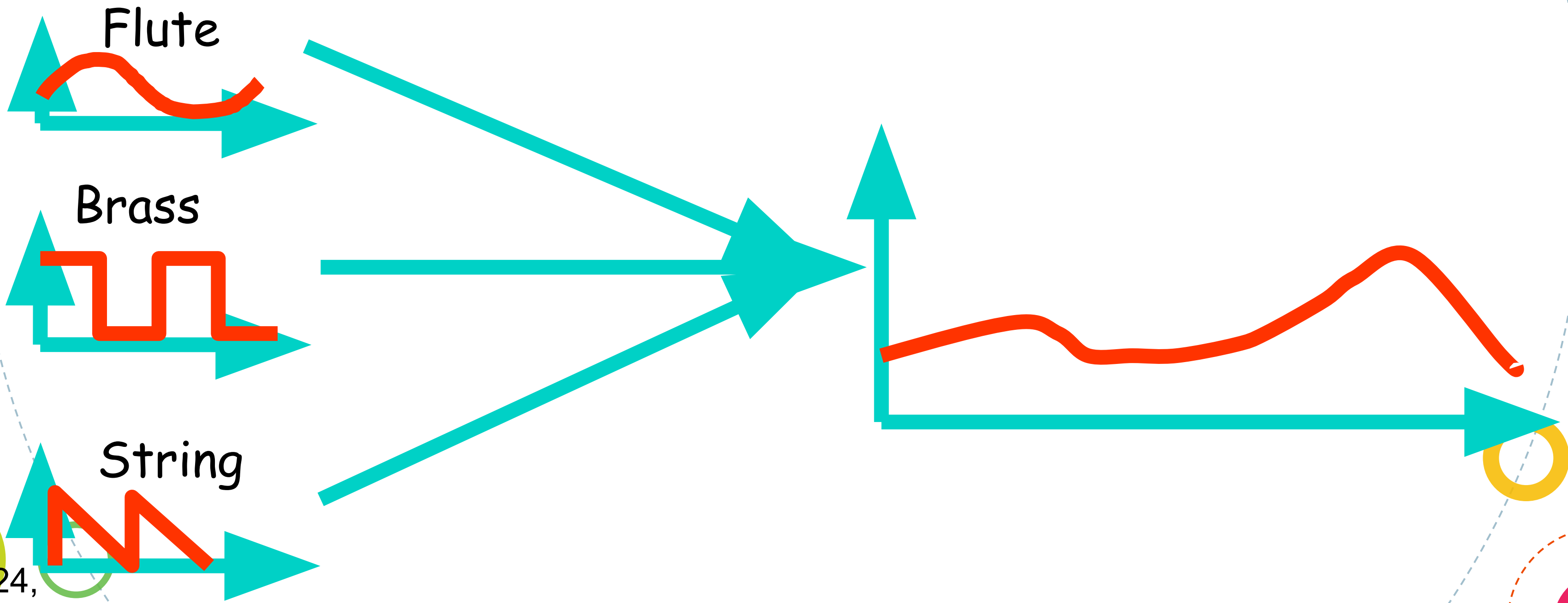
The SHAPE of the curve defines the sound character







Listening to an orchestra, you can distinguish between different instruments, although the sound is a SINGLE FUNCTION !



If the sound produced by an orchestra is the sum of different instruments, could it be possible that there are BASIC SOUNDS, that can be combined to produce every single sound ?

The answer
(Charles Fourier, 1822):

Any function that periodically repeats itself can be expressed as the sum of sines / cosines of different frequencies, each multiplied by a different coefficient

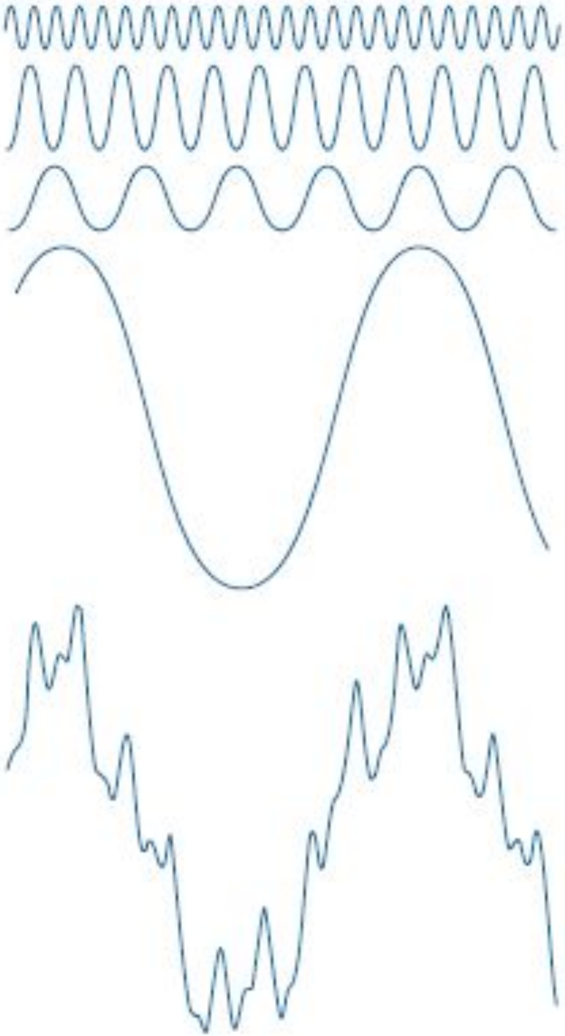
Jean Baptiste Joseph Fourier (1768-1830)

Had crazy idea (1807):

- **Any periodic function can be rewritten as a weighted sum of Sines and Cosines of different frequencies.**
- Don't believe it?
  - Neither did Lagrange, Laplace, Poisson and other big wigs
  - Not translated into English until 1878!
- But it's true!
  - called Fourier Series
  - Possibly the greatest tool used in Engineering

Fourier Transform

Even functions that are not periodic (but whose area under the curve is finite) can be expressed as the integral of sines &/or cosines multiplied by a weighting function.

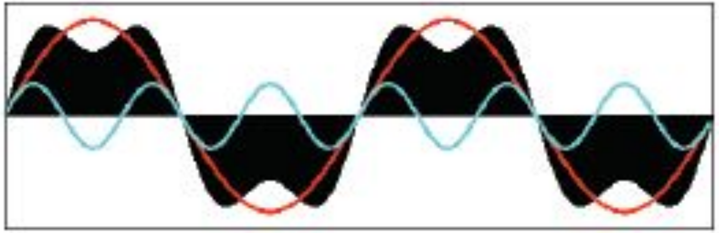


The function at the bottom is the sum of the four functions above it.

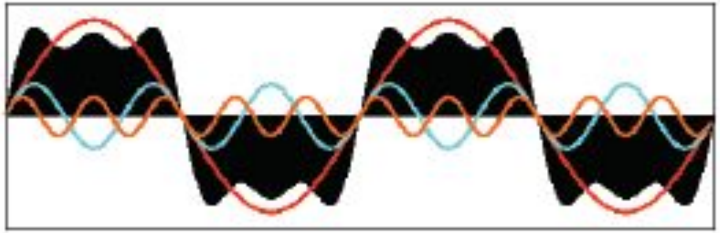
Square Wave

Frequencies: *f*

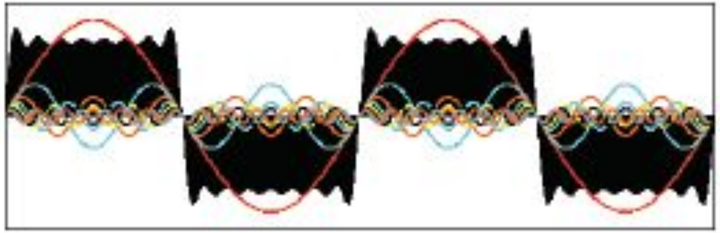
Frequencies: *f* + 3*f*



Frequencies: *f* + 3*f* + 5*f*



Frequencies: *f* + 3*f* + 5*f* + … + 15*f*



Fourier Transform & The Frequency Domain

The One-Dimensional Fourier Transform & its Inverse

Let f(x) be a continuous function of x.

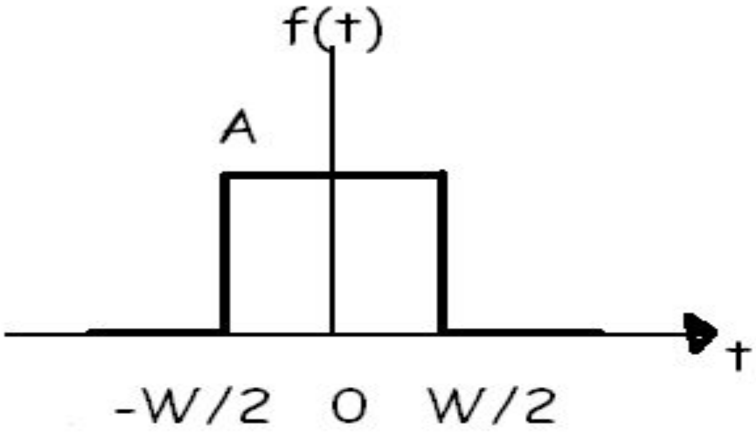
The Fourier transform of f(x) is:

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

$$= \int_{-\infty}^{\infty} f(x)[cos2\pi ux - jsin2\pi ux]dx$$

The inverse Fourier transform of f(x) is:

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$



Find its Fourier Transform.

$$F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}dt$$

$$F(u) = \int_{-\frac{W}{2}}^{\frac{W}{2}} Ae^{-j2\pi ut}dt$$

$$F(u) = \int_{-\frac{W}{2}}^{\frac{W}{2}} A e^{-j2\pi ut} dt$$

$$F(u) = \frac{-A}{j2\pi u} \left[e^{-j2\pi ut} \right]_{\frac{-W}{2}}^{\frac{W}{2}}$$

$$F(u) = \frac{-A}{j2\pi u} \left[e^{-j\pi uW} - e^{j\pi uW} \right]$$

$$F(u) = \frac{A}{j2\pi u} \left[e^{j\pi uW} - e^{-j\pi uW} \right]$$

$$F(u) = \frac{A}{j2\pi u}\left[e^{j\pi uW} - e^{-j\pi uW}\right]$$

We know that

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Therefore,

$$F(u) = \frac{A}{\pi u}\sin(\pi uW)$$

$$= AW\frac{\sin(\pi uW)}{\pi uW}$$

A *sinc* function is defined as:

$$sinc(m) = \frac{\sin \pi m}{\pi m}$$

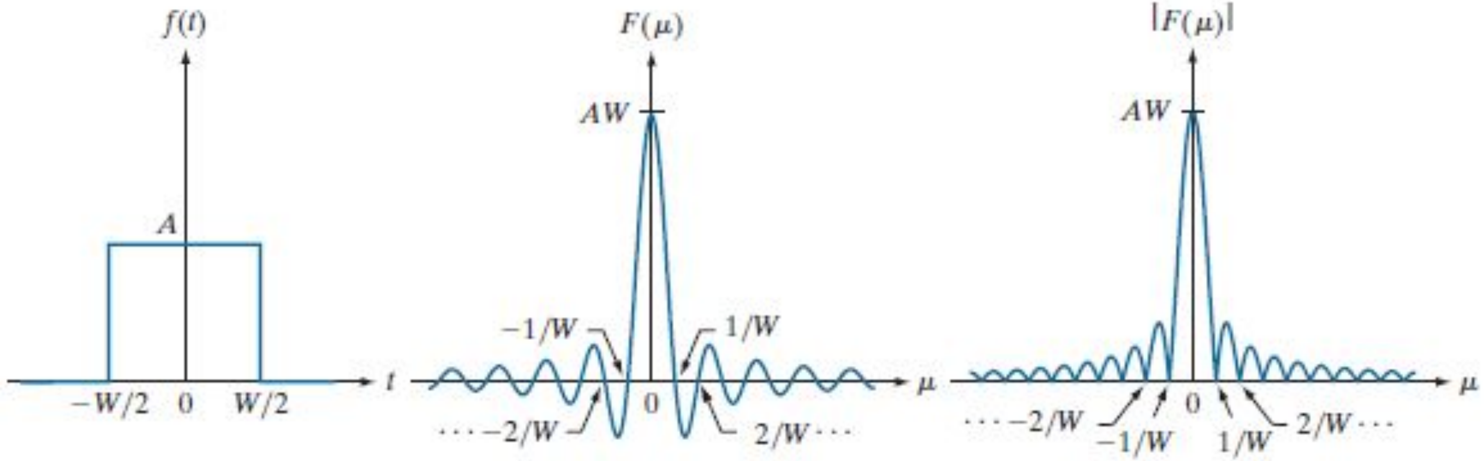
where *sinc(0) = 1* & *sinc(m) = 0* for all integer values of m.

sin 0/ 0 is indeterminate. Therefore we use L Hospital's rule & get sinc(0) = 1

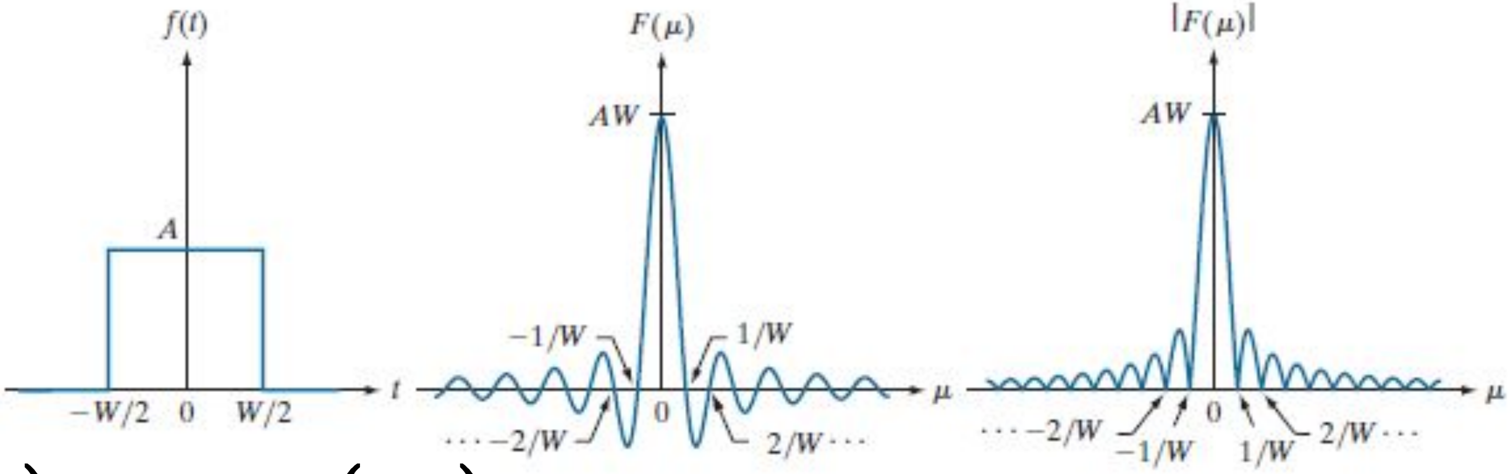
T

$$F(u) = AW \frac{\sin(\pi u W)}{\pi u W}$$

$$= AW \operatorname{sinc}(uW)$$

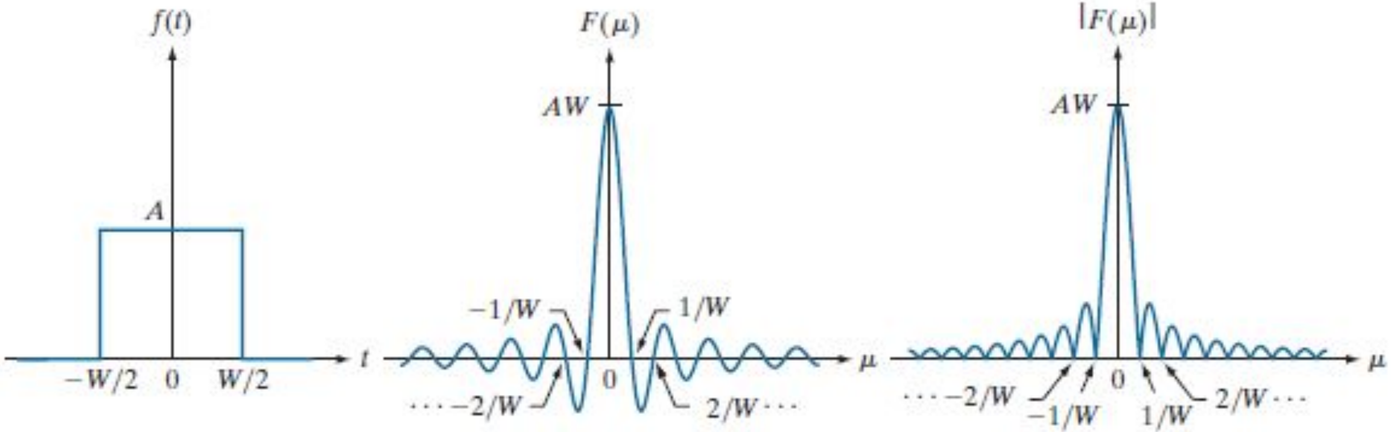


(a) A box function,
(b) its Fourier transform,
(c) its spectrum. All functions extend to infinity in both directions. Note the inverse relationship between the width, *W, of the function and the zeros of the transform.*



$$F(u) = AW \sin c(uW)$$

u	F(u)	Value of F(u)
0	AW sinc(0)	AW
1/W	AW sinc(1)	0
2/W	AW sinc(2)	0

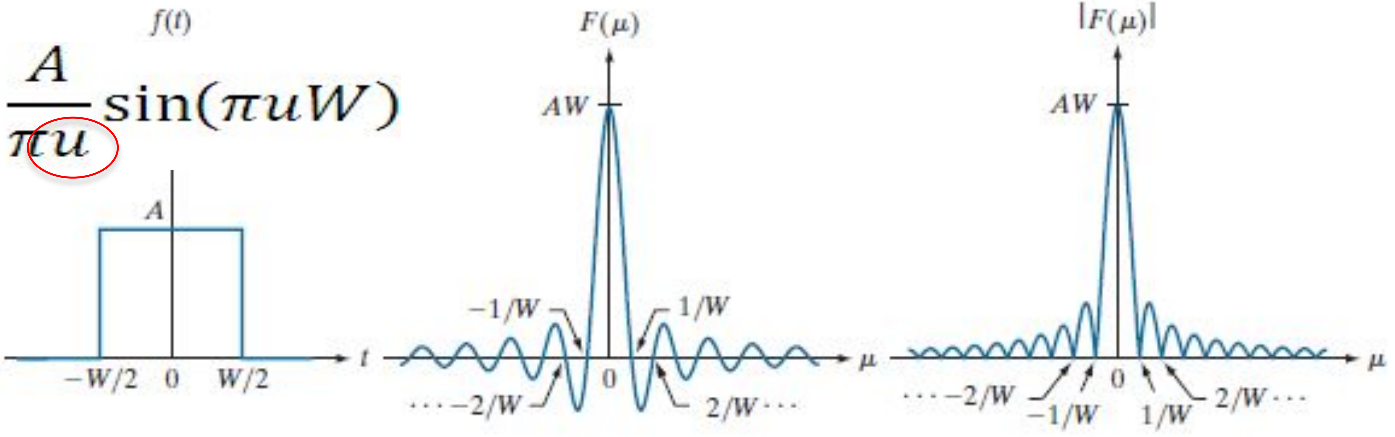


- The Fourier transform contains complex terms
- Customary for display purposes to wo[rk] with the magnitude of the transform (a real quantity)

$$|F(\mu)| = AW\left|\frac{\sin(\pi\mu W)}{(\pi\mu W)}\right|$$

- This is called the *Fourier spectrum or the frequency spectrum*:

$$F(u) = \frac{A}{\pi u}\sin(\pi u W)$$



The key properties to note are

(1) The locations of the zeros of both *F(m) and |F(m)| are* *inversely proportional to the width, W*, *of the "box"* function

(2) The height of the lobes decreases as a function of distance from the origin (Why?)

(3) The function extends to infinity for both positive and negative values of m.

The One-Dimensional Discrete Fourier Transform (DFT) & its Inverse

The Fourier transform of a discrete function of one variable, f(x), x = 0, 1, ..., M-1, is given by the following equation

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

where $j = \sqrt{-1}$

Conversely, given F(u), f(x) can be obtained by means of the *inverse* Fourier transform, i.e. we can obtain the original function back using the inverse DFT

$$f(x) = \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad \text{for } x = 0, 1, \ldots, M-1$$

$f(x) \Leftrightarrow F(u)$

***denotes a* Fourier transform pair.**

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

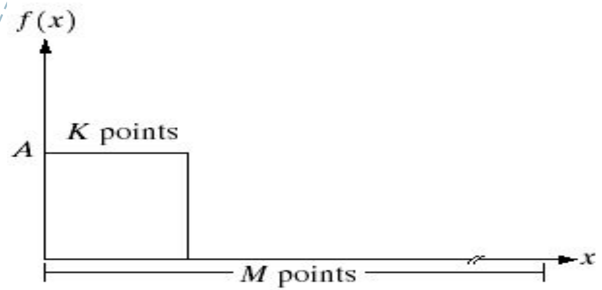
To compute F(u):

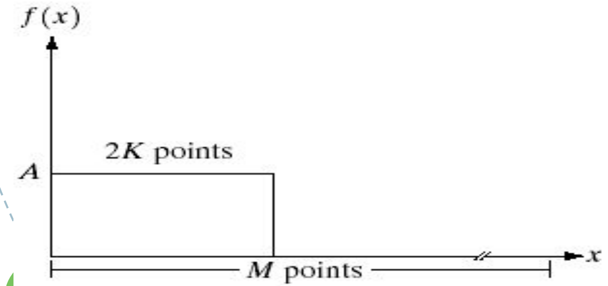
1. Substitute u = 0 in the exponential term & then sum for *all* values of x.
2. Next substitute u = 1 in the exponential & repeat the summation over all values of x.
3. Repeat this process for all M values of u in order to obtain the complete Fourier transform.

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

- Each term of the Fourier transform is composed of the sum of all values of the function f(x)
- The domain (values of u) over which the values of F(u) range is called the Frequency domain because u determines the frequency components of the transform
- Each of the M terms of F(u) are known as Frequency Component of the transform

Example: A simple one-dimensional DFT





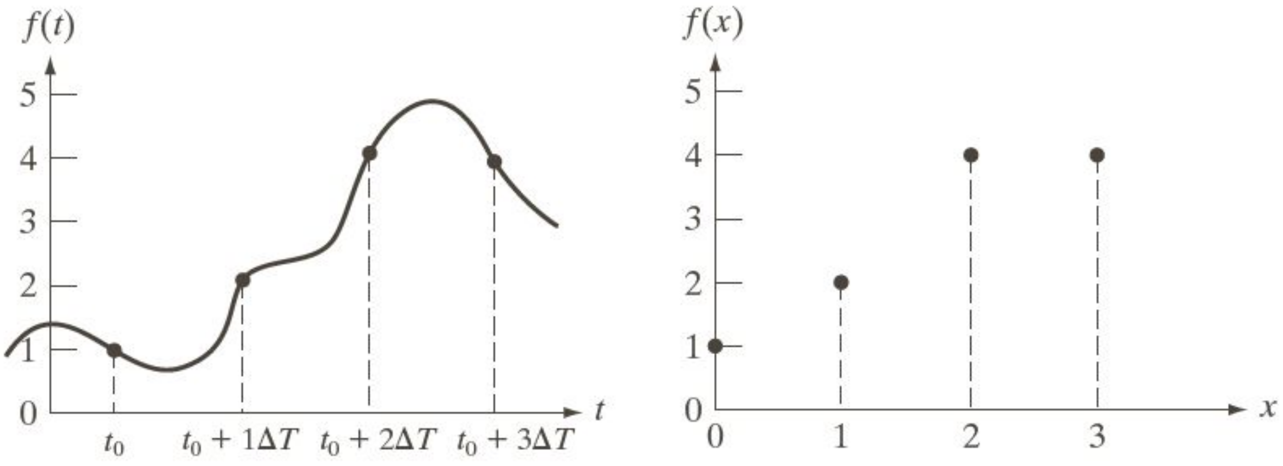
Find the value of F(0)

Find the value of F(0)

a	b
c	d

(a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.

Discrete Function & it Fourier Spectrum



- The function f(x) for x = 0, 1, …, M-1 represents M samples
- These samples are not necessarily always taken at integer values of x. They are taken at equally spaced, but otherwise arbitrary points.
- t_o represents the first point in the sequence. The first value of the sampled function is then $f(x_o)$. The next sample has been taken a fixed interval ΔT units away to give $\mathbf{f(x_1) = f(t_o + \Delta T)}$
- When we write f(x), it is understood to mean $f(x) \approx f(t_0 + x\Delta T)$

Variable u has similar interpretation

But …

The sequence always starts at true zero frequency.

Thus, the sequence for the values of u is 0, Δu, $2\Delta u$, …, $[M-1]\Delta u$

$$F(u) \triangleq f(u\Delta u)$$

Inverse relationship exits between a function & its transform. Hence

$$\Delta u = \frac{1}{M\Delta x}$$

Find the DFT of f(x) = {0, 1, 2, 1}

$M =$ 4?

Do you remember Euler's Formula:

$$e^{j\theta} = \cos\theta + j \sin\theta$$

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

Calculate F(o)

$$F(0) = \frac{1}{4}[0 + 1 + 2 + 1]$$

$$= 1$$

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

Calculate F(1)

$$F(1) = \frac{1}{4}\sum_{x=0}^{3} f(x)e^{\frac{-j2\pi ux}{M}}$$

$$= \frac{1}{4}\left[0 + 1e^{\frac{-j2\pi 1.1}{4}} + 2e^{\frac{-j2\pi 1.2}{4}} + 1e^{\frac{-j2\pi 1.3}{4}}\right]$$

$$= \frac{1}{4}\left[e^{\frac{-j\pi}{2}} + 2e^{-j\pi} + e^{\frac{-j3\pi}{2}}\right]$$

$$= \frac{1}{4}\left[\cos\left(\frac{-\pi}{2}\right) + j\sin\left(\frac{-\pi}{2}\right) + 2(\cos(-\pi) + j\sin(-\pi)) + \cos\left(\frac{-3\pi}{2}\right) + j\sin\left(\frac{-3\pi}{2}\right)\right]$$

$$= \frac{1}{4}\left[0 - j + 2(-1+0) + 0 + j\right] = \frac{-1}{2}$$

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

Calculate F(2)

$$F(2) = \frac{1}{4}\left[0 + 1e^{-\frac{j2\pi 2.1}{4}} + 2e^{-\frac{j2\pi 2.2}{4}} + 1e^{-\frac{j2\pi 2.3}{4}}\right]$$

$$= \frac{1}{4}\left[\left(\cos(\pi) - j\sin(\pi)\right) + 2\left(\cos(2\pi) - j\sin(2\pi)\right) + \left(\cos(3\pi) - j\sin(3\pi)\right)\right]$$

$$= \frac{1}{4}\left[-1 + 0 + 2\,.\,1 + 0 + -1 + 0\right]$$

$$= 0$$

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

Calculate F(3)

$$F(3) = \frac{1}{4}\left[0 + 1e^{-\frac{j2\pi 3.1}{4}} + 2e^{-\frac{j2\pi 3.2}{4}} + 1e^{-\frac{j2\pi 3.3}{4}}\right]$$

$$= \frac{1}{4}\left[\cos\left(-\frac{3\pi}{2}\right) + j\sin\left(-\frac{3\pi}{2}\right) + 2(\cos(-3\pi) + j\sin(-3\pi)) + \cos\left(-\frac{9\pi}{2}\right) + j\sin\left(-\frac{9\pi}{2}\right)\right]$$

$$= \frac{1}{4}[0 + j - 2 + 0 + 0 - j]$$

$$= \frac{-1}{2}$$

F(0) = 1	F(1) = -1/2
F(2) = 0	F(3) = -1/2

From these values, now find the values of f(x).

Angle (degrees)	0	30	45	60	90	120	135	150	180	210	225	240	270	300	315	330	360
Angle (Radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{2}$	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	UD	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	UD	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0
csc	UD	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	UD	-2	$-\sqrt{2}$	$-\frac{2}{\sqrt{3}}$	-1	$-\frac{2}{\sqrt{3}}$	$-\sqrt{2}$	-2	UD
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	UD	-2	$-\sqrt{2}$	$-\frac{2}{\sqrt{3}}$	-1	$-\frac{2}{\sqrt{3}}$	$-\sqrt{2}$	-2	UD	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
cot	UD	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	UD	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	UD

UD -> Undefined

DFT Matrix Method

Easier way to calculate DFT

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-\frac{j2\pi ux}{M}} \qquad \text{for } u = 0, 1, \ldots, M-1$$

Twiddle Factor

A new factor defined as

$$W_M = e^{\frac{-j2\pi}{M}}$$

The DFT now reduces to

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) W_M^{xu}$$

To solve this equation, form a square matrix W_M of size M x M

The equation then reduces to

$$\mathbf{F = W \cdot f}$$

If M = 4, then W_M will be a 4x4 matrix

$$W_4 = \begin{array}{c|cccc} u/x & 0 & 1 & 2 & 3 \\ \hline 0 & & & & \\ 1 & & & & \\ 2 & & & & \\ 3 & & & & \end{array}\begin{bmatrix} W_M^0 & W_M^0 & W_M^0 & W_M^0 \\ W_M^0 & W_M^1 & W_M^2 & W_M^3 \\ W_M^0 & W_M^2 & W_M^4 & W_M^6 \\ W_M^0 & W_M^3 & W_M^6 & W_M^9 \end{bmatrix}$$

$$\boxed{W_M = e^{\frac{-j2\pi}{M}}}$$

$$W_M = ??$$

$$W_M^1 = ??$$

$$W_M^1 = e^{\frac{-j2\pi.1}{4}} = \cos\left(-\frac{\pi}{2}\right) + j\sin\left(-\frac{\pi}{2}\right) = -j$$

$$W_M^0 = e^{\frac{-j2\pi.0}{4}} = 1$$

F(0) = 1   F(1) = -1/2

F(2) = 0   F(3) = -1/2

Find the DFT of f(x) = {0, 1, 2, 1}

$$\begin{bmatrix} F(0) \\ F(1) \\ F(2) \\ F(3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

What do these values indicate????

$F(0) = 1$ $F(1) = -1/2$

$F(2) = 0$ $F(3) = -1/2$

Where are the sin / cos waves?

Let's go to the basics of Fourier Transform

Any M point signal can be decomposed into M + 2 signals, half of them sine waves and half of them cosine waves.

F(0) indicates that the waves (both sine & cosine) make 0 complete cycles over M samples. F(1) indicates they make one complete cycle, F(2) means two complete cycles & so on.

- The sine and cosine waves used in the DFT are commonly called the **DFT basis functions.** The basis functions are a set of sine and cosine waves with *unity amplitude*.

- The output of the DFT is a set of numbers that represent amplitudes.

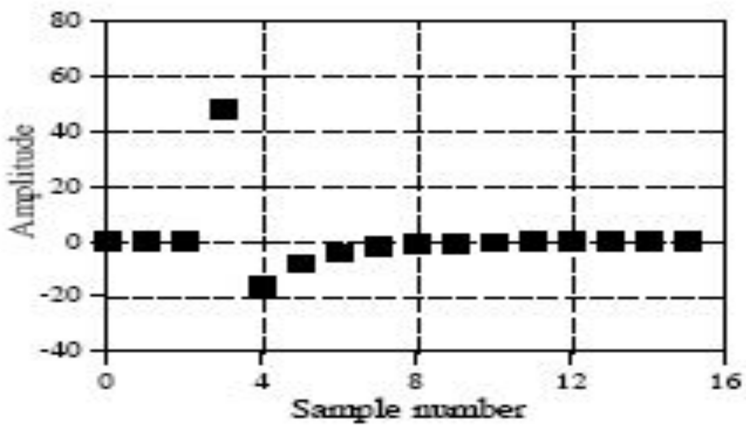
- *If you assign each amplitude (the frequency* domain) to the proper sine or cosine wave (the basis functions), the result is a set of *scaled sine and cosine waves that can be added to form the time* domain signal.

The DFT basis functions are generated from the equations:

$$\mathbf{C_u[x] = cos\ (2\Pi\ ux/M)}$$

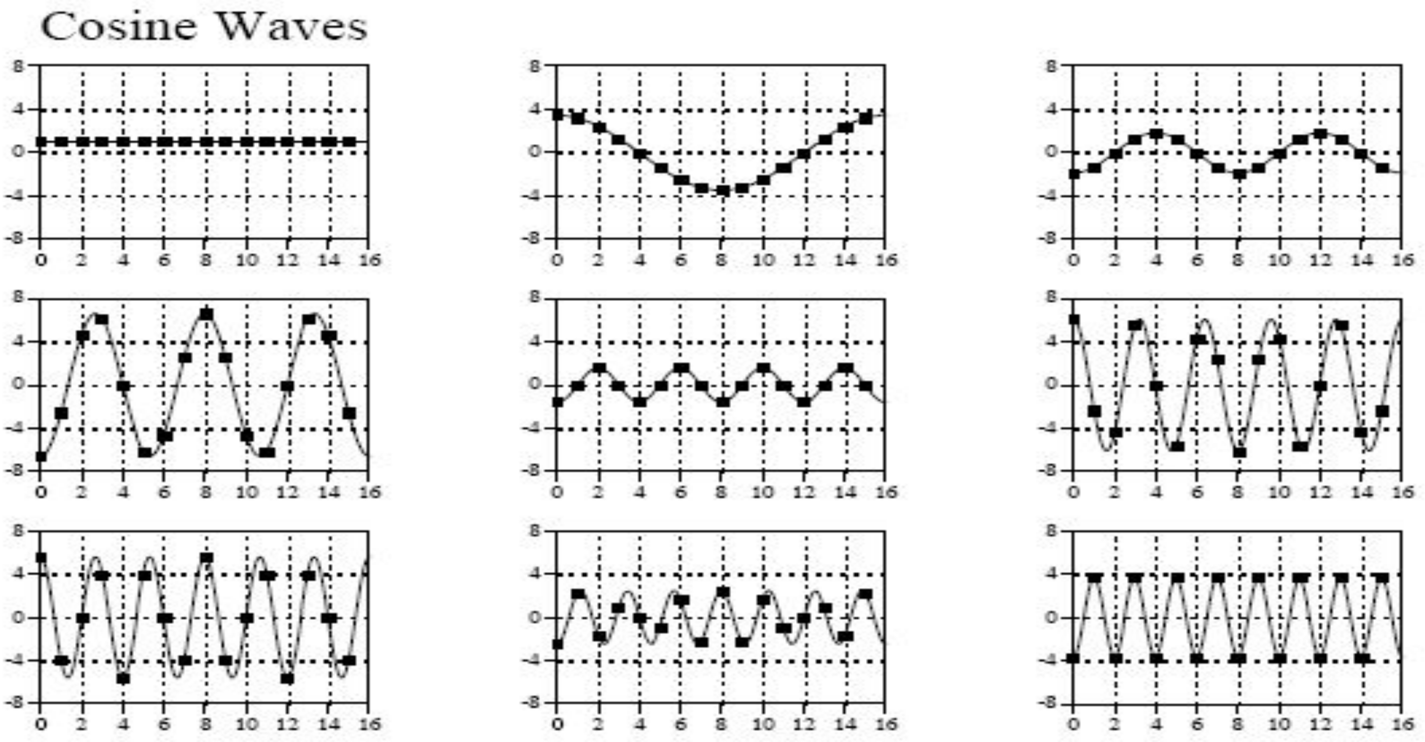
$$\mathbf{S_u[x] = sin\ (2\Pi\ ux/M)}$$

Consider a signal 16 points long.



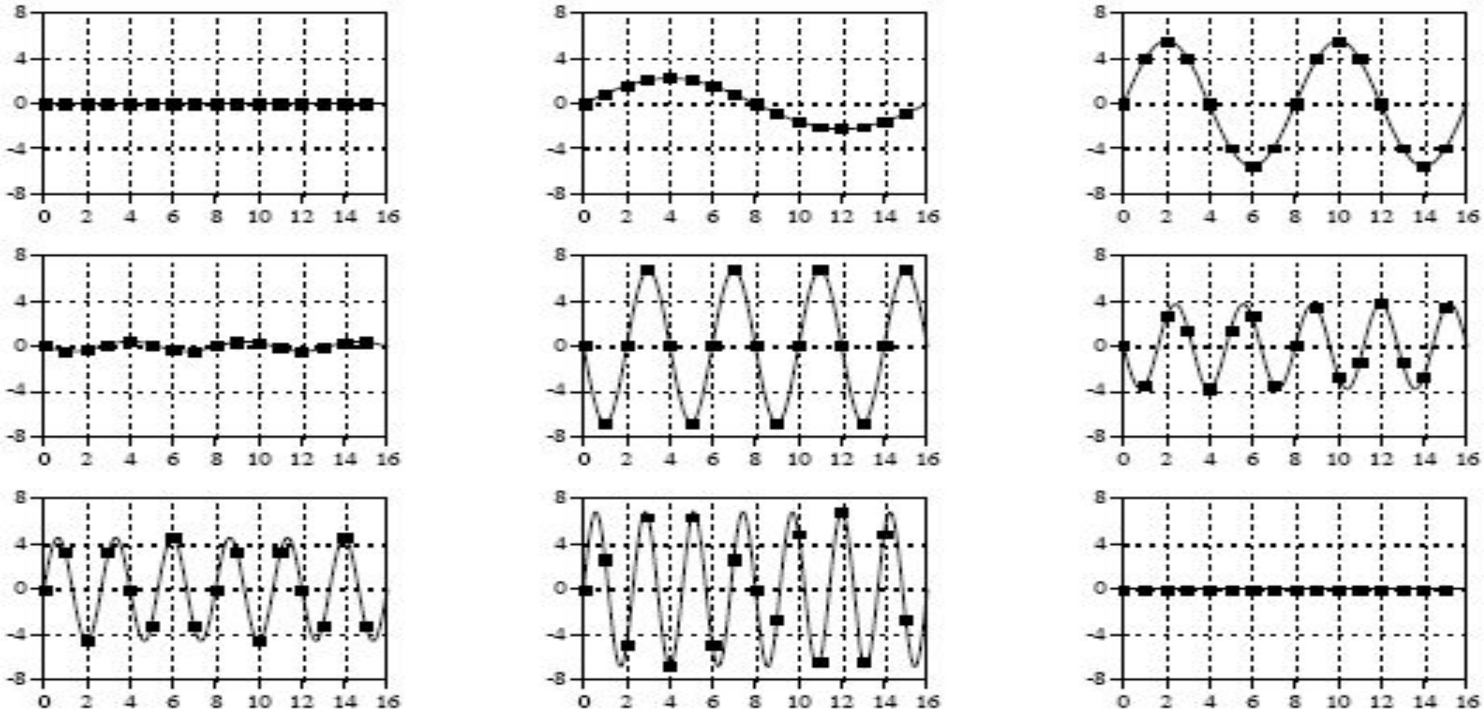
The Fourier decomposition of this signal gives nine cosine waves and nine sine waves, each with a different frequency and amplitude.

$C_u[x] = \cos(2\Pi ux/M)$



$$S_u[x] = \sin(2\Pi\, ux/M)$$

Sine Waves



When $f(x) = \{0, 1, 2, 1\}$, we got

$F(u) = \{1, -1/2, 0, -1/2\}$

From $F(u)$, now get back $f(x)$

Remember?

The DFT basis functions are generated from the equations:

$$C_u[x] = \cos(2\Pi\, ux/M)$$

$$S_u[x] = \sin(2\Pi\, ux/M)$$

u	F(u)		x_0	x_1	x_2	x_3
0	1	$C_0(x)$	1	1	1	1
		$S_0(x)$	0	0	0	0
1	-1/2	$C_1(x)$	-1/2	0	1/2	0
		$S_1(x)$	0	-1/2	0	1/2
2	0	$C_2(x)$	0	0	0	0
		$S_2(x)$	0	0	0	0
3	-1/2	$C_3(x)$	-1/2	0	1/2	0
		$S_3(x)$	0	1/2	0	-1/2
Sum			**0**	**1**	**2**	**1**

The Two-Dimensional Discrete Fourier Transform (DFT) & its Inverse

Two Dimensional Discrete Fourier Transform (DFT) Pair

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

Properties of DFT

- **Separability**
- **Translation**

The Separability Property

A 2-D DFT can be separated into two 1-D DFT

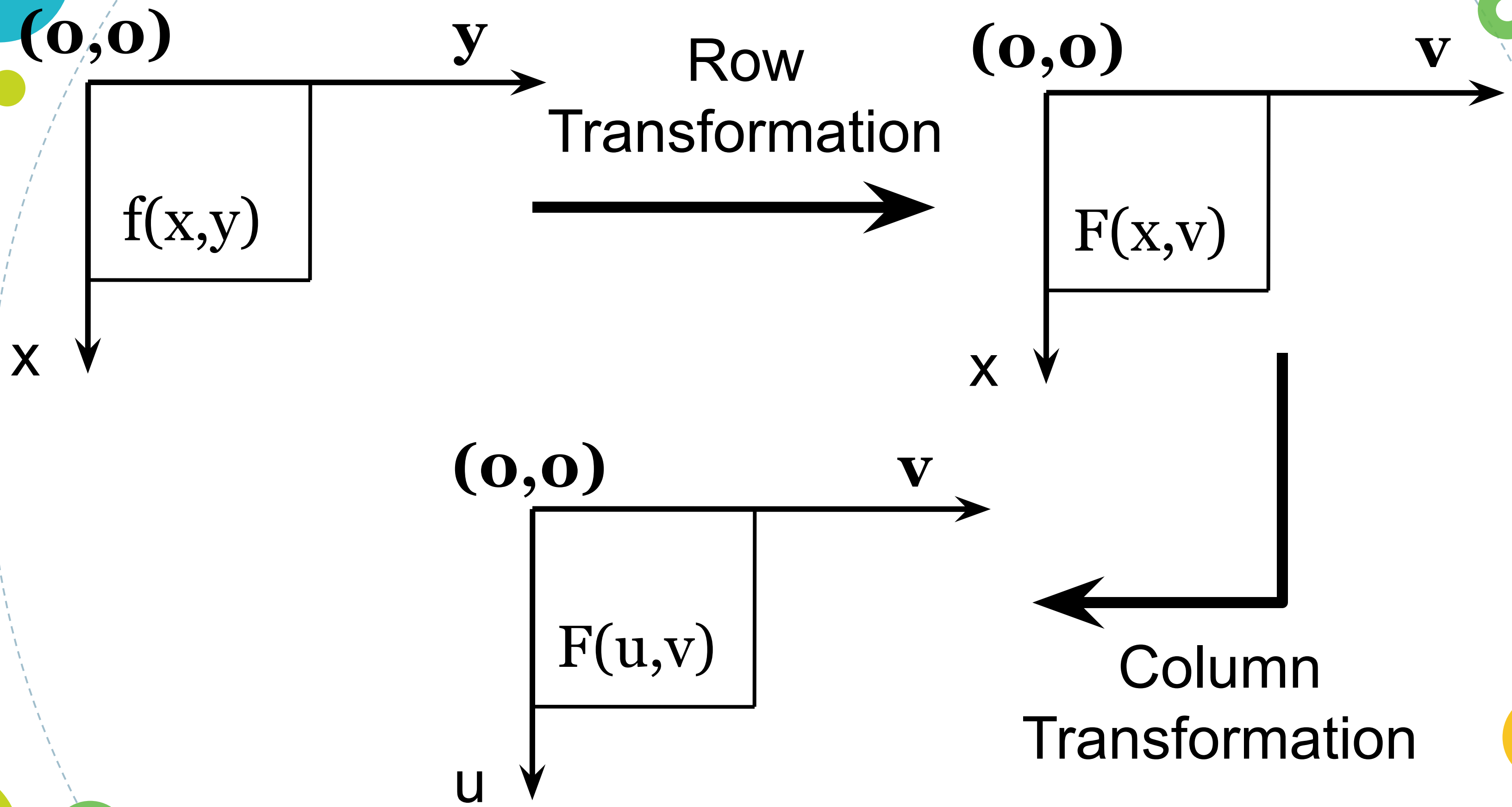
$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

We can rearrange the terms as follows

$$F(u,v) = \frac{1}{M}\sum_{x=0}^{M-1} e^{-j2\pi\frac{ux}{M}} \frac{1}{N}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\frac{vy}{N}}$$

$$F(u,v) = \frac{1}{M}\sum_{x=0}^{M-1} F(x,v)\, e^{-j2\pi\frac{ux}{M}}$$


(o,o)
y
f(x,y)
x
Row
Transformation
(o,o)
v
F(x,v)
x
Column
Transformation
(o,o)
v
F(u,v)
u

Q) Find the DFT of the following image.

0	1	2	1
1	2	3	2
2	3	4	3
1	2	3	2

Hint:
First do row-wise transformation & then column-wise.

$$F(0,v) = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 4 \\ -2 \\ 0 \\ -2 \end{bmatrix}$$

Likewise find the DFT of other rows.

$$\begin{bmatrix} 1 & \frac{-1}{2} & 0 & \frac{-1}{2} \\ 2 & \frac{-1}{2} & 0 & \frac{-1}{2} \\ 3 & \frac{-1}{2} & 0 & \frac{-1}{2} \\ 2 & \frac{-1}{2} & 0 & \frac{-1}{2} \end{bmatrix}$$

Now find the 1-D DFT along the columns of this intermediate image.

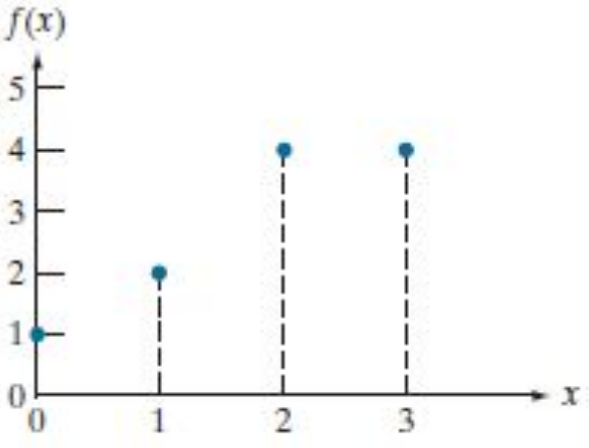
$$F(u,0) = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 8 \\ -2 \\ 0 \\ -2 \end{bmatrix}$$

Likewise find the DFT of other columns.

The final DFT of the entire image is:

$$\begin{bmatrix} 2 & \frac{-1}{2} & 0 & \frac{-1}{2} \\ \frac{-1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-1}{2} & 0 & 0 & 0 \end{bmatrix}$$



Find all F(u)s

The Translation Property

If f(x,y) is multiplied by an exponential, the original Fourier Transform, F(u,v), gets shifted in frequency by $F(u-u_o, v-v_o)$

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$\Im(f(x,y)) = F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$\Im\left(f(x,y)e^{j2\pi\left(\frac{u_0x}{M}+\frac{v_0y}{N}\right)}\right) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}e^{j2\pi\left(\frac{u_0x}{M}+\frac{v_0y}{N}\right)}$$

$$= \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{(u-u_0)x}{M}+\frac{(v-v_0)y}{N}\right)}$$

$= F(u-u_0, v-v_0)$

Thus, if

$$\Im\big(f(x,y)\big) = F(u,v)$$

Then,

$$\Im\left(f(x,y)e^{\left(\frac{u_0 x}{M}+\frac{v_0 y}{N}\right)}\right) = F(u-u_0, v-v_0)$$

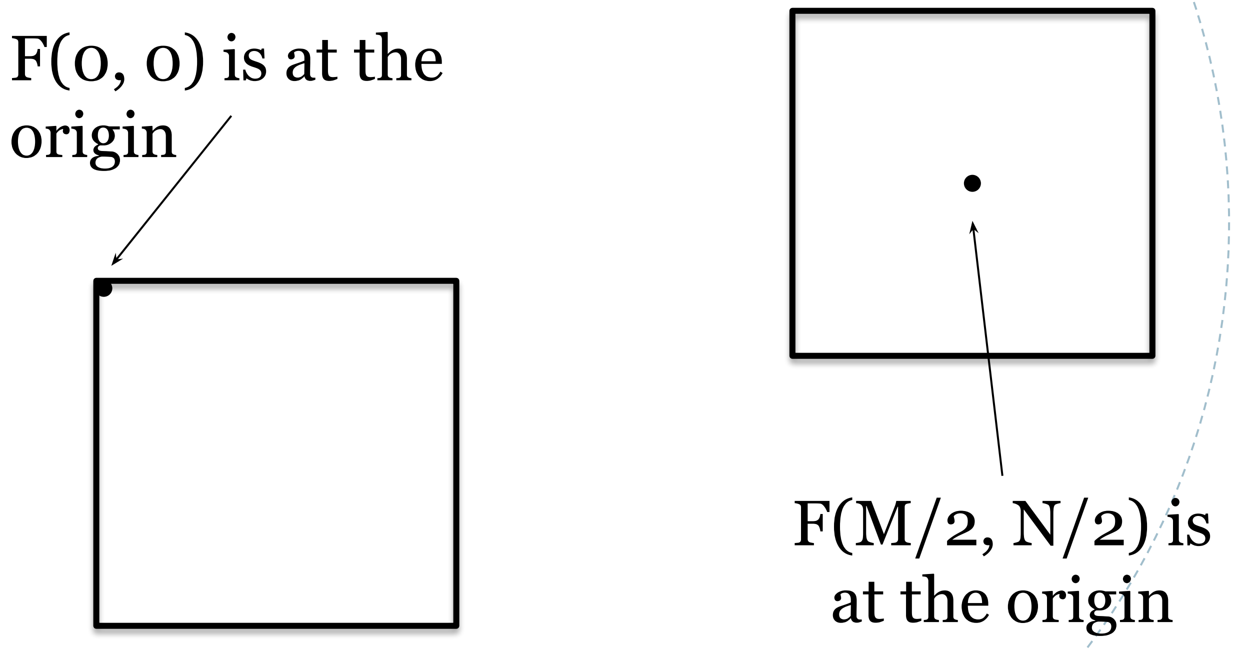
What happens when $u_0 = M/2$ & $v_0 = N/2$

Why?

For an M x N image, what are the co ordinates of the centre pixel?

Since (x + y) is an integer
So $\sin\Pi(x+y) = 0$ & $\cos\Pi(x+y)$ is 1 if (x+y) is even & -1 if it is odd

$$\therefore \Im\left(f(x,y)(-1)^{(x+y)}\right) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$



A common practice …

Multiply the i/p image function by $(-1)^{x+y}$ prior to computing the Fourier transform

This shifts the origin of F(u, v) to frequency coordinates (M/2, N/2),which is the centre of the M x N area occupied by the 2-DFT

The Average Value Property

What is the average of all the intensity values of the image?

$$\bar{f}(x,y) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

The value of transform at (u, v) = (0, 0)

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$

Which is the average of f(x, y)
f(x, y) is an image
The value of the Fourier transform at origin is equal to the average gray-level of the image

What do we mean by *frequency* in an image?

Low frequency region

High frequency region

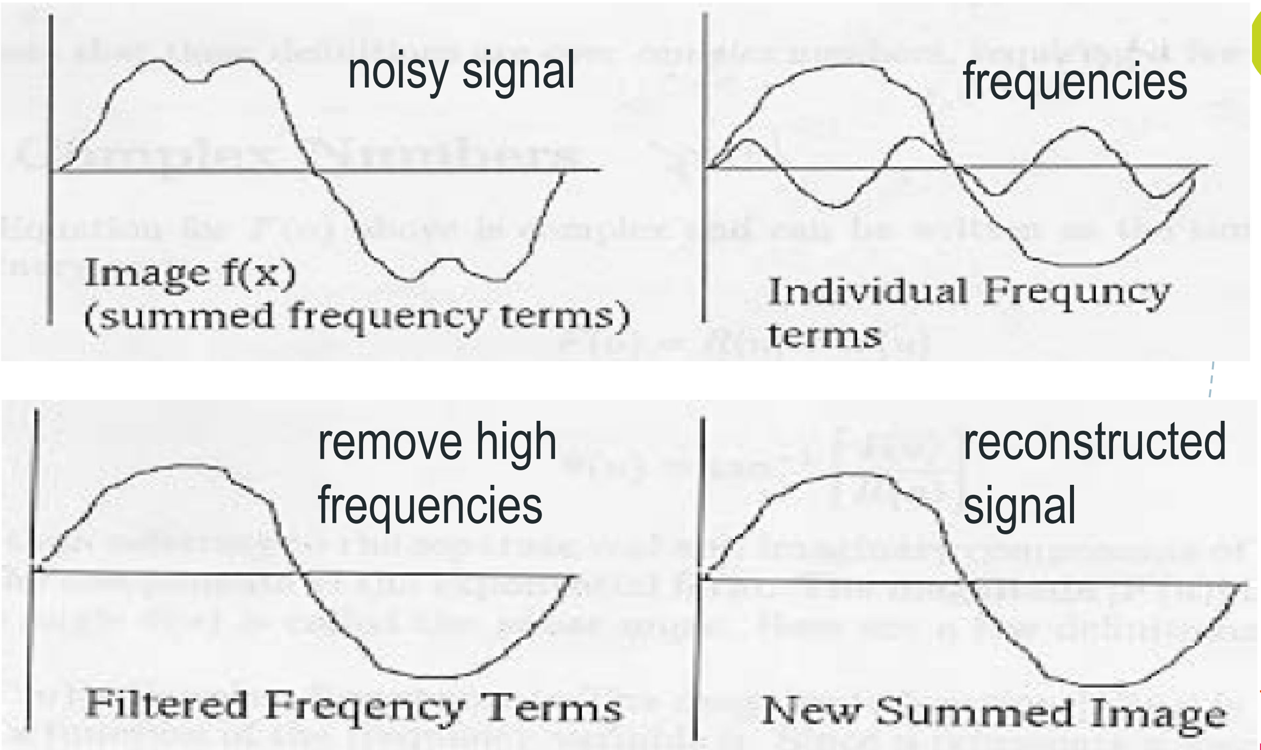
Low frequency region

Why is FT Useful?

- Remove undesirable frequencies from a signal.
- Easier and faster to perform certain operations in the **frequency** domain than in the **spatial** domain.

Example:
Removing undesirable frequencies

To remove certain frequencies, set their corresponding F(u) coefficients to zero!



Filtering In Frequency Domain

Steps for Filtering in Frequency Domain

1. Multiply the i/p image by $(-1)^{x+y}$ to centre the transform

$$\mathcal{F}\left[f(x,y)(-1)^{x+y}\right] = F\left(u-\frac{M}{2}, v-\frac{N}{2}\right)$$

denotes the Fourier transform of the argument

Compute F(u, v), the DFT of the image

3. Multiply F(u, v) by a *filter* function H(u, v)

4. Compute the inverse DFT

5. Obtain the real part of the result

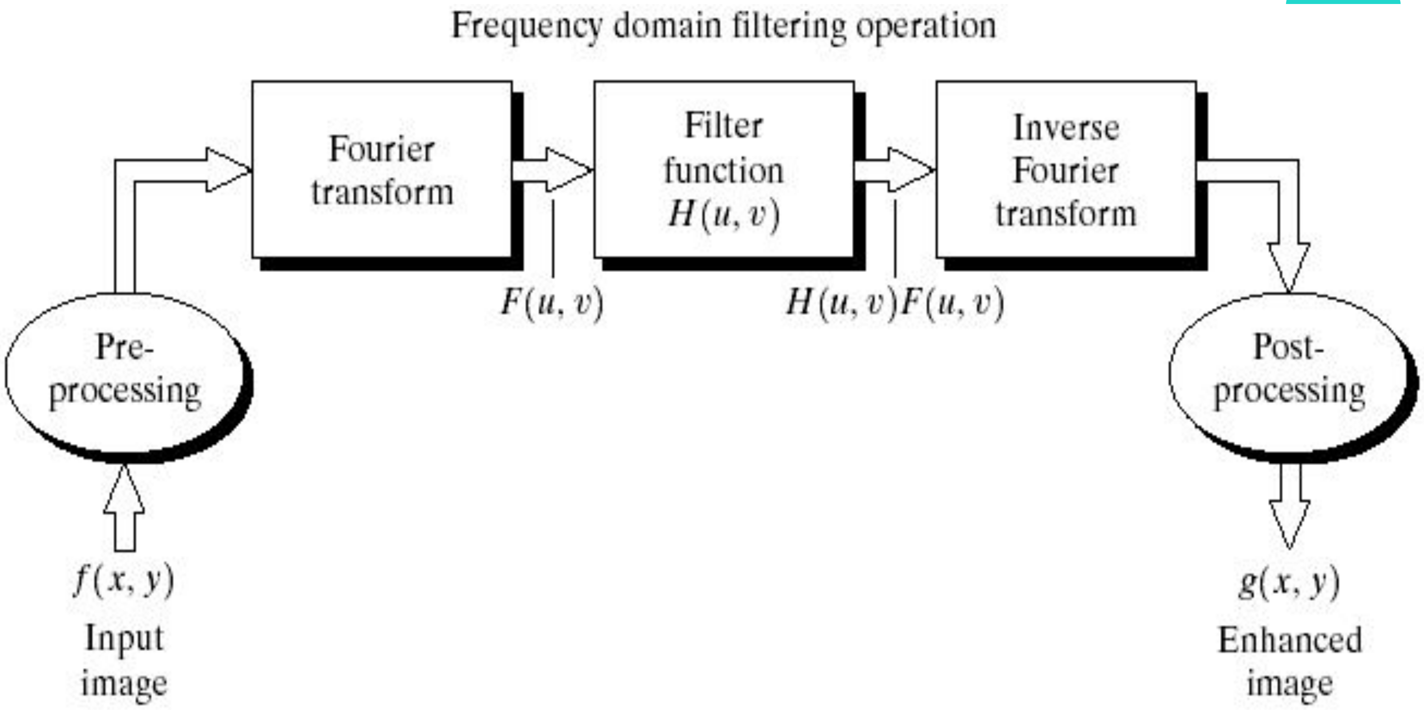
6. Multiply the result by $(-1)^{x+y}$

$$G(u, v) = H(u, v)\, F(u, v)$$

Filter

Fourier Transform of the i/p image f(x, y)

Multiplication of H & F On an element by element basis



Basic steps for filtering in the frequency domain.

Low Frequencies

Responsible for the general gray level appearance of an image over smooth areas

High Frequencies

Responsible for detail, such as edges & noise

Some Basic Filters

Filtering techniques in the frequency domain are based on modifying the Fourier transform to achieve a specific objective, and then computing the inverse DFT to get us back to the spatial domain

Frequency Domain Filtering Fundamentals

Given (a padded) digital image, f (x, y), of size P × Q pixels, the basic filtering equation has the form:

$$g(x, y) = \mathrm{Real}\{\underbrace{\mathfrak{I}^{-1}}_{\text{IDFT}}[\underbrace{H(u, v)}_{\text{Filter}}F(u, v)]\}$$

- The product H(u,v)F(u,v) is formed using element wise multiplication
- The filter transfer function modifies the transform of the input image to yield the processed output, $g(x, y)$.
- Specifying H(u,v) becomes simple by using functions that are symmetric about their center, which requires that F(u,v) be centered also.
- This is accomplished by multiplying the input image by $(-1)^{x+y}$ *prior to computing* its transform.

Notch Filter

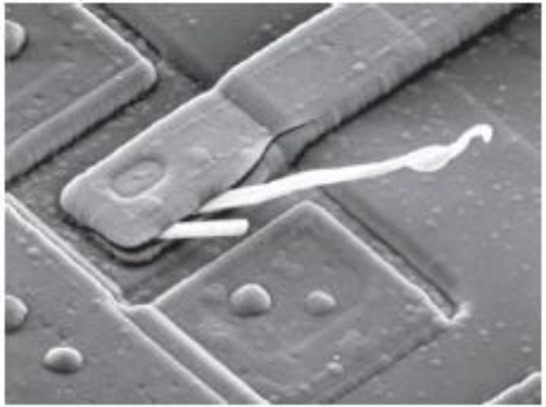
- One of the simplest filters
- Set H(u,v) to be 0 at the center of the (centered) transform, and 1's elsewhere.
- This filter rejects the dc term and "passes" (i.e., leave unchanged) all other terms of F(u,v) when the product H(u,v)F(u,v) is done.
- Remember the dc term is responsible for the average intensity of an image, so setting it to zero will reduce the average intensity of the output image to zero.
- An average of zero implies the existence of negative intensities. Therefore, all negative intensities are set to 0.

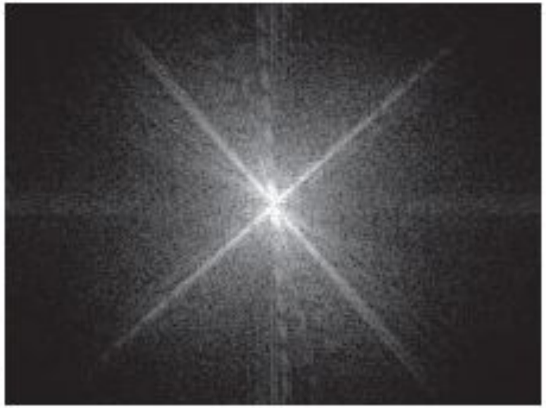
If the transform has been centred, the average value can be forced to zero by the following *filter function*:

$$H(u, v) = \begin{cases} 0 \text{ if } (u, v) = \left(\frac{P}{2}, \frac{Q}{2}\right) \\ 1 \quad \text{otherwise} \end{cases}$$

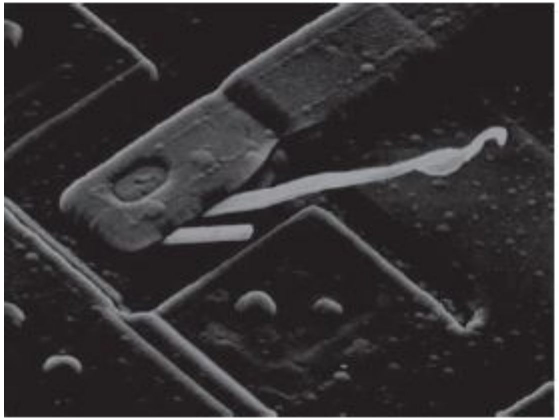
- This filter sets F(0, 0) to zero & leaves all other frequency components of the Fourier transform untouched
- The processed image can be obtained by taking the inverse Fourier transform of

$$H(u, v)\, F(u, v)$$





a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a)



Result of filtering the image with a filter transfer function that sets to 0 the dc term, $F(P/2, Q/2)$, in the centered Fourier transform, while leaving all other transform terms unchanged.

Low frequencies in the transform are related to slowly varying intensity components in an image, such as the walls of a room or a cloudless sky in an outdoor scene.

High frequencies are caused by sharp transitions in intensity, such as edges and noise.

Lowpass Filter

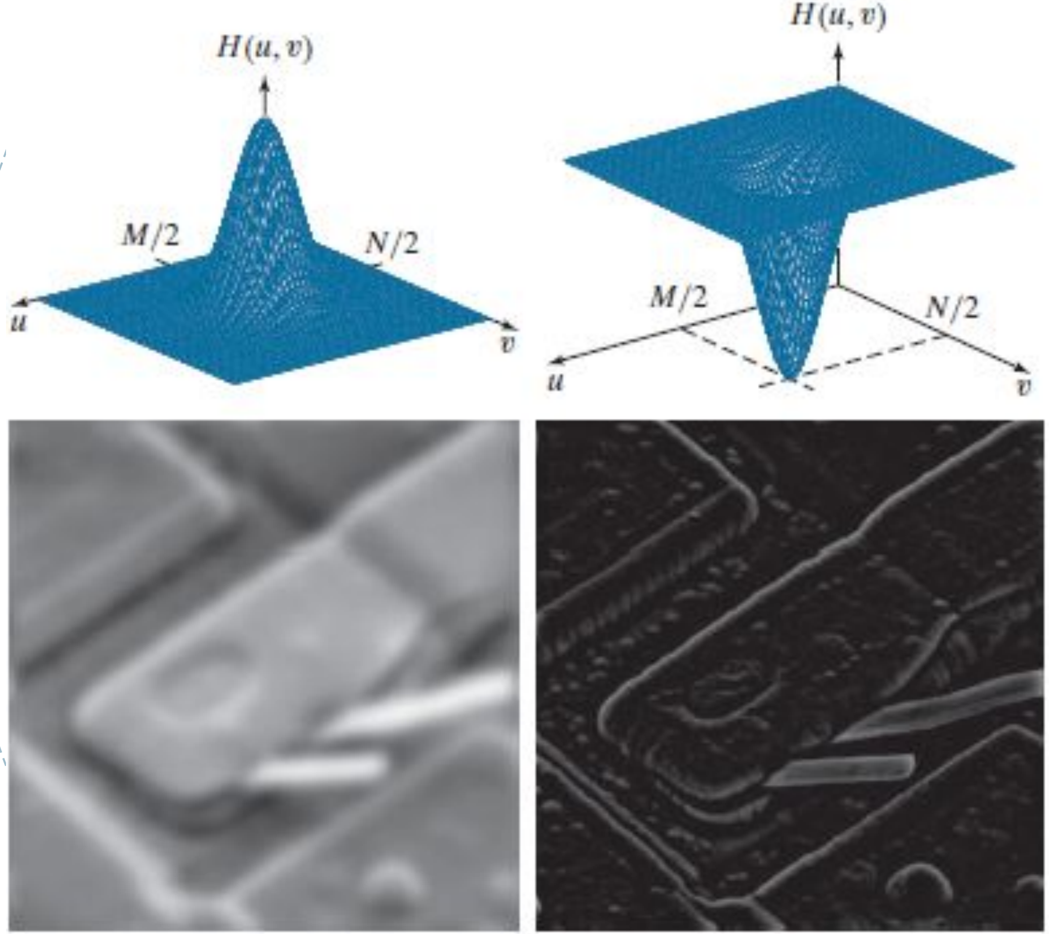
- Attenuates high frequencies
- Passes low frequencies

Lowpass filtered images have less sharp details than the original image as high frequencies have been attenuated

Highpass Filter

- Attenuates low frequencies
- Passes high frequencies

Highpass filtered images have less gray level variations in the smooth areas & emphasized transitional (e.g. edge) gray level details. Such images appears sharper.



Top row: Frequency domain filter transfer functions of
(a) a lowpass filter,
(b) a highpass filter,

Bottom row: Corresponding filtered images

Smoothing Frequency Domain Filters

Smoothing (blurring) is achieved in the frequency domain by attenuating a specified range of high frequency components in the transform of a given image

Ideal Lowpass Filters

Cuts – off all high frequency components of the Fourier transform which are at a distance greater than a specified distance D_o from the origin of the (centred) transform

If the size of the image is M x N, then the size of the transform is M x N

?

Centre of the frequency rectangle is at (u, v) = (M/2, N/2)

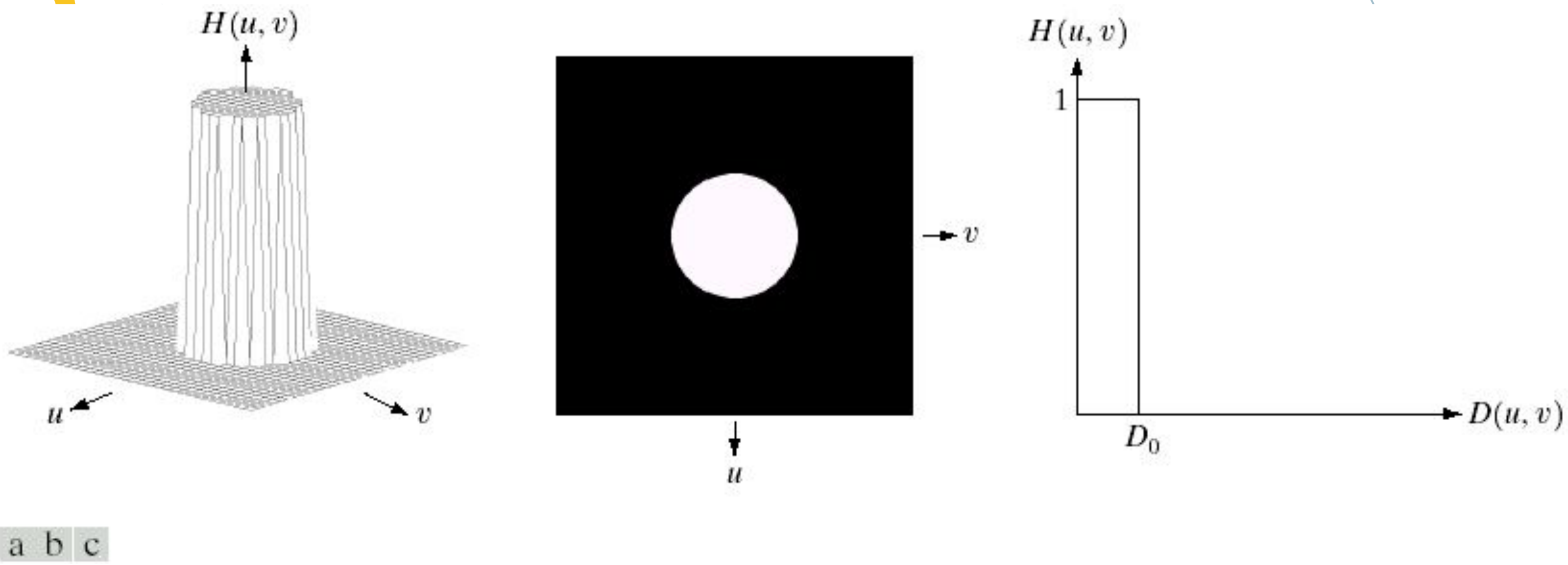
?

Transfer function of ILPF

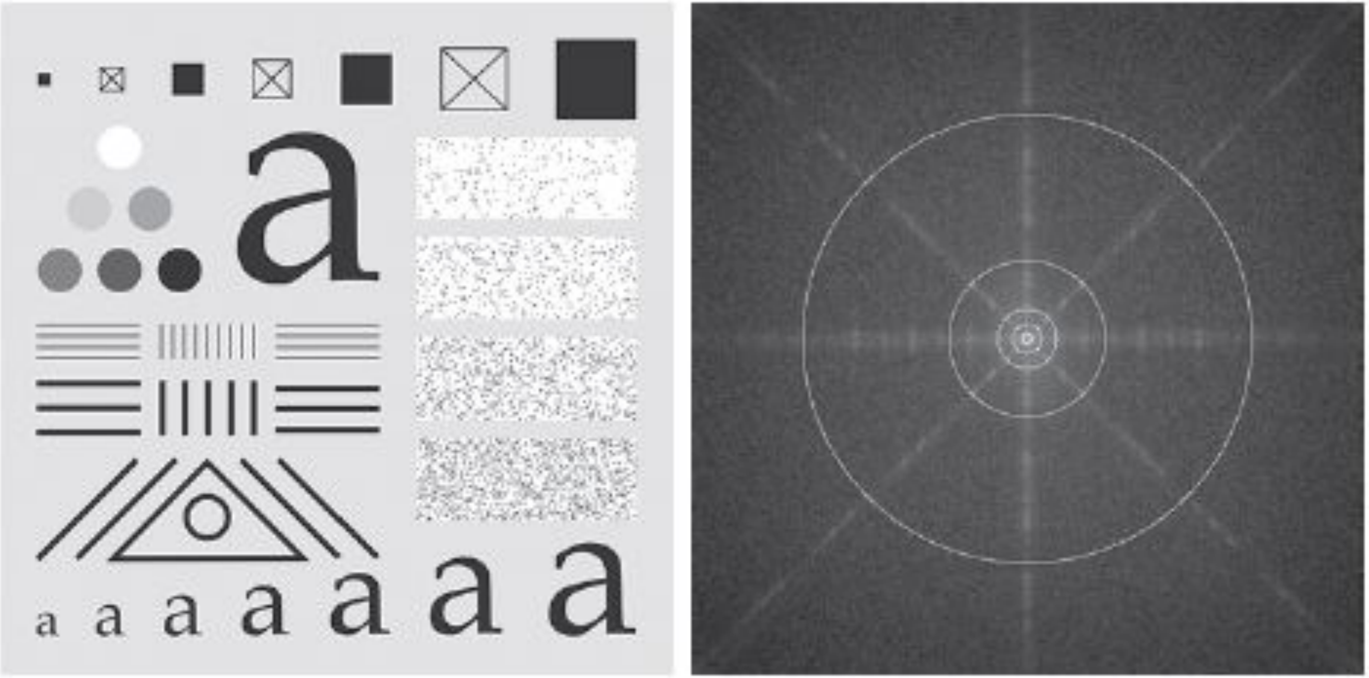
$$H(u,v) = \begin{cases} 1 \ \ if\ D(u,v) \le D_0 \\ 0 \ if\ D(u,v) > D_0 \end{cases}$$

D(u, v) is the distance between point (u, v) & the origin of the frequency rectangle. It is given as follows:

$$D(u, v) = [(u - M/2)^2 + (v - N/2)^2]^{1/2}$$



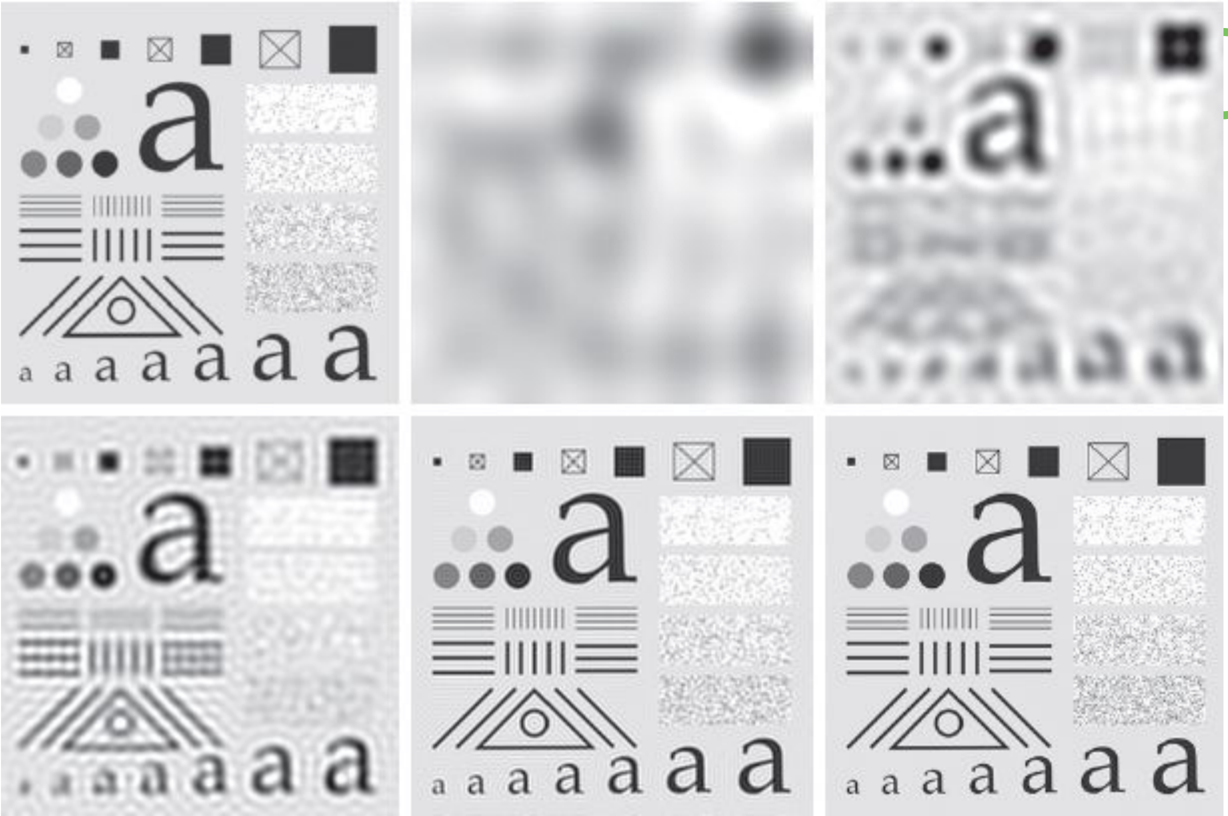
(a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.



a b

FIGURE 4.40 (a) Test pattern of size 688 × 688 pixels, and (b) its spectrum. The spectrum is double the image size as a result of padding, but is shown half size to fit. The circles have radii of 10, 30, 60, 160, and 460 pixels with respect to the full-size spectrum. The radii enclose 86.9, 92.8, 95.1, 97.6, and 99.4% of the padded image power, respectively.

Filtering with ILPF



(a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460

Note:

- When the cut off frequency loci is small, all details are lost due to blurring. Large objects appear as blobs.

- As the filter radius increases, blurring becomes less severe.

- A “ringing” effect becomes finer in texture as the amount of high frequency content removed decreases.

The “ringing” effect is a characteristic of ideal filters.

Gaussian Lowpass Filters

The transfer function of a GLPF is given as:

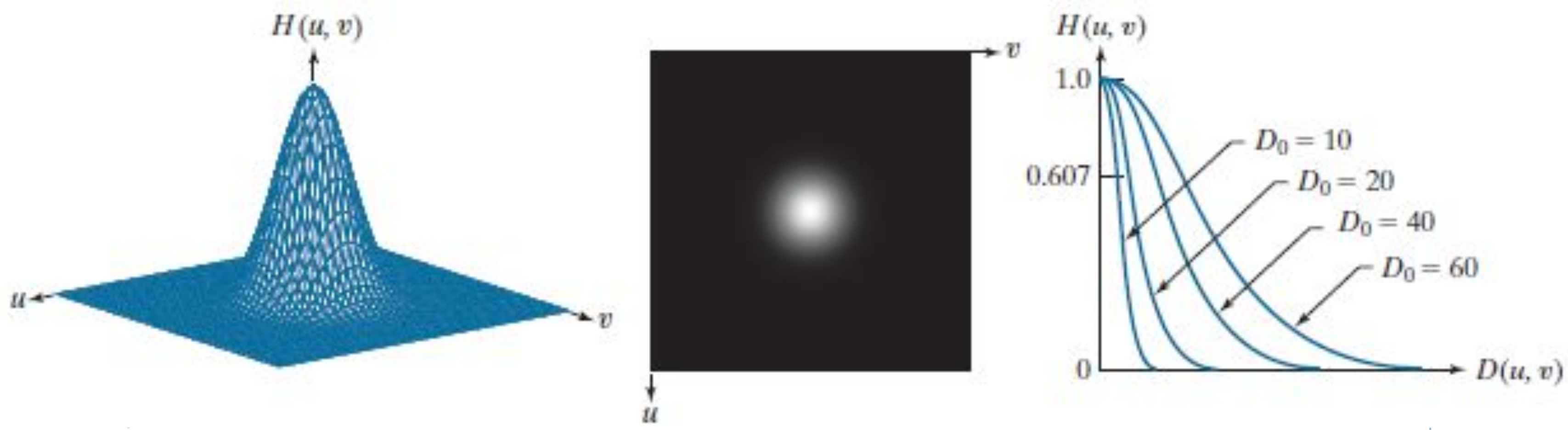
$$H(u,v) = e^{-D^2(u,v)/2\sigma^2}$$

σ is a measure of the spread of the Gaussian curve

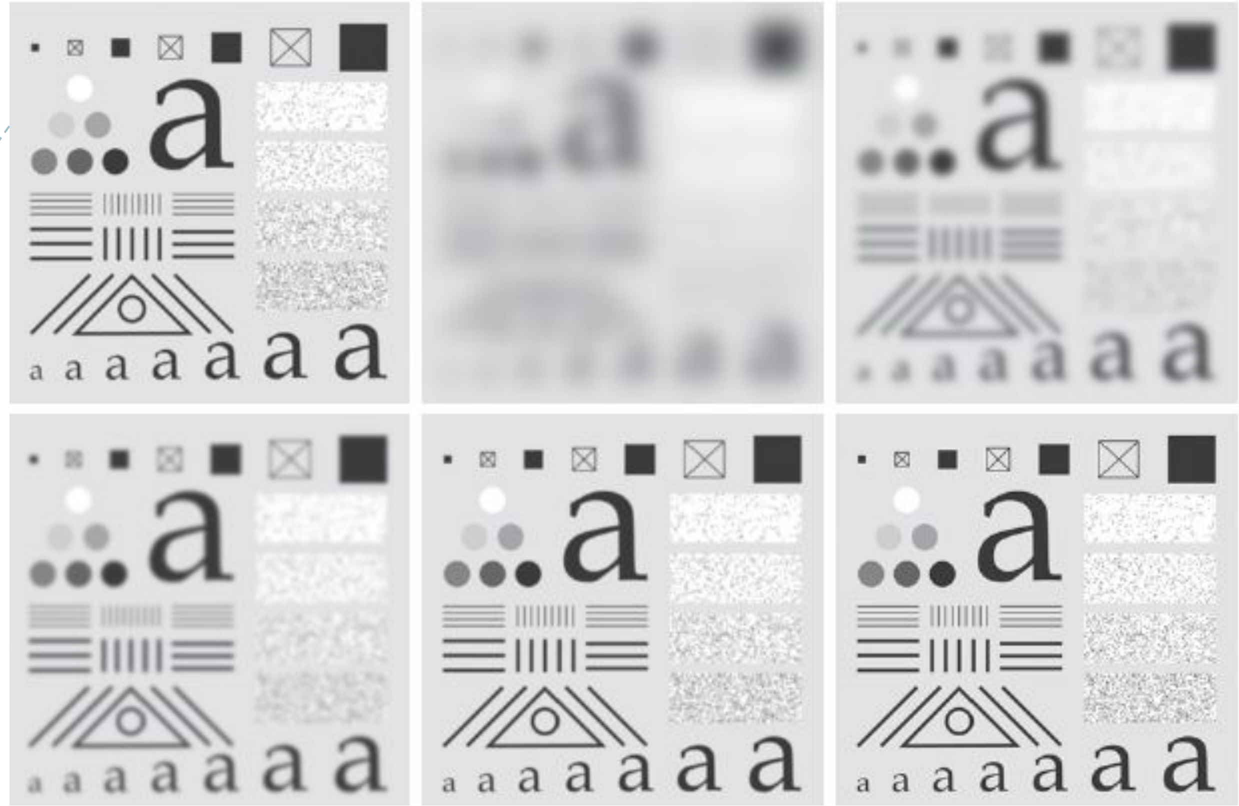
Inverse Fourier Transform of a GLPF is also Gaussian

Advantages:

- Since the transform & its inverse both are Gaussian, therefore they are real. This facilitates analysis because we do not have to be concerned with complex nos.
- Gaussian curves are intuitive & easy to manipulate.
- These functions behave reciprocally wrt one another. This helps considerably in developing a solid understanding of the properties of filtering in both spatial & frequency domains because they lend themselves to familiar analytical interpretation.



(a) Perspective plot of a GLPF transfer function.
(b) Function displayed as an image.
(c) Radial cross sections for various values of D_o .

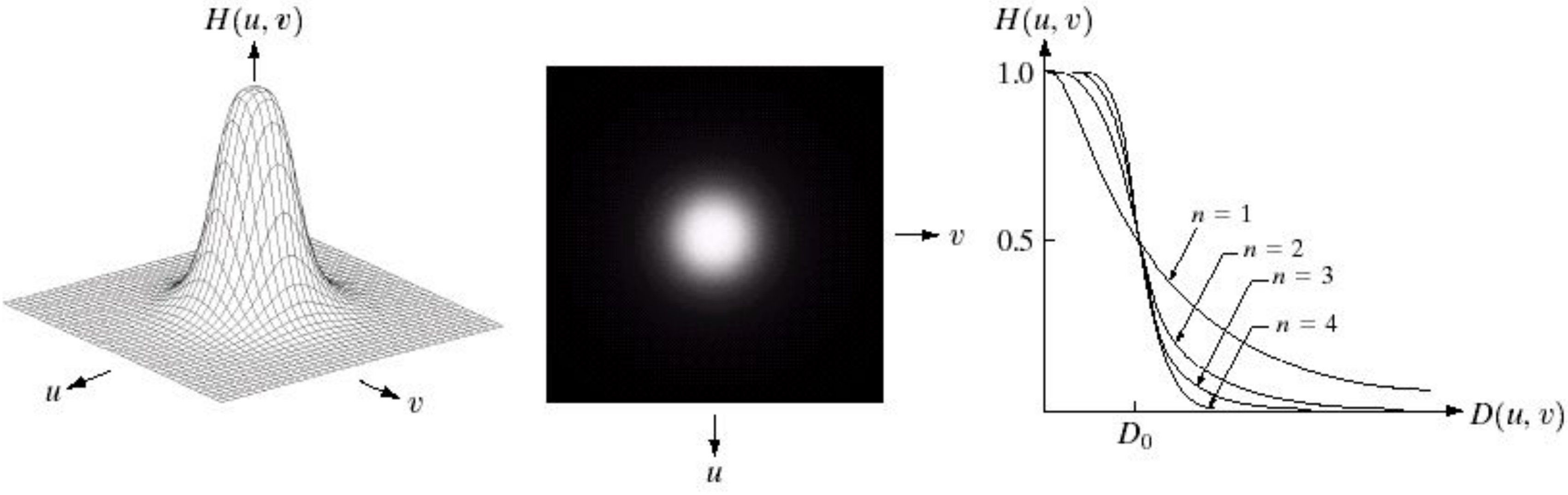


(a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460

Butterworth Lowpass Filters

The transfer function of a BLPF of order n & with a cut off frequency at a distance D_o from the origin is given as:

$$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)}{D_0}\right]^{2n}}$$

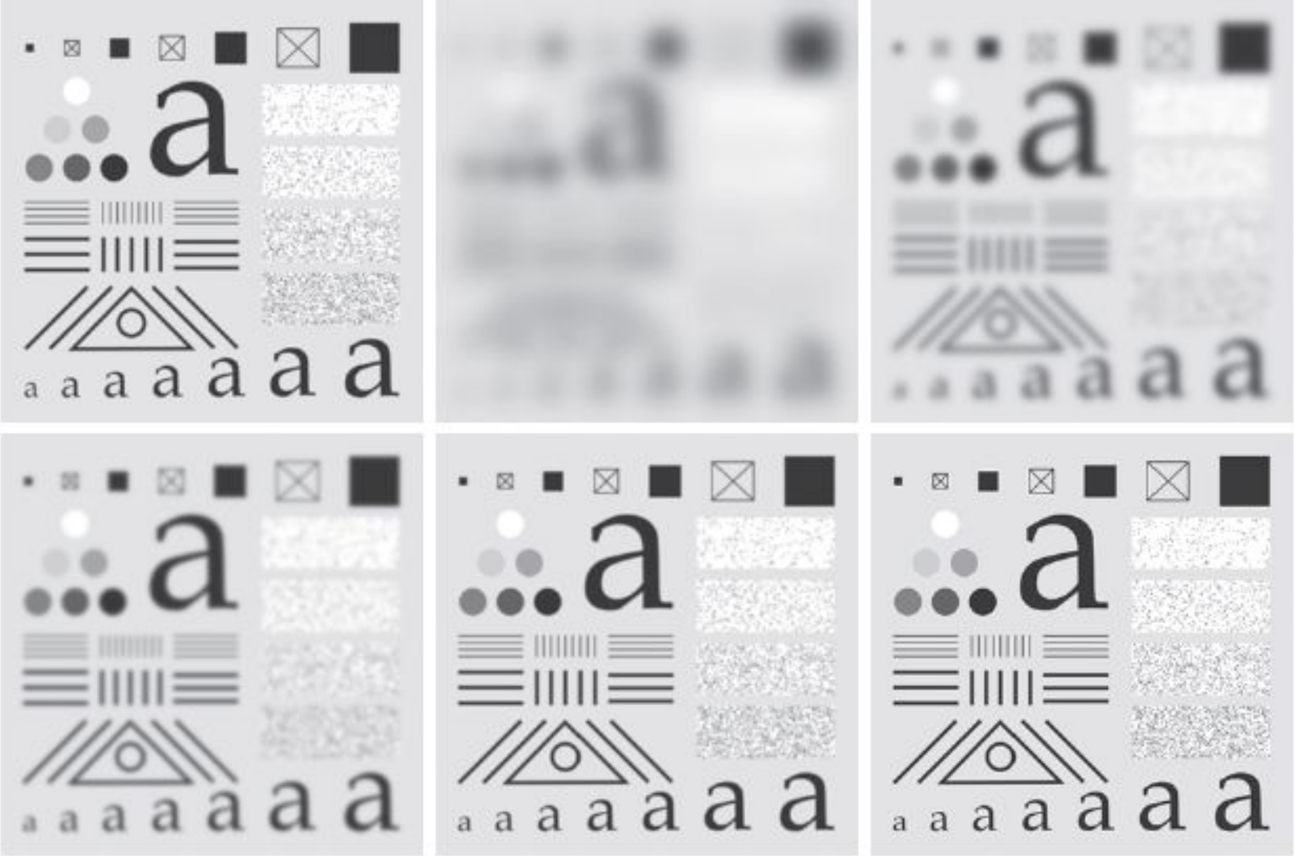


a b c

(a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Note:

- The BLPF (unlike ILPF), does not have a sharp discontinuity that establishes a clear cut off between passed & filtered frequencies.
- A BLPF of order 1 has no ringing. Ringing generally is imperceptible in filters of order of 2, but can become a significant factor in filters of higher order.



(a) Original image of size 688 × 688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460

Sharpening Frequency Domain Filters

Edges & other abrupt changes in gray levels are associated with high frequency components.

Image sharpening can be achieved in the frequency domain by a *highpass filter*ing process, which attenuates the low frequency components without disturbing high frequency components in the Fourier transform.

Lowpass Filters

Precisely Reverse operation

Highpass Filters

Hence the transfer function of Highpass filters can be obtained as follows:

$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

Ideal Highpass Filters

A 2-D IHPF is defined as

$$H(u,v) = \begin{cases} 0 \; if \; D(u,v) \leq D_0 \\ 1 \; if \; D(u,v) > D_0 \end{cases}$$

Gaussian Highpass Filters

The transfer function of a GHPF is given as:

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

σ is a measure of the spread of the Gaussian curve

Butterworth Highpass Filters

The transfer function of a BHPF of order n & with a cut off frequency at a distance D_o from the origin is given as:

If this is BLPF, what is the BHPF?

$$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)}{D_0}\right]^{2n}}$$

Module I

Introduction and Fundamentals: Motivation and Perspective, Applications, Components of Image Processing System, Element of Visual Perception, A Simple Image Model, Sampling and Quantization, Some Basic Relationships between Pixels.

Intensity Transformations and Spatial Filtering: Introduction, Some Basic Intensity Transformation Functions, Histogram Processing, Histogram Equalization, Histogram Specification, Local Enhancement, Enhancement using Arithmetic/Logic Operations – Image Subtraction, Image Averaging, Basics of Spatial Filtering, Smoothing - Mean Filter, Order Statistics Filters, Sharpening – The Laplacian.

Filtering in the Frequency Domain: Fourier Transform and the Frequency Domain, Basis of Filtering in Frequency Domain

Want big impact?
Use big image.